Gilbert Strang - Introduction to Linear Algebra
Chapters 1 - 7

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Contents

1 Chapter 1: Introduction to Vectors

1.1 Vectors and Linear Combinations

Problems 1-9 are about addition of vectors and linear combinations.

1. **Describe geometrically (line, plane, or all of R^3) all linear combinations of**

   (a) $\begin{bmatrix}1\\2\\3\end{bmatrix}$ *and* $\begin{bmatrix}3\\6\\9\end{bmatrix}$ **(b)** $\begin{bmatrix}1\\0\\0\end{bmatrix}$ *and* $\begin{bmatrix}0\\2\\3\end{bmatrix}$ **(c)** $\begin{bmatrix}2\\0\\0\end{bmatrix}$ *and* $\begin{bmatrix}0\\2\\2\end{bmatrix}$ *and* $\begin{bmatrix}2\\2\\3\end{bmatrix}$

2. **Draw $v = \begin{bmatrix}4\\1\end{bmatrix}$ and $w = \begin{bmatrix}-2\\2\end{bmatrix}$ and $v + w$ and $v - w$ in a single xy plane.**

3. **If $v + w = \begin{bmatrix}5\\1\end{bmatrix}$ and $v - w = \begin{bmatrix}1\\5\end{bmatrix}$, compute and draw the vectors v and w.**

4. **From $v = \begin{bmatrix}2\\1\end{bmatrix}$ and $w = \begin{bmatrix}1\\2\end{bmatrix}$, find the components of $3v + w$ and $cv + dw$.**

5. **Compute $u + v + w$ and $2u + 2v + w$. How do you know u, v, w lie in a plane?**

   These lie in a plane because $w = cu + dv$. Find c and d

   $$u = \begin{bmatrix}1\\2\\3\end{bmatrix}, \quad v = \begin{bmatrix}-3\\1\\-2\end{bmatrix}, \quad w = \begin{bmatrix}2\\-3\\-1\end{bmatrix}.$$

6. **Every combination of $v = (1, -2, 1)$ and $w = (0, 1, -1)$ has components that add to ______. Find c and d so that $cv + dw = (3, 3, -6)$. Why is $(3, 3, 6)$ impossible?**

7. **In the xy plane mark all nine of these linear combinations:**

   $$c\begin{bmatrix}2\\1\end{bmatrix} + d\begin{bmatrix}0\\1\end{bmatrix} \quad \textbf{with} \quad c = 0, 1, 2 \quad \textbf{and} \quad d = 0, 1, 2.$$

8. **The parallelogram in Figure 1.1 has diagonal $v + w$. What is its other diagonal? What is the sum of the two diagonals? Draw that vector sum.**

9. **If three corners of a parallelogram are $(1, 1)$, $(4, 2)$, and $(1, 3)$, what are all three of the possible fourth corners? Draw two of them.**

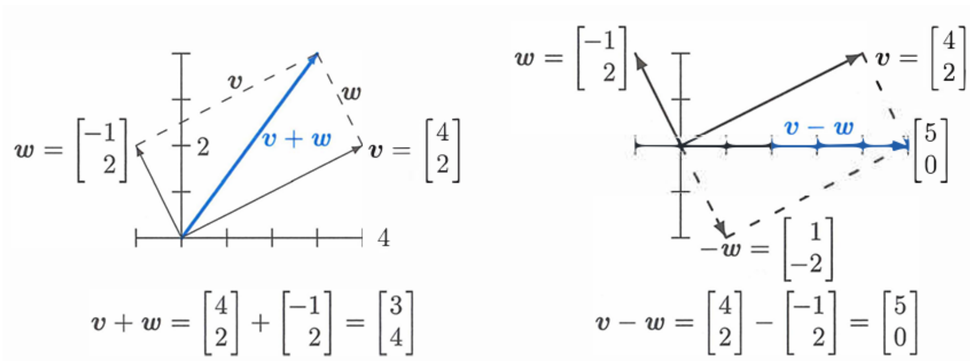


Figure 1.1: Vector addition $v + w = (3, 4)$ produces the diagonal of a parallelogram. The reverse of w is $-w$. The linear combination on the right is $v - w = (5, 0)$.

Problems 10–14 are about special vectors on cubes and clocks in Figure 1.4.

10. **Which point of the cube is $i + j$? Which point is the vector sum of $i = (1, 0, 0)$ and $j = (0, 1, 0)$ and $k = (0, 0, 1)$? Describe all points (x, y, z) in the cube.**

11. **Four corners of this unit cube are $(0,0,0), (1,0,0), (0,1,0), (0,0,1)$. What are the other four corners? Find the coordinates of the center point of the cube. The center points of the six faces are ____________. The cube has how many edges?**

12. **Review Question. In xyz space, where is the plane of all linear combinations of $\boldsymbol{i} = (1,0,0)$ and $\boldsymbol{i}+\boldsymbol{j} = (1,1,0)$?**

13. **(a) What is the sum V of the twelve vectors that go from the center of a clock to the hours 1:00, 2:00, …, 12:00?**

    (b) If the 2:00 vector is removed, why do the 11 remaining vectors add to 8:00?

    (c) What are the x, y components of that 2:00 vector $\boldsymbol{v} = (\cos\theta, \sin\theta)$?

14. **Suppose the twelve vectors start from 6:00 at the bottom instead of $(0,0)$ at the center. The vector to 12:00 is doubled to $(0,2)$. The new twelve vectors add to ____________.**

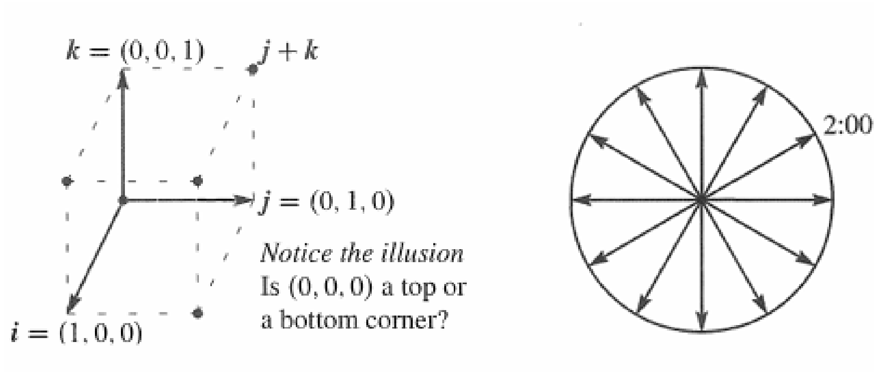


Figure 1.4: Unit cube from i,j, k and twelve clock vectors.

Problems 15–19 go further with linear combinations of v and w (Figure 1.5a).

15. **Figure 1.5a shows $\frac{1}{2}\boldsymbol{v} + \frac{1}{2}\boldsymbol{w}$. Mark the points $\frac{3}{4}\boldsymbol{v} + \frac{1}{4}\boldsymbol{w}$ and $\frac{1}{4}\boldsymbol{v} + \frac{3}{4}\boldsymbol{w}$ and $\boldsymbol{v}+\boldsymbol{w}$.**

16. **Mark the point $-v + 2w$ and any other combination $cv + dw$ with $c + d = 1$. Draw the line of all combinations that have $c + d = 1$.**

17. **Locate $\frac{1}{3}\boldsymbol{v} + \frac{1}{3}\boldsymbol{w}$ and $\frac{2}{3}\boldsymbol{v} + \frac{2}{3}\boldsymbol{w}$. The combinations $cv + cw$ fill out what line?**

18. **Restricted by $0 \le c \le 1$ and $0 \le d \le 1$, shade in all combinations $cv + dw$.**

19. **Restricted only by $c \ge 0$ and $d \ge 0$ draw the "cone" of all combinations $cv + dw$.**

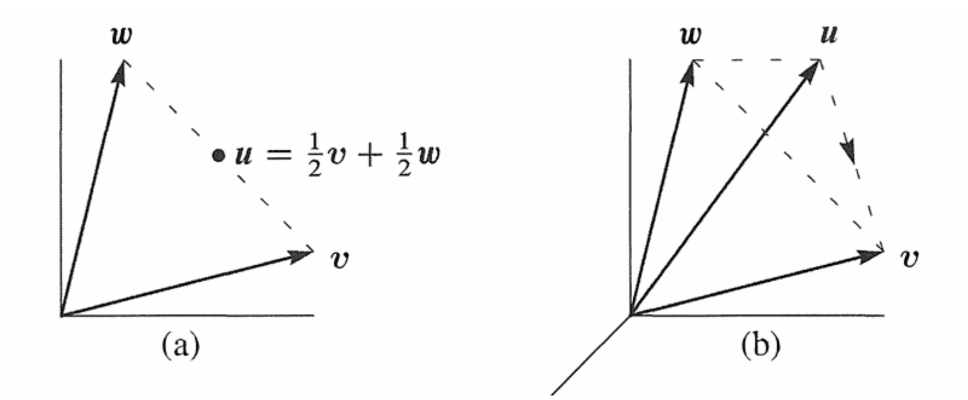


Figure 1.5: Problems 15-19 in a plane   Problems 20-25 in 3-dimensional space

Problems 20-25 deal with u, v, w in three-dimensional space (see Figure 1.5b).

20. **Locate $\frac{1}{2}\boldsymbol{v}+\frac{1}{2}\boldsymbol{w}+\frac{1}{2}\boldsymbol{v}$ and $\frac{1}{2}\boldsymbol{v}+\frac{1}{2}\boldsymbol{w}+\frac{1}{2}\boldsymbol{w}$ in Figure 1.5b. Challenge problem: Under what restrictions on c, d, e, will the combinations $c\boldsymbol{u}+d\boldsymbol{v}+e\boldsymbol{w}$ fill in the dashed triangle? To stay in the triangle, one requirement is $c \geq 0, d \geq 0, e \geq 0$.**

21. **The three sides of the dashed triangle are $\boldsymbol{v}-\boldsymbol{u}$ and $\boldsymbol{w}-\boldsymbol{v}$ and $\boldsymbol{u}-\boldsymbol{w}$. Their sum is ____________. Draw the head-to-tail addition around a plane triangle of $(3,1)$ plus $(-1,1)$ plus $(-2,-2)$.**

22. **Shade in the pyramid of combinations $c\boldsymbol{u}+d\boldsymbol{v}+e\boldsymbol{w}$ with $c \geq 0$, $d \geq 0$, $e \geq 0$ and $c+d+e \leq 1$. Mark the vector $\frac{1}{2}(\boldsymbol{u}+\boldsymbol{v}+\boldsymbol{w})$ as inside or outside this pyramid.**

23. **If you look at all combinations of those $\boldsymbol{u}, \boldsymbol{v}$, and $\boldsymbol{w}$, is there any vector that can't be produced from $c\boldsymbol{u}+d\boldsymbol{v}+e\boldsymbol{w}$? Different answer if $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are all in ____________.**

24. **Which vectors are combinations of $\boldsymbol{u}$ and $\boldsymbol{v}$, and also combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$?**

25. **Draw vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ so that their combinations $c\boldsymbol{u}+d\boldsymbol{v}+e\boldsymbol{w}$ fill only a line. Find vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ so that their combinations $c\boldsymbol{u}+d\boldsymbol{v}+e\boldsymbol{w}$ fill only a plane.**

26. **What combination $c\begin{bmatrix}2\\1\end{bmatrix}+d\begin{bmatrix}3\\1\end{bmatrix}$ produces $\begin{bmatrix}14\\8\end{bmatrix}$? Express this question as two equations for the coefficients c and d in the linear combination.**

Challenge Problems

27. **How many corners does a cube have in 4 dimensions? How many 3D faces? How many edges? A typical corner is $(0,0,1,0)$. A typical edge goes to $(0,1,0,0)$.**

28. **Find vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ so that $\boldsymbol{v}+\boldsymbol{w}=(4,5,6)$ and $\boldsymbol{v}-\boldsymbol{w}=(2,5,8)$. This is a question with ____________ unknown numbers, and an equal number of equations to find those numbers.**

29. **Find two different combinations of the three vectors $\boldsymbol{u}=(1,3)$ and $\boldsymbol{v}=(2,7)$ and $\boldsymbol{w}=(1,5)$ that produce $\boldsymbol{b}=(0,1)$. Slightly delicate question: If I take any three vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ in the plane, will there always be two different combinations that produce $\boldsymbol{b}=(0,1)$?**

30. **The linear combinations of $\boldsymbol{v}=(a,b)$ and $\boldsymbol{w}=(c,d)$ fill the plane unless ____________.
Find four vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}, \boldsymbol{z}$ with four components each so that their combinations $c\boldsymbol{u}+d\boldsymbol{v}+e\boldsymbol{w}+f\boldsymbol{z}$ produce all vectors (b_1, b_2, b_3, b_4) in four-dimensional space.**

31. **Write down three equations for c, d, e so that $c\boldsymbol{u}+d\boldsymbol{v}+e\boldsymbol{w}=\boldsymbol{b}$. Can you somehow find c, d, e for $\boldsymbol{b}$?**

$$\boldsymbol{u}=\begin{bmatrix}2\\-1\\0\end{bmatrix}, \quad \boldsymbol{v}=\begin{bmatrix}-1\\2\\-1\end{bmatrix}, \quad \boldsymbol{w}=\begin{bmatrix}0\\-1\\2\end{bmatrix}, \quad \boldsymbol{b}=\begin{bmatrix}1\\0\\0\end{bmatrix}.$$

1.2 Lengths and Dot Products

1. **Calculate the dot products $\boldsymbol{u}\cdot\boldsymbol{v}$ and $\boldsymbol{u}\cdot\boldsymbol{w}$ and $\boldsymbol{u}\cdot(\boldsymbol{v}+\boldsymbol{w})$ and $\boldsymbol{w}\cdot\boldsymbol{v}$:**

$$\boldsymbol{u}=\begin{bmatrix}-.6\\.8\end{bmatrix}, \qquad \boldsymbol{v}=\begin{bmatrix}4\\3\end{bmatrix}, \qquad \boldsymbol{w}=\begin{bmatrix}1\\2\end{bmatrix}.$$

2. **Compute the lengths $\|\boldsymbol{u}\|$ and $\|\boldsymbol{v}\|$ and $\|\boldsymbol{w}\|$ of those vectors. Check the Schwarz inequalities $|\boldsymbol{u}\cdot\boldsymbol{v}| \leq \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|$ and $\boldsymbol{v}\cdot\boldsymbol{w} \leq \|\boldsymbol{v}\|\,\|\boldsymbol{w}\|$.**

3. **Find unit vectors in the directions of $\boldsymbol{v}$ and $\boldsymbol{w}$ in Problem 1, and the cosine of the angle θ. Choose vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ that make $0°$, $90°$, and $180°$ angles with $\boldsymbol{w}$.**

4. **For any *unit* vectors $\boldsymbol{v}$ and $\boldsymbol{w}$, find the dot products (actual numbers) of**

   (a) $\boldsymbol{v}$ **and** $-\boldsymbol{v}$ **(b)** $\boldsymbol{v}+\boldsymbol{w}$ **and** $\boldsymbol{v}-\boldsymbol{w}$ **(c)** $\boldsymbol{v}-2\boldsymbol{w}$ **and** $\boldsymbol{v}+2\boldsymbol{w}$

5. **Find unit vectors $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ in the directions of $\boldsymbol{v}=(1,3)$ and $\boldsymbol{w}=(2,1,2)$. Find unit vectors $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$ that are perpendicular to $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$.**

6. **(a) Describe every vector $\boldsymbol{w} = (w_1, w_2)$ that is perpendicular to $\boldsymbol{v} = (2, -1)$.**
   (b) All vectors perpendicular to $\boldsymbol{V} = (1, 1, 1)$ lie on a ____________ in 3 dimensions.
   (c) The vectors perpendicular to both $(1, 1, 1)$ and $(1, 2, 3)$ lie on a ____________.

7. **Find the angle θ (from its cosine) between these pairs of vectors:**

   (a) $\boldsymbol{v} = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ **and** $\boldsymbol{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ **(b)** $\boldsymbol{v} = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$ **and** $\boldsymbol{w} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$

   (c) $\boldsymbol{v} = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ **and** $\boldsymbol{w} = \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$ **(d)** $\boldsymbol{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ **and** $\boldsymbol{w} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$

8. **True or false (give a reason if true or find a counterexample if false):**
   (a) If $\boldsymbol{u} = (1, 1, 1)$ is perpendicular to $\boldsymbol{v}$ and $\boldsymbol{w}$, then $\boldsymbol{v}$ is parallel to $\boldsymbol{w}$.
   (b) If $\boldsymbol{u}$ is perpendicular to $\boldsymbol{v}$ and $\boldsymbol{w}$, then $\boldsymbol{u}$ is perpendicular to $\boldsymbol{v} + 2\boldsymbol{w}$.
   (c) If $\boldsymbol{u}$ and $\boldsymbol{v}$ are perpendicular unit vectors then $\|\boldsymbol{u} - \boldsymbol{v}\| = \sqrt{2}$. Yes!

9. **The slopes of the arrows from $(0, 0)$ to (v_1, v_2) and (w_1, w_2) are v_2/v_1 and w_2/w_1. Suppose the product v_2w_2/v_1w_1 of those slopes is -1. Show that $\boldsymbol{v} \cdot \boldsymbol{w} = 0$ and the vectors are perpendicular. (The line $y = 4x$ is perpendicular to $y = -\frac{1}{4}x$.)**

10. **Draw arrows from $(0, 0)$ to the points $\boldsymbol{v} = (1, 2)$ and $\boldsymbol{w} = (-2, 1)$. Multiply their slopes. That answer is a signal that $\boldsymbol{v} \cdot \boldsymbol{w} = 0$ and the arrows are ____________.**

11. **If $\boldsymbol{v} \cdot \boldsymbol{w}$ is negative, what does this say about the angle between $\boldsymbol{v}$ and $\boldsymbol{w}$? Draw a 3-dimensional vector $\boldsymbol{v}$ (an arrow), and show where to find all $\boldsymbol{w}$'s with $\boldsymbol{v} \cdot \boldsymbol{w} < 0$.**

12. **With $\boldsymbol{v} = (1, 1)$ and $\boldsymbol{w} = (1, 5)$ choose a number c so that $\boldsymbol{w} - c\boldsymbol{v}$ is perpendicular to $\boldsymbol{v}$. Then find the formula for c starting from any nonzero $\boldsymbol{v}$ and $\boldsymbol{w}$.**

13. **Find nonzero vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ that are perpendicular to $(1, 0, 1)$ and to each other.**

14. **Find nonzero vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ that are perpendicular to $(1, 1, 1)$ and to each other.**

15. **The geometric mean of $x = 2$ and $y = 8$ is $\sqrt{xy} = 4$. The arithmetic mean is larger: $\frac{1}{2}(x + y) =$ ____________. This would come in Example 6 from the Schwarz inequality for $\boldsymbol{v} = (\sqrt{2}, \sqrt{8})$ and $\boldsymbol{w} = (\sqrt{8}, \sqrt{2})$. Find $\cos\theta$ for this $\boldsymbol{v}$ and $\boldsymbol{w}$.**

16. **How long is the vector $\boldsymbol{v} = (1, 1, \ldots, 1)$ in 9 dimensions? Find a unit vector $\boldsymbol{u}$ in the same direction as $\boldsymbol{v}$ and a unit vector $\boldsymbol{w}$ that is perpendicular to $\boldsymbol{v}$.**

17. **What are the cosines of the angles α, β, θ between the vector $(1, 0, -1)$ and the unit vectors $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ along the axes? Check the formula $\cos^2\alpha + \cos^2\beta + \cos^2\theta = 1$.**

Problems 18-28 lead to the main facts about lengths and angles in triangles.

18. **The parallelogram with sides $\boldsymbol{v} = (4, 2)$ and $\boldsymbol{w} = (-1, 2)$ is a rectangle. Check the Pythagoras formula $a^2 + b^2 = c^2$ which is for *right triangles only*:**

$$\boxed{(\textbf{length of } \boldsymbol{v})^2 + (\textbf{length of } \boldsymbol{w})^2 = (\textbf{length of } \boldsymbol{v} + \boldsymbol{w})^2.}$$

19. **(Rules for dot products) These equations are simple but useful:**

   (a) $\boldsymbol{v} \cdot \boldsymbol{w} = \boldsymbol{w} \cdot \boldsymbol{v}$ **(b)** $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w}$ **(c)** $(c\boldsymbol{v}) \cdot \boldsymbol{w} = c(\boldsymbol{v} \cdot \boldsymbol{w})$

   Use (2) with $\boldsymbol{u} = \boldsymbol{v} + \boldsymbol{w}$ to prove $\|\boldsymbol{v} + \boldsymbol{w}\|^2 = \boldsymbol{v} \cdot \boldsymbol{v} + 2\boldsymbol{v} \cdot \boldsymbol{w} + \boldsymbol{w} \cdot \boldsymbol{w}$.

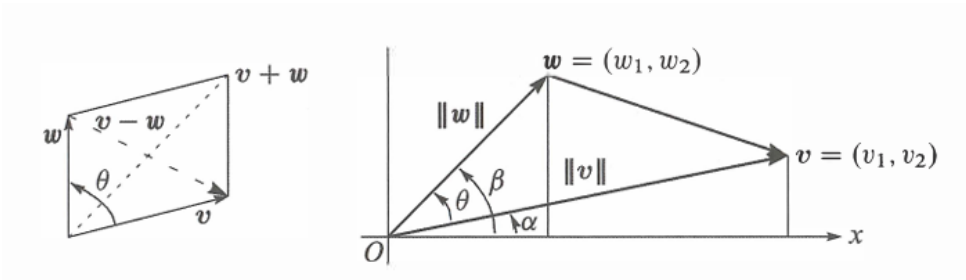
20. **The "Law of Cosines" comes from $(\boldsymbol{v} - \boldsymbol{w}) \cdot (\boldsymbol{v} - \boldsymbol{w}) = \boldsymbol{v} \cdot \boldsymbol{v} - 2\boldsymbol{v} \cdot \boldsymbol{w} + \boldsymbol{w} \cdot \boldsymbol{w}$:**

$$\textbf{Cosine Law} \qquad \|\boldsymbol{v} - \boldsymbol{w}\|^2 = \|\boldsymbol{v}\|^2 - 2\|\boldsymbol{v}\|\,\|\boldsymbol{w}\|\cos\theta + \|\boldsymbol{w}\|^2.$$

   Draw a triangle with sides $\boldsymbol{v}$ and $\boldsymbol{w}$ and $\boldsymbol{v} - \boldsymbol{w}$. Which of the angles is θ?

21. **The triangle inequality says: (length of $v + w$) $\le$ (length of v) + (length of w).**
Problem 19 found $\|v + w\|^2 = \|v\|^2 + 2v \cdot w + \|w\|^2$.
Increase by $\|v\|\,\|w\|$ to show that $v \cdot w$ can not exceed $\|v\|\,\|w\|$:

Triangle inequality $\quad \|v+w\|^2 \le (\|v\|+\|w\|)^2$ **or** $\boxed{\|v + w\| \le \|v\| + \|w\|}$.



22. **The Schwarz inequality $|v \cdot w| \le \|v\|\,\|w\|$ by algebra instead of trigonometry:**
   (a) **Multiply out both sides of $(v_1w_1 + v_2w_2)^2 \le (v_1^2 + v_2^2)(w_1^2 + w_2^2)$.**
   (b) **Show that the difference between those two sides equals $(v_1w_2 - v_2w_1)^2$. This cannot be negative since it is a square—so the inequality is true.**

23. **The figure shows that $\cos\alpha = v_1/\|v\|$ and $\sin\alpha = v_2/\|v\|$. Similarly $\cos\beta$ is ______ and $\sin\beta$ is ______. The angle θ is $\beta - \alpha$. Substitute into the trigonometry formula $\cos\beta\cos\alpha + \sin\beta\sin\alpha$ for $\cos(\beta - \alpha)$ to find $\cos\theta = v \cdot w/\|v\|\,\|w\|$.**

24. **One-line proof of the inequality $|u \cdot U| \le 1$ for unit vectors (u_1, u_2) and (U_1, U_2):**

$$|u \cdot U| \le |u_1||U_1| + |u_2||U_2| \le \frac{u_1^2 + U_1^2}{2} + \frac{u_2^2 + U_2^2}{2} = 1.$$

Put $(u_1, u_2) = (.6, .8)$ and $(U_1, U_2) = (.8, .6)$ in that whole line and find $\cos\theta$.

25. **Why is $|\cos\theta|$ never greater than 1 in the first place?**

26. **(Recommended) Draw a parallelogram.**

27. **Parallelogram with two sides v and w. Show that the squared diagonal lengths $\|v+w\|^2+\|v-w\|^2$ add to the sum of four squared side lengths $2\|v\|^2 + 2\|w\|^2$.**

28. **If $v = (1, 2)$ draw all vectors $w = (x, y)$ in the xy-plane with $v \cdot w = x + 2y = 5$. Why do those w's lie along a line? Which is the shortest w?**

29. **(Recommended) If $\|v\| = 5$ and $\|w\| = 3$, what are the smallest and largest possible values of $\|v - w\|$? What are the smallest and largest possible values of $v \cdot w$?**

Challenge Problems

30. **Can three vectors in the xy-plane have $u \cdot v < 0$ and $v \cdot w < 0$ and $u \cdot w < 0$?**
I don't know how many vectors in xyz space can have all negative dot products. (Four of those vectors in the plane would certainly be impossible….)

31. **Pick any numbers that add to $x + y + z = 0$. Find the angle between your vector $v = (x, y, z)$ and the vector $w = (z, x, y)$. Challenge question: Explain why $v \cdot w/\|v\|\,\|w\|$ is always $-\frac{1}{2}$.**

32. **How could you prove $\sqrt[3]{xyz} \le \frac{1}{3}(x + y + z)$ (geometric mean $\le$ arithmetic mean)?**

33. **Find 4 perpendicular unit vectors of the form $\left(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}\right)$: Choose $+$ or $-$.**

34. **Using $v = \text{randn}(3, 1)$ in MATLAB, create a random unit vector $u = v/\|v\|$.**
Using $V = \text{randn}(3, 30)$ create 30 more random unit vectors U_j.
What is the average size of the dot products $|u \cdot U_j|$?
In calculus, the average is $\displaystyle\int_0^\pi |\cos\theta|\, d\theta/\pi = 2/\pi$.

1.3 Matrices

1. **Find the linear combination $3s_1 + 4s_2 + 5s_3 = \boldsymbol{b}$. Then write $\boldsymbol{b}$ as a matrix-vector multiplication $S\boldsymbol{x}$, with $3, 4, 5$ in $\boldsymbol{x}$. Compute the three dot products (row of S) $\cdot\boldsymbol{x}$:**

$$\boldsymbol{s}_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \quad \boldsymbol{s}_2 = \begin{bmatrix}0\\1\\1\end{bmatrix}, \quad \boldsymbol{s}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix} \quad \textbf{go into the columns of } S.$$

2. **Solve these equations $S\boldsymbol{y} = \boldsymbol{b}$ with $\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3$ in the columns of S:**

$$\begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix} \quad \textbf{and} \quad \begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} = \begin{bmatrix}1\\4\\9\end{bmatrix}.$$

S is a sum matrix. The sum of the first 3 odd numbers is ____________.

3. **Solve these three equations for y_1, y_2, y_3 in terms of c_1, c_2, c_3:**

$$S\boldsymbol{y} = \boldsymbol{c} \qquad \begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} = \begin{bmatrix}c_1\\c_2\\c_3\end{bmatrix}.$$

Write the solution $\boldsymbol{y}$ as a matrix $\boldsymbol{A} = S^{-1}$ times the vector $\boldsymbol{c}$. Are the columns of S independent or dependent?

4. **Find a combination $x_1\boldsymbol{w}_1 + x_2\boldsymbol{w}_2 + x_3\boldsymbol{w}_3$ that gives the zero vector with $x_1 = 1$:**

$$\boldsymbol{w}_1 = \begin{bmatrix}1\\2\\3\end{bmatrix} \qquad \boldsymbol{w}_2 = \begin{bmatrix}4\\5\\6\end{bmatrix} \qquad \boldsymbol{w}_3 = \begin{bmatrix}7\\8\\9\end{bmatrix}$$

Those vectors are (independent) (dependent). The three vectors lie in a ____________. The matrix W with those three columns is not invertible.

5. **The rows of that matrix W produce three vectors (I write them as columns):**

$$\boldsymbol{r}_1 = \begin{bmatrix}1\\4\\7\end{bmatrix} \qquad \boldsymbol{r}_2 = \begin{bmatrix}2\\5\\8\end{bmatrix} \qquad \boldsymbol{r}_3 = \begin{bmatrix}3\\6\\9\end{bmatrix}$$

Linear algebra says that these vectors must also lie in a plane. There must be many combinations with $y_1\boldsymbol{r}_1 + y_2\boldsymbol{r}_2 + y_3\boldsymbol{r}_3 = \boldsymbol{0}$. Find two sets of y's.

6. **Which numbers c give dependent columns so a combination of columns equals zero?**

$$\begin{bmatrix}1&1&0\\3&2&1\\7&4&c\end{bmatrix} \quad \begin{bmatrix}1&0&c\\1&1&0\\0&1&1\end{bmatrix} \quad \begin{bmatrix}c&c&c\\2&1&5\\3&3&6\end{bmatrix} \quad \begin{array}{l}\textbf{maybe}\\\textbf{always}\\\textbf{independent for } c \neq 0\end{array}$$

7. **If the columns combine into $A\boldsymbol{x} = \boldsymbol{0}$, then each of the rows has $\boldsymbol{r} \cdot \boldsymbol{x} = 0$:**

$$\begin{bmatrix}a_1 & a_2 & a_3\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \quad \textbf{By rows} \quad \begin{bmatrix}\boldsymbol{r}_1\cdot\boldsymbol{x}\\\boldsymbol{r}_2\cdot\boldsymbol{x}\\\boldsymbol{r}_3\cdot\boldsymbol{x}\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}.$$

The three rows also lie in a plane. Why is that plane perpendicular to $\boldsymbol{x}$?

8. **Moving to a 4×4 difference equation $A\boldsymbol{x} = \boldsymbol{b}$, find the four components x_1, x_2, x_3, x_4. Then write this solution as $\boldsymbol{x} = A^{-1}\boldsymbol{b}$ to find the inverse matrix:**

$$A\boldsymbol{x} = \begin{bmatrix}1&0&0&0\\-1&1&0&0\\0&-1&1&0\\0&0&-1&1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} = \begin{bmatrix}b_1\\b_2\\b_3\\b_4\end{bmatrix} = \boldsymbol{b}.$$

9. What is the cyclic 4 by 4 difference matrix C? It will have 1 and -1 in each row and each column. Find all solutions $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$ to $C\boldsymbol{x} = \mathbf{0}$. The four columns of C lie in a "three-dimensional hyperplane" inside four-dimensional space.

10. A forward difference matrix Δ is upper triangular:

$$\Delta \boldsymbol{z} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} z_2 - z_1 \\ z_3 - z_2 \\ 0 - z_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \boldsymbol{b}.$$

Find z_1, z_2, z_3 from b_1, b_2, b_3. What is the inverse matrix in $\boldsymbol{z} = \Delta^{-1}\boldsymbol{b}$?

11. Show that the forward differences $(t+1)^2 - t^2$ are $2t+1 =$ odd numbers. As in calculus, the difference $(t+1)^n - t^n$ will begin with the derivative of t^n, which is __________________.

12. The last lines of the Worked Example say that the 4×4 centered difference matrix in (16) is invertible. Solve $C\boldsymbol{x} = (b_1, b_2, b_3, b_4)$ to find its inverse in $\boldsymbol{x} = C^{-1}\boldsymbol{b}$.

Challenge Problems

13. The very last words say that the 5×5 centered difference matrix is not invertible. Write down the 5 equations $C\boldsymbol{x} = \boldsymbol{b}$. Find a combination of left sides that gives zero. What combination of b_1, b_2, b_3, b_4, b_5 must be zero? (The 5 columns lie on a "4-dimensional hyperplane" in 5-dimensional space. Hard to visualize.)

14. If (a, b) is a multiple of (c, d) with $abcd \neq 0$, show that (a, c) is a multiple of (b, d). This is surprisingly important: two columns are falling on one line. You could use numbers first to see how a, b, c, d are related. The question will lead to:

$$\textbf{If } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \textbf{ has dependent rows, then it also has dependent columns.}$$

2 Chapter 2: Solving Linear Equations

2.1 Vectors and Linear Equations

1. **With $A = I$ (the identity matrix) draw the planes in the row picture. Three sides of a box meet at the solution $\boldsymbol{x} = (x, y, z) = (2, 3, 4)$:**

$$\begin{aligned} 1x + 0y + 0z &= 2 \\ 0x + 1y + 0z &= 3 \\ 0x + 0y + 1z &= 4 \end{aligned} \quad \textbf{or} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

   Draw the vectors in the column picture. Two times column 1 plus three times column 2 plus four times column 3 equals the right side $\boldsymbol{b}$.

2. **If the equations in Problem 1 are multiplied by $2, 3, 4$ they become $D\boldsymbol{x} = \boldsymbol{B}$:**

$$\begin{aligned} 2x + 0y + 0z &= 4 \\ 0x + 3y + 0z &= 9 \\ 0x + 0y + 4z &= 16 \end{aligned} \quad \textbf{or} \quad D\boldsymbol{x} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 16 \end{bmatrix} = \boldsymbol{B}.$$

   Why is the row picture the same? Is the solution $\boldsymbol{X}$ the same as x? What is changed in the column picture—the columns or the right combination to give $\boldsymbol{B}$?

3. **If equation 1 is added to equation 2, which of these are changed: the planes in the row picture, the vectors in the column picture, the coefficient matrix, the solution? The new equations in Problem 1 would be $x = 2$, $x + y = 5$, $z = 4$.**

4. **Find a point with $z = 2$ on the intersection line of the planes $x + y + 3z = 6$ and $x - y + z = 4$. Find the point with $z = 0$. Find a third point halfway between.**

5. **The first of these equations plus the second equals the third:**

$$\begin{aligned} x + y + z &= 2 \\ x + 2y + z &= 3 \\ 2x + 3y + 2z &= 5. \end{aligned}$$

   The first two planes meet along a line. The third plane contains that line, because if x, y, z satisfy the first two equations then they also ________________. The equations have infinitely many solutions (the whole line L). Find three solutions on L.

6. **Move the third plane in Problem 5 to a parallel plane $2x + 3y + 2z = 9$. Now the three equations have no solution—why not? The first two planes meet along the line L, but the third plane doesn't ________________ that line.**

7. **In Problem 5 the columns are $(1, 1, 2)$ and $(1, 2, 3)$ and $(1, 1, 2)$. This is a "singular case" because the third column is ________________. Find two combinations of the columns that give $\boldsymbol{b} = (2, 3, 5)$. This is only possible for $\boldsymbol{b} = (4, 6, c)$ if $c =$ __________.**

8. **Normally 4 "planes" in 4-dimensional space meet at a ________________. Normally 4 column vectors in 4-dimensional space can combine to produce $\boldsymbol{b}$. What combination of $(1, 0, 0, 0)$, $(1, 1, 0, 0)$, $(1, 1, 1, 0)$, $(1, 1, 1, 1)$ produces $\boldsymbol{b} = (3, 3, 3, 2)$? What 4 equations for x, y, z, t are you solving?**

Problems 9–14 are about multiplying matrices and vectors.

9. **Compute each Ax by dot products of the rows with the column vector:**

$$\textbf{(a)} \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \qquad \textbf{(b)} \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

10. Compute each Ax in Problem 9 as a combination of the columns:

$$9\textbf{(a) becomes} \quad Ax = 2\begin{bmatrix}1\\-2\\-4\end{bmatrix} + 2\begin{bmatrix}2\\3\\1\end{bmatrix} + 3\begin{bmatrix}4\\1\\2\end{bmatrix} = \begin{bmatrix}\ \\ \ \\ \ \end{bmatrix}.$$

How many separate multiplications for Ax, when the matrix is "3 by 3"?

11. Find the two components of Ax by rows or by columns:

$$\begin{bmatrix}2&3\\5&1\end{bmatrix}\begin{bmatrix}4\\2\end{bmatrix}, \qquad \begin{bmatrix}3&6\\6&12\end{bmatrix}\begin{bmatrix}2\\-1\end{bmatrix}, \qquad \begin{bmatrix}1&2&4\\2&0&1\end{bmatrix}\begin{bmatrix}3\\1\\1\end{bmatrix}.$$

12. Multiply A times x to find three components of Ax:

$$\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}, \qquad \begin{bmatrix}2&1&3\\1&2&3\\3&3&6\end{bmatrix}\begin{bmatrix}1\\1\\-1\end{bmatrix}, \qquad \begin{bmatrix}2&1\\1&2\\3&3\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}.$$

13. (a) A matrix with m rows and n columns multiplies a vector with ____________ components to produce a vector with ____________ components.

(b) The planes from the m equations $Ax = \boldsymbol{b}$ are in ____________-dimensional space. The combination of the columns of A is in ____________-dimensional space.

14. Write $2x + 3y + z + 5t = 8$ as a matrix A (how many rows?) multiplying the column vector $\boldsymbol{x} = (x, y, z, t)$ to produce $\boldsymbol{b}$. The solutions $\boldsymbol{x}$ fill a plane or a "hyperplane" in 4-dimensional space. The plane is 3-dimensional with no 4D volume.

Problems 15–22 ask for matrices that act in special ways on vectors.

15. (a) What is the 2×2 identity matrix? I times $\begin{bmatrix}x\\y\end{bmatrix}$ equals $\begin{bmatrix}x\\y\end{bmatrix}$.

(b) What is the 2×2 exchange matrix? P times $\begin{bmatrix}x\\y\end{bmatrix}$ equals $\begin{bmatrix}y\\x\end{bmatrix}$.

16. (a) What 2×2 matrix R rotates every vector by $90°$? R times $\begin{bmatrix}x\\y\end{bmatrix}$ is $\begin{bmatrix}-y\\x\end{bmatrix}$.

(b) What 2×2 matrix R^2 rotates every vector by $180°$?

17. Find the matrix P that multiplies (x, y, z) to give (y, x, z). Find the matrix Q that multiplies (y, x, z) to bring back (x, y, z).

18. What 2×2 matrix E subtracts the first component from the second component? What 3×3 matrix does the same?

$$E\begin{bmatrix}3\\5\end{bmatrix} = \begin{bmatrix}3\\2\end{bmatrix} \quad \textbf{and} \quad E\begin{bmatrix}3\\5\\7\end{bmatrix} = \begin{bmatrix}3\\2\\7\end{bmatrix}.$$

19. What 3×3 matrix E multiplies (x, y, z) to give $(x, y, z + x)$? What matrix E^{-1} multiplies (x, y, z) to give $(x, y, z - x)$? If you multiply $(3, 4, 5)$ by E and then multiply by E^{-1}, the two results are ____________ and ____________.

20. (a) What 2×2 matrix P_1 projects the vector (x, y) onto the x axis to produce $(x, 0)$?

(b) What matrix P_2 projects onto the y axis to produce $(0, y)$? If you multiply $(5, 7)$ by P_1 and then multiply by P_2, you get ____________ and ____________.

21. What 2×2 matrix R rotates every vector through $45°$? The vector $(1, 0)$ goes to $(\sqrt{2}/2, \sqrt{2}/2)$. The vector $(0, 1)$ goes to $(-\sqrt{2}/2, \sqrt{2}/2)$. Those determine the matrix. Draw these particular vectors in the xy plane and find R.

22. **Write the dot product of $(1,4,5)$ and (x,y,z) as a matrix multiplication Ax. The matrix A has one row. The solutions to $Ax = 0$ lie on a ____________ perpendicular to the vector ____________. The columns of A are only in ____________-dimensional space.**

23. **In MATLAB notation, write the commands that define this matrix A and the column vectors x and b. What command would test whether or not $Ax = b$?**

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}.$$

24. **The MATLAB commands `A = eye(3)` and `v = [3;5]'` produce the 3×3 identity matrix and the column vector $(3,4,5)$. What are the outputs from `A*v` and `v'*A`? (Computer not needed!) If you ask for `v*A`, what happens?**

25. **If you multiply the 4×4 all-ones matrix $A = \text{ones}(4)$ and the column vector $\boldsymbol{v} = \text{ones}(4,1)$, what is $A\boldsymbol{v}$? (Computer not needed.) If you multiply $B = \text{eye}(4) + \text{ones}(4)$ times $\boldsymbol{w} = \text{zeros}(4,1) + 2 * \text{ones}(4,1)$, what is $B * \boldsymbol{w}$?**

26. **Draw the row and column pictures for the equations $x - 2y = 0$, $x + y = 6$.**

27. **For two linear equations in three unknowns x, y, z, the row picture will show (2 or 3) (lines or planes) in (2 or 3)-dimensional space. The column picture is in (2 or 3)-dimensional space. The solutions normally lie on a ________________.**

28. **For four linear equations in two unknowns x and y, the row picture shows four ________________. The column picture is in ________________-dimensional space. The equations have no solution unless the vector on the right side is a combination of ________________.**

29. **Start with the vector $\boldsymbol{u}_0 = (1,0)$. Multiply again and again by the same "Markov matrix"**

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}.$$

The next three vectors are $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$:

$$\boldsymbol{u}_1 = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} .8 \\ .2 \end{bmatrix}, \qquad \boldsymbol{u}_2 = A\boldsymbol{u}_1 = \underline{\qquad\qquad}, \qquad \boldsymbol{u}_3 = A\boldsymbol{u}_2 = \underline{\qquad\qquad}.$$

What property do you notice for all four vectors $\boldsymbol{u}_0, \boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$?

Challenge Problems

30. **Continue Problem 29 from $\boldsymbol{u}_0 = (1,0)$ to $\boldsymbol{u}_7$, and also from $\boldsymbol{v}_0 = (0,1)$ to $\boldsymbol{v}_7$. What do you notice about $\boldsymbol{u}_7$ and $\boldsymbol{v}_7$? Here are two MATLAB codes, with `while` and `for`:**

```
u = [1; 0]; A = [.8 .3; .2 .7];
x = u; k = [0 : 7];
while size(x,2) ¡= 7
    u = A*u; x = [x u];
end
plot(k, x)
```

```
v = [0; 1]; A = [.8 .3; .2 .7];
x = v; k = [0 : 7];
for j = 1 : 7
    v = A*v; x = [x v];
end
plot(k, x)
```

The $\boldsymbol{u}$'s and $\boldsymbol{v}$'s are approaching a steady state vector $\boldsymbol{s}$. Guess that vector and check that $A\boldsymbol{s} = \boldsymbol{s}$. If you start with $\boldsymbol{s}$, you stay with $\boldsymbol{s}$.

31. **Invent a 3×3 magic matrix M_3 with entries $1, 2, \ldots, 9$. All rows and columns and diagonals add to 15. The first row could be $8, 3, 4$. What is M_3 times $(1,1,1)$? What is M_4 times $(1,1,1,1)$ if a 4×4 magic matrix has entries $1, \ldots, 16$?**

32. **Suppose $\boldsymbol{u}$ and $\boldsymbol{v}$ are the first two columns of a 3×3 matrix A. Which third columns $\boldsymbol{w}$ would make this matrix singular? Describe a typical column picture of $A\boldsymbol{x} = \boldsymbol{b}$ in that singular case, and a typical row picture (for a random $\boldsymbol{b}$).**

33. **Multiplication by A is a "linear transformation". Those words mean:**

If w is a combination of u and v, then Aw is the same combination of Au and Av.

It is this "linearity" $A\boldsymbol{w} = cA\boldsymbol{u} + dA\boldsymbol{v}$ that gives us the name "linear algebra".

Problem: If $\boldsymbol{u} = \begin{bmatrix}1\\0\end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix}0\\1\end{bmatrix}$, then Au and Av are the columns of A.

Combine $\boldsymbol{w} = c\boldsymbol{u} + d\boldsymbol{v}$. If $\boldsymbol{w} = \begin{bmatrix}5\\7\end{bmatrix}$, how is Aw connected to Au and Av?

34. **Start from the four equations $-x_{i+1} + 2x_i - x_{i-1} = i$ (for $i = 1, 2, 3, 4$ with $x_0 = x_5 = 0$). Write those equations in their matrix form $A\boldsymbol{x} = \boldsymbol{b}$. Can you solve them for x_1, x_2, x_3, x_4?**

35. **A 9×9 Sudoku matrix S has the numbers $1, \ldots, 9$ in every row and every column, and in every 3 by 3 block. For the all-ones vector $\boldsymbol{x} = (1, \ldots, 1)$, what is $S\boldsymbol{x}$?**
A better question is: Which row exchanges will produce another Sudoku matrix? Also, which exchanges of block rows give another Sudoku matrix?
Section 2.7 will look at all possible permutations (reorderings) of the rows. I can see 6 orders for the first 3 rows, all giving Sudoku matrices. Also 6 permutations of the next 3 rows, and of the last 3 rows. And 6 block permutations of the block rows?

2.2 The Idea of Elimination

Problems 1–10 are about elimination on 2×2 systems.

1. **What multiple ℓ_{21} of equation 1 should be subtracted from equation 2?**

$$\begin{aligned} 2x + 3y &= 1 \\ 10x + 9y &= 11 \end{aligned}$$

After elimination, write down the upper triangular system and circle the two pivots. The numbers 1 and 11 don't affect the pivots—use them now in back substitution.

2. **Solve the triangular system of Problem 1 by back substitution, y before x. Verify that x times $(2, 10)$ plus y times $(3, 9)$ equals $(1, 11)$. If the right side changes to $(4, 44)$, what is the new solution?**

3. **What multiple of equation 1 should be subtracted from equation 2?**

$$\begin{aligned} 2x - 4y &= 6 \\ -x + 5y &= 0 \end{aligned}$$

After this elimination step, solve the triangular system. If the right side changes to $(-6, 0)$, what is the new solution?

4. **What multiple ℓ of equation 1 should be subtracted from equation 2 to remove c?**

$$\begin{aligned} ax + by &= f \\ cx + dy &= g \end{aligned}$$

The first pivot is a (assumed nonzero). Elimination produces what formula for the second pivot? What is y? The second pivot is missing when $ad = bc$: singular.

5. **Choose a right side which gives no solution and another right side which gives infinitely many solutions. What are two of those solutions?**

Singular system

$$\begin{aligned} 3x + 2y &= 10 \\ 6x + 4y &= \end{aligned}$$

6. **Choose a coefficient b that makes this system singular. Then choose a right side g that makes it solvable. Find two solutions in that singular case.**

$$\begin{aligned} 2x + by &= 16 \\ 4x + 8y &= g \end{aligned}$$

7. **For which numbers a does elimination break down (1) permanently (2) temporarily?**

$$\begin{aligned} ax + 3y &= -3 \\ 4x + 6y &= 6 \end{aligned}$$

Solve for x and y after fixing the temporary breakdown by a row exchange.

8. **For which three numbers k does elimination break down? Which is fixed by a row exchange? In each case, is the number of solutions 0 or 1 or ∞?**

$$\begin{aligned} kx + 3y &= 6 \\ 3x + ky &= -6 \end{aligned}$$

9. **What test on b_1 and b_2 decides whether these two equations allow a solution? How many solutions will they have? Draw the column picture for $\boldsymbol{b} = (1, 2)$ and $(1, 0)$.**

$$\begin{aligned} 3x - 2y &= b_1 \\ 6x - 4y &= b_2 \end{aligned}$$

10. **In the xy plane, draw the lines $x+y = 5$ and $x+2y = 6$ and the equation $y =$ ____ that comes from elimination. The line $5x - 4y = c$ will go through the solution of these equations if $c =$ ____.**

Problems 11–20 study elimination on 3×3 systems (and possible failure).

11. **(Recommended) A system of linear equations can't have exactly two solutions. Why?**
(a) If (x, y, z) and (X, Y, Z) are two solutions, what is another solution?
(b) If 25 planes meet at two points, where else do they meet?

12. **Reduce this system to upper triangular form by two row operations:**

$$\begin{aligned} 2x + 3y + z &= 8 \\ 4x + 7y + 5z &= 20 \\ -2y + 2z &= 0 \end{aligned}$$

Circle the pivots. Solve by back substitution for z, y, x.

13. **Apply elimination (circle the pivots) and back substitution to solve**

$$\begin{aligned} 2x - 3y \phantom{{}+z} &= 3 \\ 4x - 5y + z &= 7 \\ 2x - y - 3z &= 5 \end{aligned}$$

List the three row operations: Subtract ____ times row ____ from row ____.

14. **Which number d forces a row exchange, and what is the triangular system (not singular) for that d? Which d makes this system singular (no third pivot)?**

$$\begin{aligned} 2x + 5y + z &= 0 \\ 4x + dy + z &= 2 \\ y - z &= 3 \end{aligned}$$

15. **Which number b leads later to a row exchange? Which b leads to a missing pivot? In that singular case find a nonzero solution x, y, z.**

$$\begin{aligned} x + by \phantom{{}-z} &= 0 \\ x - 2y - z &= 0 \\ + y + z &= 0 \end{aligned}$$

16. **(a) Construct a 3×3 system that needs two row exchanges to reach a triangular form and a solution.**

    (b) Construct a 3×3 system that needs a row exchange to keep going, but breaks down later.

17. **If rows 1 and 2 are the same, how far can you get with elimination (allowing row exchange)? If columns 1 and 2 are the same, which pivot is missing?**

$$\begin{array}{cccc} \textit{Equal} & \begin{aligned} 2x - y + z &= 0 \\ 2x - y + z &= 0 \\ 4x + y + z &= 2 \end{aligned} & \begin{aligned} 2x + 2y + z &= 0 \\ 4x + 4y + z &= 0 \\ 6x + 6y + z &= 2 \end{aligned} & \textit{Equal} \\ \textit{rows} & & & \textit{columns} \end{array}$$

18. **Construct a 3×3 example that has 9 different coefficients on the left side, but rows 2 and 3 become zero in elimination. How many solutions to your system with $\boldsymbol{b} = (1, 10, 100)$ and how many with $\boldsymbol{b} = (0, 0, 0)$?**

19. **Which number q makes this system singular and which right side t gives it infinitely many solutions? Find the solution that has $z = 1$.**

$$\begin{aligned} x + 4y - 2z &= 1 \\ x + 7y - 6z &= 6 \\ 3y + qz &= t \end{aligned}$$

20. **Three planes can fail to have an intersection point, even if no planes are parallel. The system is singular if row 3 of A is a ______ of the first two rows. Find a third equation that can't be solved together with $x + y + z = 0$ and $x - 2y - z = 1$.**

21. **Find the pivots and the solution for both systems ($A\boldsymbol{x} = \boldsymbol{b}$ and $K\boldsymbol{x} = \boldsymbol{b}$):**

$$\begin{aligned} 2x + y &= 0 \\ x + 2y + z &= 0 \\ y + 2z + t &= 0 \\ z + 2t &= 5 \end{aligned} \qquad \begin{aligned} 2x - y &= 0 \\ -x + 2y - z &= 0 \\ -y + 2z - t &= 0 \\ -z + 2t &= 5 \end{aligned}$$

22. **If you extend Problem 21 following the $1, 2, 1$ pattern or the $-1, 2, -1$ pattern, what is the fifth pivot? What is the nth pivot? K is my favorite matrix.**

23. **If elimination leads to $x + y = 1$ and $2y = 3$, find three possible original problems.**

24. **For which two numbers a will elimination fail on $A = \begin{bmatrix} a & 2 \\ a & a \end{bmatrix}$?**

25. **For which three numbers a will elimination fail to give three pivots?**

$$A = \begin{bmatrix} a & 2 & 3 \\ a & a & 4 \\ a & a & a \end{bmatrix} \quad \textbf{is singular for three values of } a.$$

26. **Look for a matrix that has row sums 4 and 8, and column sums 2 and s:**

$$\textbf{Matrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad \begin{cases} a + b = 4 \\ a + c = 2 \\ c + d = 8 \\ b + d = s \end{cases}$$

    The four equations are solvable only if $s =$ ____. Then find two different matrices that have the correct row and column sums. *Extra credit:* Write down the 4×4 system $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{x} = (a, b, c, d)$ and make A triangular by elimination.

27. Elimination in the usual order gives what matrix U and what solution to this "lower triangular" system? We are really solving by *forward substitution*:

$$\begin{aligned} 3x \qquad\quad &= 3 \\ 6x + 2y \qquad &= 8 \\ 9x - 2y + z &= 9 \end{aligned}$$

28. Create a MATLAB command `A(2,:) = ...` for the new row 2, to subtract 3 times row 1 from the existing row 2 if the matrix A is already known.

Challenge Problems

29. Find experimentally the average 1st and 2nd and 3rd pivot sizes from MATLAB's `[L,U] = lu(rand(3))`. The average size `abs(U(1,1))` is above $\frac{1}{2}$ because `lu` picks the largest available pivot in column 1. Here $A =$ `rand(3)` has random entries between 0 and 1.

30. If the last corner entry is $A(5,5) = 11$ and the last pivot of A is $U(5,5) = 4$, what different entry $A(5,5)$ would have made A singular?

31. Suppose elimination takes A to U without row exchanges. Then row j of U is a combination of which rows of A? If $Ax = 0$, is $Ux = 0$? If $Ax = b$, is $Ux = b$? If A starts out lower triangular, what is the upper triangular U?

32. Start with 100 equations $Ax = 0$ for 100 unknowns $x = (x_1, \ldots, x_{100})$. Suppose elimination reduces the 100th equation to $0 = 0$, so the system is "singular".

    (a) Elimination takes linear combinations of the rows. So this singular system has the singular property: Some linear combination of the 100 rows is ______.

    (b) Singular systems $Ax = 0$ have infinitely many solutions. This means that some linear combination of the 100 ____ is ______.

    (c) Invent a 100 by 100 singular matrix with no zero entries.

    (d) For your matrix, describe in words the row picture and the column picture of $Ax = 0$. Not necessary to draw 100-dimensional space.

2.3 Elimination Using Matrices

Problems 1–15 are about elimination matrices.

1. Write down the 3 by 3 matrices that produce these elimination steps:

    (a) E_{21} subtracts 5 times row 1 from row 2.

    (b) E_{32} subtracts -7 times row 2 from row 3.

    (c) P exchanges rows 1 and 2, then rows 2 and 3.

2. In Problem 1, applying E_{21} and then E_{32} to $\boldsymbol{b} = (1, 0, 0)$ gives $E_{32}E_{21}\boldsymbol{b} =$ ______. Applying E_{32} before E_{21} gives $E_{21}E_{32}\boldsymbol{b} =$ ______. When E_{32} comes first, row ____ feels no effect from row ____.

3. Which three matrices E_{21}, E_{31}, E_{32} put A into triangular form U?

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix} \quad \text{and} \quad E_{32}E_{31}E_{21}A = U.$$

   Multiply those E's to get one matrix M that does elimination: $MA = U$.

4. Include $\boldsymbol{b} = (1, 0, 0)$ as a fourth column in Problem 3 to produce $[A \mid \boldsymbol{b}]$. Carry out the elimination steps on this augmented matrix to solve $Ax = \boldsymbol{b}$.

5. Suppose $a_{33} = 7$ and the third pivot is 5. If you change a_{33} to 11, the third pivot is ______. If you change a_{33} to ______, there is no third pivot.

6. **If every column of A is a multiple of $(1,1,1)$, then Ax is always a multiple of $(1,1,1)$. Do a 3 by 3 example. How many pivots are produced by elimination?**

7. **Suppose E subtracts 7 times row 1 from row 3.**

   (a) **To invert that step you should ____ 7 times row ____ to row ____.**

   (b) **What "inverse matrix" E^{-1} takes that reverse step (so $E^{-1}E = I$)?**

   (c) **If the reverse step is applied first (and then E) show that $EE^{-1} = I$.**

8. **The determinant of $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\det M = ad - bc$. Subtract ℓ times row 1 from row 2 to produce a new M^*. Show that $\det M^* = \det M$ for every ℓ. When $\ell = c/a$, the product of pivots equals the determinant: $(a)(d - \ell b)$ equals $ad - bc$.**

9. (a) **E_{21} subtracts row 1 from row 2 and then P_{23} exchanges rows 2 and 3. What matrix $M = P_{23}E_{21}$ does both steps at once?**

   (b) **P_{23} exchanges rows 2 and 3 and then E_{31} subtracts row 1 from row 3. What matrix $M = E_{31}P_{23}$ does both steps at once? Explain why the M's are the same but the E's are different.**

10. (a) **What 3 by 3 matrix E_{13} will add row 3 to row 1?**

    (b) **What matrix adds row 1 to row 3 and *at the same time* row 3 to row 1?**

    (c) **What matrix adds row 1 to row 3 and *then* adds row 3 to row 1?**

11. **Create a matrix that has $a_{11} = a_{22} = a_{33} = 1$ but elimination produces two negative pivots without row exchanges. (The first pivot is 1.)**

12. **Multiply these matrices:**

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 4 & 0 \end{bmatrix}.$$

13. **Explain these facts. If the third column of B is all zero, the third column of EB is all zero (for any E). If the third row of B is all zero, the third row of EB might *not* be zero.**

14. **This 4 by 4 matrix will need elimination matrices E_{21} and E_{32} and E_{43}. What are those matrices?**

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}.$$

15. **Write down the 3 by 3 matrix that has $a_{ij} = 2i - 3j$. This matrix has $a_{32} = 0$, but elimination still needs E_{32} to produce a zero in the $(3,2)$ position. Which previous step destroys the original zero and what is E_{32}?**

Problems 16–23 are about creating and multiplying matrices.

16. **Write these ancient problems in a 2 by 2 matrix form $Ax = \boldsymbol{b}$ and solve them:**

    (a) **X is twice as old as Y and their ages add to 33.**

    (b) **$(x,y) = (2,5)$ and $(3,7)$ lie on the line $y = mx + c$. Find m and c.**

17. **The parabola $y = a + bx + cx^2$ goes through the points $(x,y) = (1,4)$ and $(2,8)$ and $(3,14)$. Find and solve a matrix equation for the unknowns (a,b,c).**

18. **Multiply these matrices in the orders EF and FE:**

$$E = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix}, \qquad F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix}.$$

Also compute $E^2 = EE$ and $F^3 = FFF$. You can guess P^{100}.

19. Multiply these row exchange matrices in the orders PQ and QP and P^2:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{and} \quad Q = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Find another non-diagonal matrix whose square is $M^2 = I$.

20. (a) Suppose all columns of B are the same. Then all columns of EB are the same, because each one is E times ______.

(b) Suppose all rows of B are $[\,1 \;\; 2 \;\; 4\,]$. Show by example that all rows of EB are *not* $[\,1 \;\; 2 \;\; 4\,]$. It is true that those rows are ________.

21. If E adds row 1 to row 2 and F adds row 2 to row 1, does EF equal FE?

22. The entries of A and $\boldsymbol{x}$ are a_{ij} and x_j. So the first component of $A\boldsymbol{x}$ is $\sum a_{1j}x_j = a_{11}x_1 + \cdots + a_{1n}x_n$. If E_{21} subtracts row 1 from row 2, write a formula for

(a) the third component of $E_{21}A\boldsymbol{x}$

(b) the $(2,1)$ entry of $E_{21}A$

(c) the $(2,1)$ entry of $E_{21}(E_{21}A)$

(d) the first component of $(E_{21}E_{21})A\boldsymbol{x}$

23. The elimination matrix $E = \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}$ subtracts 2 times row 1 of A from row 2 of A. The result is EA. What is the effect of $E(AE)$? In the opposite order AE, we are subtracting 2 times ____ of A from ____. (Do examples.)

Problems 24–27 include the column $\boldsymbol{b}$ in the augmented matrix $[\,A \mid \boldsymbol{b}\,]$.

24. Apply elimination to the 2 by 3 augmented matrix $[\,A \mid \boldsymbol{b}\,]$. What is the triangular system $U\boldsymbol{x} = \boldsymbol{c}$? What is the solution $\boldsymbol{x}$?

$$A\boldsymbol{x} = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 17 \end{bmatrix}.$$

25. Apply elimination to the 3 by 4 augmented matrix $[\,A \mid \boldsymbol{b}\,]$. How do you know this system has no solution? Change the last number 6 so there is a solution.

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}.$$

26. The equations $A\boldsymbol{x} = \boldsymbol{b}$ and $A\boldsymbol{x}^* = \boldsymbol{b}^*$ have the same matrix A. What double augmented matrix should you use in elimination to solve both equations at once? Solve both of these equations by working on a 2 by 4 matrix:

$$\begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \textbf{and} \quad \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

27. Choose the numbers a, b, c, d in this augmented matrix so that there is (a) no solution (b) infinitely many solutions.

$$[A \mid \boldsymbol{b}] = \begin{bmatrix} 1 & 2 & 3 & a \\ 0 & 4 & 5 & b \\ 0 & 0 & d & c \end{bmatrix}.$$

Which of the numbers $a, b, c,$ or d have no effect on the solvability?

28. If $AB = I$ and $BC = I$ use the associative law to prove $A = C$.

Challenge Problems

29. Find the triangular matrix E that reduces "Pascal's matrix" to a smaller Pascal:

$$\textbf{Elimination on column 1} \quad E\begin{bmatrix}1&0&0&0\\1&1&0&0\\1&2&1&0\\1&3&3&1\end{bmatrix}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&1&1&0\\0&1&2&1\end{bmatrix}.$$

Which matrix M (multiplying several E's) reduces Pascal all the way to I? Pascal's triangular matrix is exceptional, all of its multipliers are $\ell_{ij}=1$.

30. Write $M=\begin{bmatrix}3&4\\6&7\end{bmatrix}$ as a product of many factors $A=\begin{bmatrix}1&0\\1&1\end{bmatrix}$ and $B=\begin{bmatrix}1&1\\0&1\end{bmatrix}$.

(a) What matrix E subtracts row 1 from row 2 to make row 2 of EM smaller?

(b) What matrix F subtracts row 2 of EM from row 1 to reduce row 1 of FEM?

(c) Continue E's and F's until (many E's and F's times) M is (A or B).

(d) E and F are the inverses of A and B! Moving all E's and F's to the right side will give you the desired result M = product of A's and B's. This is possible for integer matrices $M=\begin{bmatrix}a&b\\c&d\end{bmatrix}>0$ that have $ad-bc=1$.

31. Find elimination matrices E_{21} then E_{32} then E_{43} to change K into U:

$$E_{43}\,E_{32}\,E_{21}\begin{bmatrix}1&0&0&0\\-a&1&0&0\\0&-b&1&0\\0&0&-c&1\end{bmatrix}=I.$$

Apply those three steps to the identity matrix I, to multiply $E_{43}E_{32}E_{21}$.

2.4 Rules for Matrix Operations

Problems 1–16 are about the laws of matrix multiplication.

1. A is 3 by 5, B is 5 by 3, C is 5 by 1, and D is 3 by 1. All entries are 1. Which of these matrix operations are allowed, and what are the results?

$$BA \qquad AB \qquad ABD \qquad DC \qquad A(B+C).$$

2. What rows or columns or matrices do you multiply to find

(a) the second column of AB?

(b) the first row of AB?

(c) the entry in row 3, column 5 of AB?

(d) the entry in row 1, column 1 of CDE?

3. Add AB to AC and compare with $A(B+C)$:

$$A=\begin{bmatrix}1&5\\2&3\end{bmatrix} \quad \textbf{and} \quad B=\begin{bmatrix}0&2\\0&1\end{bmatrix} \quad \textbf{and} \quad C=\begin{bmatrix}3&1\\0&0\end{bmatrix}.$$

4. In Problem 3, multiply A times BC. Then multiply AB times C.

5. Compute A^2 and A^3. Make a prediction for A^5 and A^n:

$$A=\begin{bmatrix}1&b\\0&1\end{bmatrix} \quad \textbf{and} \quad A=\begin{bmatrix}2&2\\0&0\end{bmatrix}.$$

6. **Show that $(A+B)^2$ is different from $A^2+2AB+B^2$, when**

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}.$$

Write down the correct rule for $(A+B)(A+B) = A^2 + \underline{\qquad} + B^2$.

7. **True or false. Give a specific example when false:**

   (a) **If columns 1 and 3 of B are the same, so are columns 1 and 3 of AB.**

   (b) **If rows 1 and 3 of B are the same, so are rows 1 and 3 of AB.**

   (c) **If rows 1 and 3 of A are the same, so are rows 1 and 3 of ABC.**

   (d) $(AB)^2 = A^2B^2$.

8. **How is each row of DA and EA related to the rows of A, when**

$$D = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} \quad \textbf{and} \quad E = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}?$$

How is each column of AD and AE related to the columns of A?

9. **Row 1 of A is added to row 2. This gives EA below. Then column 1 of EA is added to column 2 to produce $(EA)F$:**

$$EA = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ a+c & b+d \end{bmatrix}, \quad (EA)F = (EA) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & a+b \\ a+c & a+c+b+d \end{bmatrix}.$$

   (a) **Do those steps in the opposite order. First add column 1 of A to column 2 by AF, then add row 1 of AF to row 2 by $E(AF)$.**

   (b) **Compare with $(EA)F$. What law is obeyed by matrix multiplication?**

10. **Row 1 of A is again added to row 2 to produce EA. Then F adds row 2 of EA to row 1. The result is $F(EA)$:**

$$F(EA) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ a+c & b+d \end{bmatrix} = \begin{bmatrix} 2a+c & 2b+d \\ a+c & b+d \end{bmatrix}.$$

   (a) **Do those steps in the opposite order: first add row 2 to row 1 by FA, then add row 1 of FA to row 2.**

   (b) **What law is or is not obeyed by matrix multiplication?**

11. **This fact still amazes me. If you do a row operation on A and then a column operation, the result is the same as if you did the column operation first. (Try it.) Why is this true?**

12. **(3 by 3 matrices) Choose the only B so that for every matrix A**

   (a) $BA = 4A$

   (b) $BA = 4B$

   (c) **BA has rows 1 and 3 of A reversed and row 2 unchanged**

   (d) **All rows of BA are the same as row 1 of A.**

13. **Suppose $AB = BA$ and $AC = CA$ for these two particular matrices B and C:**

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \textbf{commutes with} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \textbf{and} \quad C = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Prove that $a = d$ and $b = c = 0$. Then A is a multiple of I. The only matrices that commute with B and C and all other 2 by 2 matrices are $A =$ multiple of I.

14. **Which of the following matrices are guaranteed to equal $(A-B)^2$: $A^2 - B^2$, $(B-A)^2$, $A^2 - 2AB + B^2$, $A(A-B) - B(A-B)$, $A^2 - AB - BA + B^2$?**

15. **True or false:**

(a) **If A^2 is defined then A is necessarily square.**

(b) **If AB and BA are defined then A and B are square.**

(c) **If AB and BA are defined then AB and BA are square.**

(d) **If $AB = B$ then $A = I$.**

16. **If A is m by n, how many separate multiplications are involved when**

(a) **A multiplies a vector x with n components?**

(b) **A multiplies an n by p matrix B?**

(c) **A multiplies itself to produce A^2? Here $m = n$.**

17. **For $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 6 & 0 \end{bmatrix}$, compute these answers *and nothing more*:**

(a) **column 2 of AB**

(b) **row 2 of AB**

(c) **row 2 of $AA = A^2$**

(d) **row 2 of $AAA = A^3$.**

Problems 18–20 use a_{ij} for the entry in row i, column j of A.

18. **Write down the 3 by 3 matrix A whose entries are**

(a) $a_{ij} = \min$ **of i and j**

(b) $a_{ij} = (-1)^{i+j}$

(c) $a_{ij} = i/j$.

19. **What words would you use to describe each of these classes of matrices? Give a 3 by 3 example in each class. Which matrix belongs to all four classes?**

(a) $a_{ij} = 0$ **if** $i \neq j$

(b) $a_{ij} = 0$ **if** $i < j$

(c) $a_{ij} = a_{ji}$

(d) $a_{ij} = a_{1j}$.

20. **The entries of A are a_{ij}. Assuming that zeros don't appear, what is**

(a) **the first pivot?**

(b) **the multiplier ℓ_{31} of row 1 to be subtracted from row 3?**

(c) **the new entry that replaces a_{32} after that subtraction?**

(d) **the second pivot?**

Problems 21–24 involve powers of A.

21. **Compute A^2, A^3, A^4 and also Av, A^2v, A^3v, A^4v for**

$$A = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{v} = \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix}.$$

22. **By trial and error find real nonzero 2 by 2 matrices such that**

$$A^2 = -I \qquad BC = 0 \qquad DE = -ED \quad \textbf{(not allowing } DE = 0\textbf{)}.$$

23. (a) **Find a nonzero matrix A for which $A^2 = 0$.**

(b) **Find a matrix that has $A^2 \neq 0$ but $A^3 = 0$.**

24. By experiment with $n = 2$ and $n = 3$ predict A^n for these matrices:

$$A_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \quad \textbf{and} \quad A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \textbf{and} \quad A_3 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}.$$

Problems 25–31 use column-row multiplication and block multiplication.

25. Multiply A times I using columns of A (3 by 3) times rows of I.

26. Multiply AB using columns times rows:

$$AB = \begin{bmatrix} 1 & 0 \\ 2 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 3 & 0 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 3 & 0 \end{bmatrix} + \underline{\qquad} = \underline{\qquad}.$$

27. Show that the product of upper triangular matrices is always upper triangular:

$$AB = \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{bmatrix} \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{bmatrix} = \begin{bmatrix} \underline{\quad} & \underline{\quad} & \underline{\quad} \\ 0 & \underline{\quad} & \underline{\quad} \\ 0 & 0 & \underline{\quad} \end{bmatrix}.$$

Proof using dot products (Row times column) (Row 2 of A) $\cdot$ (column 1 of B) $= 0$. Which other dot products give zeros?
Proof using full matrices (Column times row) Draw x's and 0's in (column 2 of A) times (row 2 of B). Also show (column 3 of A) times (row 3 of B).

28. Draw the cuts in A (2 by 3) and B (3 by 4) and AB to show how each of the four multiplication rules is really a block multiplication:

(1) Matrix A times columns of B	**Columns of AB**
(2) Rows of A times the matrix B	**Rows of AB**
(3) Rows of A times columns of B	**Inner products (numbers in AB)**
(4) Columns of A times rows of B	**Outer products (matrices add to AB)**

29. Which matrices E_{21} and E_{31} produce zeros in the $(2,1)$ and $(3,1)$ positions of $E_{21}A$ and $E_{31}A$?

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -2 & 0 & 1 \\ 8 & 5 & 3 \end{bmatrix}.$$

Find the single matrix $E = E_{31}E_{21}$ that produces both zeros at once. Multiply EA.

30. Block multiplication says that column 1 is eliminated by

$$EA = \begin{bmatrix} 1 & 0 \\ -c/a & I \end{bmatrix} \begin{bmatrix} a & b \\ c & D \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & D - cb/a \end{bmatrix}.$$

In Problem 29, what numbers go into c and D and what is $D - cb/a$?

31. With $i^2 = -1$, the product of $(A + iB)$ and $(x + iy)$ is $Ax + iBx + iAy - By$. Use blocks to separate the real part without i from the imaginary part that multiplies i:

$$\begin{bmatrix} A & -B \\ ? & ? \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} Ax - By \\ ? \end{bmatrix} \quad \begin{matrix} \textbf{real part} \\ \textbf{imaginary part} \end{matrix}$$

(Fill in the ?'s.)

32. (Very important) Suppose you solve $Ax = \boldsymbol{b}$ for three special right sides $\boldsymbol{b}$:

$$A\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \textbf{and} \quad A\boldsymbol{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \textbf{and} \quad A\boldsymbol{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

If the three solutions $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3$ are the columns of a matrix X, what is A times X?

33. If the three solutions in Question 32 are $\boldsymbol{x}_1 = (1, 1, 1)$ and $\boldsymbol{x}_2 = (0, 1, 1)$ and $\boldsymbol{x}_3 = (0, 0, 1)$, solve $A\boldsymbol{x} = \boldsymbol{b}$ when $\boldsymbol{b} = (3, 5, 8)$. Challenge problem: What is A?

34. Find all matrices $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ that satisfy $A \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A$.

35. Suppose a "circle graph" has 4 nodes connected (in both directions) by edges around a circle. What is its adjacency matrix S from Worked Example 2.4 A? What is S^2? Find all the 2-step paths predicted by S^2.

Challenge Problems

36. Practical question Suppose A is m by n, B is n by p, and C is p by q. Then the multiplication count is mnp for $AB + mpq$ for $(AB)C$. The same matrix comes from A times BC with $mnq + npq$ separate multiplications. Notice npq for BC.

  (a) If A is 2 by 4, B is 4 by 7, and C is 7 by 10, do you prefer $(AB)C$ or $A(BC)$?

  (b) With N-component vectors, would you choose $(\boldsymbol{u}^T\boldsymbol{v})\boldsymbol{w}^T$ or $\boldsymbol{u}^T(\boldsymbol{v}\boldsymbol{w}^T)$?

  (c) Divide by mnp to show that $(AB)C$ is faster when $n^{-1} + q^{-1} < m^{-1} + p^{-1}$.

37. To prove that $(AB)C = A(BC)$, use the column vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n$ of B. First suppose that C has only one column $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_n$.
AB has columns $A\boldsymbol{b}_1, \ldots, A\boldsymbol{b}_n$ and then $(AB)C$ equals $c_1 A\boldsymbol{b}_1 + \cdots + c_n A\boldsymbol{b}_n$.
BC has one column $c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$ and then $A(BC)$ equals $A(c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n)$.
Linearity gives equality of those two sums. This proves $(AB)C = A(BC)$. The same is true for all other ______ of C. Therefore $(AB)C = A(BC)$. Apply to inverses: If $BA = I$ and $AC = I$, prove that the left-inverse B equals the right-inverse C.

38. (a) Suppose A has rows $\boldsymbol{a}_1^T, \ldots, \boldsymbol{a}_m^T$. Why does A^TA equal $\boldsymbol{a}_1\boldsymbol{a}_1^T + \cdots + \boldsymbol{a}_m\boldsymbol{a}_m^T$?

  (b) If C is a diagonal matrix with $c_1, \ldots, c_m$ on its diagonal, find a similar sum of columns times rows for A^TCA. First do an example with $m = n = 2$.

2.5 Inverse Matrices

1. Find the inverses (directly or from the 2 by 2 formula) of A, B, C:

$$A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}.$$

2. For these "permutation matrices" find P^{-1} by trial and error (with 1's and 0's):

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

3. Solve for the first column (x, y) and second column (t, z) of A^{-1}:

$$\begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix} \begin{bmatrix} t \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

4. Show that $\begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}$ is not invertible by trying to solve $AA^{-1} = I$ for column 1 of A^{-1}:

$$\begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \left(\textbf{For a different } A, \textbf{ could column 1 of } A^{-1} \textbf{ be possible to find but not column 2?}\right)$$

5. Find an upper triangular U (not diagonal) with $U^2 = I$ which gives $U = U^{-1}$.

6. (a) If A is invertible and $AB = AC$, prove quickly that $B = C$.

(b) **If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, find two different matrices such that $AB = AC$.**

7. **(Important) If A has row 1 + row 2 = row 3, show that A is not invertible:**
   (a) **Explain why $Ax = (0, 0, 1)$ cannot have a solution. Add eqn 1 + eqn 2.**
   (b) **Which right sides (b_1, b_2, b_3) might allow a solution to $Ax = b$?**
   (c) **In elimination, what happens to equation 3?**
8. **If A has column 1 + column 2 = column 3, show that A is not invertible:**
   (a) **Find a nonzero solution x to $Ax = \mathbf{0}$. The matrix is 3 by 3.**
   (b) **Elimination keeps column 1 + column 2 = column 3. Explain why there is no third pivot.**
9. **Suppose A is invertible and you exchange its first two rows to reach B. Is the new matrix B invertible? How would you find B^{-1} from A^{-1}?**
10. **Find the inverses (in any legal way) of**

$$A = \begin{bmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 4 & 0 & 0 \\ 5 & 0 & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 3 & 2 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 0 & 0 & 6 & 5 \\ 0 & 0 & 7 & 6 \end{bmatrix}.$$

11. (a) **Find invertible matrices A and B such that $A + B$ is not invertible.**
    (b) **Find singular matrices A and B such that $A + B$ is invertible.**
12. **If the product $C = AB$ is invertible (A and B are square), then A itself is invertible. Find a formula for A^{-1} that involves C^{-1} and B.**
13. **If the product $M = ABC$ of three square matrices is invertible, then B is invertible. (So are A and C.) Find a formula for B^{-1} that involves M^{-1} and A and C.**
14. **If you add row 1 of A to row 2 to get B, how do you find B^{-1} from A^{-1}?**
    Notice the order. The inverse of $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} A$ is ________.
15. **Prove that a matrix with a column of zeros cannot have an inverse.**
16. **Multiply $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ times $\begin{bmatrix} -d & -b \\ c & a \end{bmatrix}$. What is the inverse of each matrix if $ad \neq bc$?**
17. (a) **What 3 by 3 matrix E has the same effect as these three steps? Subtract row 1 from row 2, subtract row 1 from row 3, then subtract row 2 from row 3.**
    (b) **What single matrix L has the same effect as these three reverse steps? Add row 2 to row 3, add row 1 to row 3.**
18. **If B is the inverse of A^2, show that AB is the inverse of A.**
19. **Find the numbers a and b that give the inverse of $5 \cdot \mathbf{eye}(4) - \mathbf{ones}(4, 4)$:**

$$\begin{bmatrix} -1 & 4 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}.$$

    What are a and b in the inverse of $6 \cdot \mathbf{eye}(5) - \mathbf{ones}(5, 5)$?
20. **Show that $A = 4 \cdot \mathbf{eye}(4) - \mathbf{ones}(4, 4)$ is not invertible: Multiply $A \cdot \mathbf{ones}(4, 1)$.**
21. **There are sixteen 2 by 2 matrices whose entries are 1's and 0's. How many of them are invertible?**

Questions 22–28 are about the Gauss-Jordan method for calculating A^{-1}.

22. Change I into A^{-1} as you reduce A to I (by row operations):

$$[A \mid I] = \begin{bmatrix} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{bmatrix} \quad \textbf{and} \quad [A \mid I] = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 3 & 9 & 0 & 1 \end{bmatrix}.$$

23. Follow the 3 by 3 text example but with plus signs in A. Eliminate above and below the pivots to reduce $[A \mid I]$ to $[I \mid A^{-1}]$:

$$[A \mid I] = \begin{bmatrix} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}.$$

24. Use Gauss-Jordan elimination on $[U \mid I]$ to find the upper triangular U^{-1}:

$$UU^{-1} = I \qquad \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

25. Find A^{-1} and B^{-1} (if they exist) by elimination on $[A \mid I]$ and $[B \mid I]$:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

26. What three matrices E_{21} and E_{12} and D^{-1} reduce $A = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$ to the identity matrix? Multiply $D^{-1}E_{12}E_{21}$ to find A^{-1}.

27. Invert these matrices A by the Gauss-Jordan method starting with $[A \mid I]$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

28. Exchange rows and continue with Gauss-Jordan to find A^{-1}:

$$[A \mid I] = \begin{bmatrix} 0 & 2 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{bmatrix}.$$

29. True or false (with a counterexample if false and a reason if true):

(a) A 4 by 4 matrix with a row of zeros is not invertible.

(b) Every matrix with 1's down the main diagonal is invertible.

(c) If A is invertible then A^{-1} and A^2 are invertible.

30. (Recommended) Prove that A is invertible if $a \neq 0$ and $a \neq b$ (find the pivots or A^{-1}). Then find three numbers c so that C is not invertible:

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix} \quad \textbf{and} \quad C = \begin{bmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{bmatrix}.$$

31. This matrix has a remarkable inverse. Find A^{-1} by elimination on $[A \mid I]$. Extend to a 5 by 5 "alternating matrix" and guess its inverse; then multiply to confirm.

Invert $A = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **and solve** $A\boldsymbol{x} = (1,1,1,1)$.

32. Suppose the matrices P and Q have the same rows as I but in any order. They are "permutation matrices". Show that $P - Q$ is singular by solving $(P - Q)\boldsymbol{x} = \mathbf{0}$.

33. **Find and check the inverses (assuming they exist) of these block matrices:**

$$\begin{bmatrix} I & 0 \\ C & I \end{bmatrix} \quad \begin{bmatrix} A & 0 \\ C & D \end{bmatrix} \quad \begin{bmatrix} 0 & I \\ I & D \end{bmatrix}.$$

34. **Could a 4 by 4 matrix A be invertible if every row contains the numbers $0, 1, 2, 3$ in some order? What if every row of B contains $0, 1, 2, -3$ in some order?**

35. **In the Worked Example 2.5 C, the triangular Pascal matrix L has $L^{-1} = LDL$, where the diagonal matrix D has alternating entries $1, -1, 1, -1$. Then $LDLD = I$, so the inverse of $LD =$ pascal$(4, 1)$.**
So what is the inverse of $LD =$ pascal$(4, 1)$?

36. **The Hilbert matrices have $H_{ij} = 1/(i+j-1)$. Ask MATLAB for the exact 6 by 6 inverse inv$(H(6))$. Then ask for it to compute inv$(H(6))$. How can these be different, when the computer never makes mistakes?**

37. **(a) Use inv(P) to invert MATLAB's 4 by 4 symmetric matrix $P =$ pascal(4).**
(b) Create Pascal's lower triangular $L =$ abs(pascal$(4, 1)$) and test $P = LL^T$.

38. **If $A =$ ones(4) and $b =$ rand$(4, 1)$, how does MATLAB tell you that $Ax = \boldsymbol{b}$ has no solution? For the special $b =$ ones$(4, 1)$, which solution to $Ax = \boldsymbol{b}$ is found by $A\backslash b$?**

Challenge Problems

39. **(Recommended) A is a 4 by 4 matrix with 1 's on the diagonal and -a, -b, -c on the diagonal above. Find A^{-1} for this bidiagonal matrix.**

40. **Suppose E_1, E_2, E_3 are 4 by 4 identity matrices, except E_1 has a, b, c in column 1 and E_2 has d, e in column 2 and E_3 has f in column 3 (below the 1's). Multiply $L = E_1E_2E_3$ to show that all these nonzeros are copied into L.**
$E_1E_2E_3$ is in the *opposite* order for elimination (because E_3 is acting first). But $E_1E_2E_3 = L$ is in the *correct* order for invert elimination and recover A.

41. **Second difference matrices have beautiful inverses if they start with $T_{11} = 1$ (instead of $K_{11} = 2$). Here is the 3 by 3 tridiagonal matrix T and its inverse:**

$$T = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \qquad T^{-1} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

One approach is Gauss-Jordan elimination on $[T \mid I]$. I would rather write T as the product of first differences L times U. The inverses of L and U in Worked Example 2.5 A are sum matrices, so here are $T = LU$ and $T^{-1} = U^{-1}L^{-1}$:

$$T = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \qquad T^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Question. (4 by 4) What are the pivots of T? What is its 4 by 4 inverse? The reverse order UL gives what matrix T^{**}? What is the inverse of T^{**}?

42. **Here are two more difference matrices, both important. But are they invertible?**

$$\textbf{Cyclic } C = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \qquad \textbf{Free ends } F = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

43. **Elimination for a block matrix: When you multiply the first block row $[A\ B]$ by CA^{-1} and subtract from the second row $[C\ D]$, the "Schur complement" S appears:**

$$\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & S \end{bmatrix}, \qquad \textbf{where } A \textbf{ and } D \textbf{ are square,} \quad S = D - CA^{-1}B.$$

Multiply on the right to subtract $A^{-1}B$ times block column 1 from block column 2:

$$\begin{bmatrix} A & B \\ 0 & S \end{bmatrix} \begin{bmatrix} I & -A^{-1}B \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & S \end{bmatrix}.$$

Find S for

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 2 & 3 & 3 \\ 4 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}.$$

The block pivots are A and S. If they are invertible, so is $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$.

44. **How does the identity $A(I+BA) = (I+AB)A$ connect the inverses of $I+BA$ and $I+AB$? Those are both invertible or both singular: not obvious.**

2.6 Elimination = Factorization: $A = LU$

1. **(Important) Forward elimination changes $\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \boldsymbol{x} = \boldsymbol{b}$ to a triangular $\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \boldsymbol{x} = \boldsymbol{c}$:**

$$\begin{aligned} x + y &= 5 \\ x + 2y &= 7 \end{aligned} \quad \longrightarrow \quad \begin{aligned} x + y &= 5 \\ y &= 2 \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 5 \\ 1 & 2 & 7 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 5 \\ 0 & 1 & 2 \end{bmatrix}.$$

That step subtracted $\ell_{21} =$ ____ times row 1 from row 2. The reverse step adds ℓ_{21} times row 1 to row 2. The matrix for that reverse step is $L =$ ________. Multiply this L times the triangular system $\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \boldsymbol{x}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ to get ________ $=$ ________. In letters, L multiplies $U\boldsymbol{x} = \boldsymbol{c}$ to give ________.

2. **Write down the 2 by 2 triangular systems $L\boldsymbol{c} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{c}$ from Problem 1. Check that $\boldsymbol{c} = (5, 2)$ solves the first one. Find $\boldsymbol{x}$ that solves the second one.**

3. **(Move to 3 by 3) Forward elimination changes $A\boldsymbol{x} = \boldsymbol{b}$ to a triangular $U\boldsymbol{x} = \boldsymbol{c}$:**

$$\begin{aligned} x + y + z &= 5 \\ x + y + 2z &= 7 \\ x + 3y + 6z &= 11 \end{aligned} \quad \longrightarrow \quad \begin{aligned} x + y + z &= 5 \\ y + 2z &= 2 \\ 2y + 5z &= 6 \end{aligned} \quad \longrightarrow \quad \begin{aligned} x + y + z &= 5 \\ y + 2z &= 2 \\ z &= 2 \end{aligned}$$

The equation $z = 2$ in $U\boldsymbol{x} = \boldsymbol{c}$ comes from the original $x + 3y + 6z = 11$ in $A\boldsymbol{x} = \boldsymbol{b}$ by subtracting $\ell_{31} =$ ____ times equation 1 and $\ell_{32} =$ ____ times the final equation 2. Reverse that to recover $\begin{bmatrix} 1 & 3 & 6 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ in the last row of A and $\boldsymbol{b}$ from the final $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$ in U and $\boldsymbol{c}$:
Row 3 of $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix} = (\ell_{31}$ Row $1 + \ell_{32}$ Row $2 + 1$ Row $3)$ of $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix}$.
In matrix notation this is multiplication by L. So $A = LU$ and $\boldsymbol{b} = L\boldsymbol{c}$.

4. **What are the 3 by 3 triangular systems $L\boldsymbol{c} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{c}$ from Problem 3? Check that $\boldsymbol{c} = (5, 2, 2)$ solves the first one. Which $\boldsymbol{x}$ solves the second one?**

5. **What matrix E puts A into triangular form $EA = U$? Multiply by $E^{-1} = L$ to factor A into LU:**

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix}.$$

6. **What two elimination matrices E_{21} and E_{32} put A into upper triangular form $E_{32}E_{21}A = U$? Multiply by E_{21}^{-1} and E_{32}^{-1} to factor A into $LU = E_{21}^{-1}E_{32}^{-1}U$:**

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}.$$

7. **What three elimination matrices E_{21}, E_{31}, E_{32} put A into its upper triangular form $E_{32}E_{31}E_{21}A = U$? Multiply by $E_{21}^{-1}, E_{31}^{-1}, E_{32}^{-1}$ to factor A into L times U:**

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}, \qquad L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}.$$

8. **This is the problem that shows how the inverses E_{ij}^{-1} multiply to give L. You see this best when A is already lower triangular with 1's on the diagonal. Then $U = I$!**

$$A = L = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}.$$

The elimination matrices E_{21}, E_{31}, E_{32} contain $-a$ then $-b$ then $-c$.

(a) **Multiply $E_{32}E_{31}E_{21}$ to find the single matrix E that produces $EA = I$.**

(b) **Multiply $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$ to bring back L. The multipliers a, b, c are mixed up in E but perfect in L.**

9. **When zero appears in a pivot position, $A = LU$ is not possible! (We are requiring nonzero zeros in U.) Show directly why these equations are both impossible:**

$$\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \ell & 1 \end{bmatrix} \begin{bmatrix} d & e \\ 0 & f \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \ell & 1 & 0 \\ m & n & 1 \end{bmatrix} \begin{bmatrix} d & e & g \\ 0 & f & h \\ 0 & 0 & i \end{bmatrix}.$$

These matrices need a row exchange. That uses a "permutation matrix" P.

10. **Which number c leads to zero in the second pivot position? A row exchange is needed and $A = LU$ will not be possible. Which c produces zero in the third pivot position? Then a row exchange can't help and elimination fails:**

$$A = \begin{bmatrix} 1 & c & 0 \\ 2 & 4 & 1 \\ 3 & 5 & 1 \end{bmatrix}.$$

11. **What are L and D (the diagonal pivot matrix) for this matrix A? What is U in $A = LU$ and what is the new U in $A = LDU$?**

$$\textbf{Already triangular} \qquad A = \begin{bmatrix} 2 & 4 & 8 \\ 0 & 3 & 9 \\ 0 & 0 & 7 \end{bmatrix}.$$

12. **A and B are symmetric across the diagonal (for $4 = 4$). Find their triple factorizations LDU and say how U is related to L for these symmetric matrices:**

$$\textbf{Symmetric} \qquad A = \begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 1 & 4 & 0 \\ 4 & 12 & 4 \\ 0 & 4 & 0 \end{bmatrix}.$$

13. **(Recommended) Compute L and U for the symmetric matrix A:**

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}.$$

Find four conditions on a, b, c, d to get $A = LU$ with four pivots.

14. This nonsymmetric matrix will have the same L as in Problem 13:

$$\textbf{Find } L \textbf{ and } U \textbf{ for} \qquad A = \begin{bmatrix} a & r & r & r \\ a & b & s & s \\ a & b & c & t \\ a & b & c & d \end{bmatrix}.$$

Find the four conditions on a, b, c, d, r, s, t to get $A = LU$ with four pivots.

Problems 15–16 use L and U (without needing A) to solve $A\boldsymbol{x} = \boldsymbol{b}$.

15. Solve the triangular system $L\boldsymbol{c} = \boldsymbol{b}$ to find $\boldsymbol{c}$. Then solve $U\boldsymbol{x} = \boldsymbol{c}$ to find $\boldsymbol{x}$:

$$L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 2 \\ 11 \end{bmatrix}.$$

For safety multiply LU and solve $A\boldsymbol{x} = \boldsymbol{b}$ as usual. Circle $\boldsymbol{c}$ when you see it.

16. Solve $L\boldsymbol{c} = \boldsymbol{b}$. Then solve $U\boldsymbol{x} = \boldsymbol{c}$. What was A?

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

17. (a) When you apply the usual elimination steps to L, what matrix do you reach?

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}.$$

(b) When you apply the same steps to I, what matrix do you get?

(c) When you apply the same steps to LU, what matrix do you get?

18. If $A = LDU$ and also $A = L_1D_1U_1$ with all factors invertible, then $L = L_1$ and $D = D_1$ and $U = U_1$. *"The three factors are unique."*
Derive the equation $L_1^{-1}LD = D_1U_1U^{-1}$. Are the two sides triangular or diagonal? Deduce $L = L_1$ and $U = U_1$ (they all have diagonal 1's). Then $D = D_1$.

19. Tridiagonal matrices have zero entries except on the main diagonal and the two adjacent diagonals. Factor these into $A = LU$ and $A = LDL^T$:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{bmatrix}.$$

20. When A is tridiagonal, its L and U factors have only two nonzero diagonals. How would you take advantage of knowing the zeros in T, in a code for Gaussian elimination? Find L and U.

$$\textbf{Tridiagonal} \qquad T = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 3 & 4 \end{bmatrix}.$$

21. If A and B have nonzeros in the positions marked by x, which zeros (marked by 0) stay zero in their factors L and U?

$$A = \begin{bmatrix} x & x & x & x \\ x & x & x & 0 \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix} \qquad B = \begin{bmatrix} x & x & x & 0 \\ x & x & 0 & x \\ x & 0 & x & x \\ 0 & x & x & x \end{bmatrix}.$$

22. Suppose you eliminate upwards (almost unheard of). Use the last row to produce zeros in the last column (the pivot is 1). Then use the second row to produce zero above the second pivot. Find the factors in the unusual order $A = UL$.

$$\textbf{Upper times lower} \qquad A = \begin{bmatrix} 5 & 3 & 1 \\ 3 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

23. Easy but important. If A has pivots $5, 9, 3$ with no row exchanges, what are the pivots for the upper left 2 by 2 submatrix A_2 (without row 3 and column 3)?

Challenge Problems

24. **Which invertible matrices allow $A = LU$ (elimination without row exchanges)? *Good question!* Look at each of the square upper left submatrices A_k of A.**

    All upper left k by k submatrices A_k must be invertible(sizes $k = 1, \ldots, n$).

    Explain that answer: A_k factors into ________ because $LU = \begin{bmatrix} L_k & 0 \\ * & * \end{bmatrix} \begin{bmatrix} U_k & * \\ 0 & * \end{bmatrix}$.

25. **For the 6 by 6 second difference constant-diagonal matrix K, put the pivots and multipliers into $K = LU$. (L and U will have only two nonzero diagonals, because K has three.) Find a formula for the i, j entry of L^{-1}, by software like MATLAB using inv(L) or by looking for a nice pattern.**

$$-1, 2, -1 \text{ matrix} \quad K = \begin{bmatrix} 2 & -1 & & & & \\ -1 & 2 & -1 & & & \\ & \ddots & \ddots & \ddots & & \\ & & -1 & 2 & -1 & \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{bmatrix} = \textbf{toeplitz}([2 \ -1\ 0\ 0\ 0\ 0]).$$

26. **If you print K^{-1}, it doesn't look so good (6 by 6). But if you print $7K^{-1}$, that matrix looks wonderful. Write down $7K^{-1}$ by hand, following this pattern:**

    1. **Row 1 and column 1 are $(6, 5, 4, 3, 2, 1)$.**
    2. **On and above the main diagonal, row i is i times row 1.**
    3. **On and below the main diagonal, column j is j times column 1.**

    Multiply K times that $7K^{-1}$ to produce $7I$. Here is $4K^{-1}$ for $n = 3$:

$$\begin{array}{l} \textit{3 by 3 case} \\ \textit{The determinant} \\ \textit{of this } K \textit{ is } 4. \end{array} \quad (K)(4K^{-1}) = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & & \\ & 4 & \\ & & 4 \end{bmatrix}.$$

2.7 Transposes and Permutations

Questions 1–7 are about the rules for transpose matrices.

1. **Find A^{T} and A^{-1} and $(A^{-1})^{\mathrm{T}}$ and $(A^{\mathrm{T}})^{-1}$ for**

$$A = \begin{bmatrix} 1 & 0 \\ 9 & 3 \end{bmatrix} \quad \textbf{and also} \quad A = \begin{bmatrix} 1 & c \\ c & 0 \end{bmatrix}.$$

2. **Verify that $(AB)^{\mathrm{T}}$ equals $B^{\mathrm{T}}A^{\mathrm{T}}$ but those are different from $A^{\mathrm{T}}B^{\mathrm{T}}$:**

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad AB = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

    Show also that AA^{T} is different from $A^{\mathrm{T}}A$. But both of those matrices are ____________.

3. **(a) The matrix $((AB)^{-1})^{\mathrm{T}}$ comes from $(A^{-1})^{\mathrm{T}}$ and $(B^{-1})^{\mathrm{T}}$. In what order?**

    (b) If U is upper triangular then $(U^{-1})^{\mathrm{T}}$ is ____________ triangular.

4. **Show that $A^2 = 0$ is possible but $A^{\mathrm{T}}A = 0$ is not possible (unless $A =$ zero matrix).**

5. **(a) The row vector x^{T} times A times the column y produces what number?**

$$x^{\mathrm{T}}Ay = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \underline{\qquad\qquad}.$$

    (b) This is the row $x^{\mathrm{T}}A =$ ____________ times the column $y = (0, 1, 0)$.

(c) This is the row $x^{\mathrm{T}} = \begin{bmatrix} 0 & 1 \end{bmatrix}$ times the column $Ay =$ ____________.

6. The transpose of a block matrix $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ is $M^{\mathrm{T}} =$ ____________. Test an example. Under what conditions on A, B, C, D is the block matrix symmetric?

7. True or false:

   (a) The block matrix $\begin{bmatrix} 0 & A \\ A & 0 \end{bmatrix}$ is automatically symmetric.

   (b) If A and B are symmetric then their product AB is symmetric.

   (c) If A is not symmetric then A^{-1} is not symmetric.

   (d) When A, B, C are symmetric, the transpose of ABC is CBA.

Questions 8–15 are about permutation matrices.

8. Why are there $n!$ permutation matrices of order n?

9. If P_1 and P_2 are permutation matrices, so is P_1P_2. This still has the rows of I in some order. Give examples with $P_1P_2 \neq P_2P_1$ and $P_3P_4 = P_4P_3$.

10. There are 12 "even" permutations of $(1,2,3,4)$, with an even number of exchanges. Two of them are $(1,2,3,4)$ with no exchanges and $(4,3,2,1)$ with two exchanges. List the other ten. Instead of writing 4 by 4 matrix, just order the numbers.

11. Which permutation makes PA upper triangular? Which permutations make P_1AP_2 lower triangular? Multiplying A on the right by P_2 exchanges the ____________ of A.

$$A = \begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}.$$

12. Explain why the dot product of x and y equals the dot product of Px and Py. Then $(Px)^{\mathrm{T}}(Py) = x^{\mathrm{T}}y$ tells us that $P^{\mathrm{T}}P = I$ for any permutation. With $x = (1,2,3)$ and $y = (1,4,2)$ choose P to show that $Px \cdot Py$ is not always $x \cdot y$.

13. (a) Find a 3 by 3 permutation matrix with $P^3 = I$ (but not $P = I$).

    (b) Find a 4 by 4 permutation $\bar{P}$ with $\bar{P}^4 \neq I$.

14. If P has 1's on the antidiagonal from $(1,n)$ to $(n,1)$, describe PAP. Note $P = P^{\mathrm{T}}$.

15. All row exchange matrices are symmetric: $P^{\mathrm{T}} = P$. Then $P^{\mathrm{T}}P = I$ becomes $P^2 = I$. Other permutation matrices may or may not be symmetric.

    (a) If P sends row 1 to row 4, then P^{T} sends row ____________ to row ____________. When $P^{\mathrm{T}} = P$ the row exchanges come in pairs with no overlap.

    (b) Find a 4 by 4 example with $P^{\mathrm{T}} = P$ that moves all four rows.

Questions 16–21 are about symmetric matrices and their factorizations.

16. If $A = A^{\mathrm{T}}$ and $B = B^{\mathrm{T}}$, which of these matrices are certainly symmetric?

    (a) $A^2 - B^2$

    (b) $(A+B)(A-B)$

    (c) ABA

    (d) $ABAB$

17. Find 2 by 2 symmetric matrices $S = S^{\mathrm{T}}$ with these properties:

    (a) S is not invertible.

    (b) S is invertible but cannot be factored into LU (row exchanges needed).

    (c) S can be factored into LDL^{T} but not into LL^{T} (because of negative D).

18. **(a)** **How many entries of S can be chosen independently, if $S = S^{\mathrm{T}}$ is 5 by 5?**

(b) **How do L and D (still 5 by 5) give the same number of choices in LDL^{T}?**

(c) **How many entries can be chosen if A is skew-symmetric? ($A^{\mathrm{T}} = -A$).**

19. **Suppose A is rectangular (m by n) and S is symmetric (m by m).**

(a) **Transpose $A^{\mathrm{T}}SA$ to show its symmetry. What shape is this matrix?**

(b) **Show why $A^{\mathrm{T}}A$ has no negative numbers on its diagonal.**

20. **Factor these symmetric matrices into $S = LDL^{\mathrm{T}}$. The pivot matrix D is diagonal:**

$$S = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & b \\ b & c \end{bmatrix}, \quad S = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

21. **After elimination clears out column 1 below the first pivot, find the symmetric 2 by 2 matrix that appears in the lower right corner:**

$$\textbf{Start from } S = \begin{bmatrix} 2 & 4 & 8 \\ 4 & 3 & 9 \\ 8 & 9 & 0 \end{bmatrix} \quad \textbf{and} \quad S = \begin{bmatrix} 1 & b & c \\ b & d & e \\ c & e & f \end{bmatrix}.$$

Questions 22–24 are about the factorizations $PA = LU$ and $A = L_1P_1U_1$.

22. **Find the $PA = LU$ factorizations (and check them) for**

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

23. **Find a 4 by 4 permutation matrix (call it A) that needs 3 row exchanges to reach the end of elimination. For this matrix, what are its factors $P, L,$ and U?**

24. **Factor the following matrix into $PA = LU$. Factor it also into $A = L_1P_1U_1$ (hold the exchange of row 3 until 3 times row 1 is subtracted from row 2):**

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 & 8 \\ 2 & 1 & 1 \end{bmatrix}.$$

25. **Prove that the identity matrix cannot be the product of three row exchanges (or five). It can be the product of two exchanges (or four).**

26. **(a)** **Choose E_{21} to remove the 3 below the first pivot. Then multiply $E_{21}SE_{21}^{\mathrm{T}}$ to remove both 3's:**

$$S = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 11 & 4 \\ 0 & 4 & 9 \end{bmatrix} \quad \textbf{is going toward} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) **Choose E_{32} to remove the 4 below the second pivot. Then S is reduced to D by $E_{32}E_{21}SE_{21}^{\mathrm{T}}E_{32}^{\mathrm{T}} = D$. Invert the E's to find L in $S = LDL^{\mathrm{T}}$.**

27. **If every row of a 4 by 4 matrix contains the numbers $0, 1, 2, 3$ in some order, can the matrix be symmetric?**

28. **Prove that no reordering of rows and reordering of columns can transpose a typical matrix. (Watch the diagonal entries.)**

The next three questions are about applications of the identity $(Ax)^{\mathrm{T}}y = x^{\mathrm{T}}(A^{\mathrm{T}}y)$.

29. Wires go between Boston, Chicago, and Seattle. Those cities are at voltages x_B, x_C, x_S. With unit resistances between cities, the currents between cities are in y:

$$y = Ax \quad \textbf{is} \quad \begin{bmatrix} y_{BC} \\ y_{CS} \\ y_{BS} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_B \\ x_C \\ x_S \end{bmatrix}.$$

(a) Find the total currents $A^{\mathrm{T}}y$ out of the three cities.

(b) Verify that $(Ax)^{\mathrm{T}}y$ agrees with $x^{\mathrm{T}}(A^{\mathrm{T}}y)$—six terms in both.

30. Producing x_1 trucks and x_2 planes needs $x_1 + 50x_2$ tons of steel, $40x_1 + 1000x_2$ pounds of rubber, and $2x_1 + 50x_2$ months of labor. If the unit costs y_1, y_2, y_3 are \$700 per ton, \$3 per pound, and \$3000 per month, what are the values of one truck and one plane? Those are the components of $A^{\mathrm{T}}y$.

31. Ax gives the amounts of steel, rubber, and labor to produce x in Problem 31. Find A. Then $Ax \cdot y$ is the ___________ of inputs while $x \cdot A^{\mathrm{T}}y$ is the value of ___________.

32. The matrix P that multiplies (x, y, z) to give (z, x, y) is also a rotation matrix. Find P and P^3. The rotation axis $a = (1, 1, 1)$ doesn't move, it equals Pa. What is the angle of rotation from $v = (2, 3, -5)$ to $Pv = (-5, 2, 3)$?

33. Write $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ as the product ES of an elementary row operation matrix E and a symmetric matrix S.

34. Here is a new factorization of A into LS: triangular (with 1's) times symmetric:

Start from $A = LDU$. Then A equals $L(U^{\mathrm{T}})^{-1}$ times $S = U^{\mathrm{T}}DU$.

Why is $L(U^{\mathrm{T}})^{-1}$ triangular? Its diagonal is all 1's. Why is $U^{\mathrm{T}}DU$ symmetric?

35. A group of matrices includes AB and A^{-1} if it includes A and B. "Products and inverses stay in the group." Which of these sets are groups?
Lower triangular matrices L with 1's on the diagonal, symmetric matrices S, positive matrices M, diagonal invertible matrices D, permutation matrices P, matrices with $Q^{\mathrm{T}} = Q^{-1}$. Invent two more matrix groups.

Challenge Problems

36. A square northwest matrix B is zero in the southeast corner, below the antidiagonal that connects $(1, n)$ to $(n, 1)$. Will B^{T} and B^2 be northwest matrices? Will B^{-1} be northwest or southeast? What is the shape of BC = northwest times southeast?

37. If you take powers of a permutation matrix P, why is some P^k eventually equal to I? Find a 5 by 5 permutation P so that the smallest power to equal I is P^6.

38. (a) Write down any 3 by 3 matrix M. Split M into $S + A$ where $S = S^{\mathrm{T}}$ is symmetric and $A = -A^{\mathrm{T}}$ is anti-symmetric.

(b) Find formulas for S and M^{T}. We want $M = S + A$.

39. Suppose Q^{T} equals Q^{-1} (transpose equals inverse, so $Q^{\mathrm{T}}Q = I$).

(a) Show that the columns $q_1, \ldots, q_n$ are unit vectors: $\|q_i\|^2 = 1$.

(b) Show that every two columns of Q are perpendicular: $q_1^{\mathrm{T}}q_2 = 0$.

(c) Find a 2 by 2 example with first entry $q_{11} = \cos\theta$.

3 Chapter 3: Vector Spaces and Subspaces

3.1 Spaces of Vectors

The first problems 1–8 are about vector spaces in general. The vectors in those spaces are not necessarily column vectors. In the definition of a *vector space*, vector addition $x+y$ and scalar multiplication cx must obey the following eight rules:

(1) $x+y=y+x$

(2) $x+(y+z)=(x+y)+z$

(3) **There is a unique "zero vector" such that $x+0=x$ for all x**

(4) **For each x there is a unique vector $-x$ such that $x+(-x)=0$**

(5) **1 times x equals x**

(6) $(c_1c_2)x=c_1(c_2x)$

(7) $c(x+y)=cx+cy$

(8) $(c_1+c_2)x=c_1x+c_2x$

(1) to (4) about $x+y$ (5) to (6) about cx (7) to (8) connects them

1. **Suppose $(x_1,x_2)+(y_1,y_2)$ is defined to be (x_1+y_2,x_2+y_1). With the usual multiplication $cx=(cx_1,cx_2)$, which of the eight conditions are not satisfied?**

2. **Suppose the multiplication cx is defined to produce $(cx_1,0)$ instead of (cx_1,cx_2). With the usual addition in $\mathbb{R}^2$, are the eight conditions satisfied?**

3. **(a) Which rules are broken if we keep only the positive numbers $x>0$ in $\mathbb{R}^1$? Every c must be allowed. The half-line is not a subspace.**

   (b) The positive numbers with $x+y$ and cx redefined to equal the usual xy and x^c do satisfy the eight rules. Test rule 7 when $c=3$, $x=2$, $y=1$. (Then $x+y=2$ and $cx=8$.) Which number acts as the "zero vector"?

4. **The matrix $A=\begin{bmatrix}2&-2\\2&-2\end{bmatrix}$ is a "vector" in the space M of all 2 by 2 matrices. Write down the zero vector in this space, the vector $\frac{1}{2}A$, and the vector $-A$. What matrices are in the smallest subspace containing A?**

5. **Describe a subspace of M that contains $A=\begin{bmatrix}1&0\\0&0\end{bmatrix}$ but not $B=\begin{bmatrix}0&0\\0&1\end{bmatrix}$.**

   (a) If a subspace of M contains A and B, must it contain 0?

   (b) Describe a subspace of M that contains no nonzero diagonal matrices.

6. **The functions $f(x)=x^2$ and $g(x)=5x$ are "vectors" in F. This is the vector space of all real functions. (The functions are defined for $-\infty<x<\infty$.) The combination $3f(x)-4g(x)$ is the function $h(x)=$ ________________.**

7. **Which rule is broken if multiplying $f(x)$ by c gives the function $f(cx)$? Keep the usual addition, $f(x)+g(x)$.**

8. **If the sum of the "vectors" $f(x)$ and $g(x)$ is defined to be the function $f(g(x))$, then the "zero vector" is $g(x)=x$. Keep the usual scalar multiplication $cf(x)$ and find two rules that are broken.**

Questions 9–18 are about the "subspace requirements": $x+y$ and cx and then all linear combinations $cx+dy$ stay in the subspace.

9. **One requirement can be met while the other fails. Show this by finding**

   (a) A set of vectors in $\mathbb{R}^2$ for which $x+y$ stays in the set but $\frac{1}{2}x$ may be outside.

(b) A set of vectors in $\mathbb{R}^2$ (other than two quarter-planes) for which every cx stays in the set but $x + y$ may be outside.

10. Which of the following subsets of $\mathbb{R}^3$ are actually subspaces?

(a) The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$.

(b) The plane of vectors with $b_1 = 1$.

(c) The vectors with $b_1 b_2 b_3 = 0$.

(d) All linear combinations of $v = (1, 4, 0)$ and $w = (2, 2, 2)$.

(e) All vectors that satisfy $b_1 + b_2 + b_3 = 0$.

(f) All vectors with $b_1 \le b_2 \le b_3$.

11. Describe the smallest subspace of the matrix space M that contains

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ **and** $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ **(b)** $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ **(c)** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ **and** $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

12. Let P be the plane in $\mathbb{R}^3$ with equation $x + y - 2z = 4$. The origin $(0, 0, 0)$ is not in P! Find two vectors in P and check that their sum is not in P.

13. Let P_0 be the plane through $(0, 0, 0)$ parallel to the previous plane P. What is the equation for P_0? Find two vectors in P_0 and check that their sum is in P_0.

14. The subspaces of $\mathbb{R}^3$ are planes, lines, $\mathbb{R}^3$ itself, or $\mathbb{Z}$ containing only $(0, 0, 0)$.

(a) Describe the three types of subspaces of $\mathbb{R}^2$.

(b) Describe all subspaces of D, the space of 2 by 2 diagonal matrices.

15. (a) The intersection of two planes through $(0, 0, 0)$ is probably a ____________ in $\mathbb{R}^3$ but it could be a ____________. It can't be $\mathbb{Z}$!

(b) The intersection of a plane through $(0, 0, 0)$ with a line through $(0, 0, 0)$ is probably a ____________ but it could be a ____________.

(c) If S and T are subspaces of $\mathbb{R}^5$, prove that their intersection S $\cap$ T is a subspace of $\mathbb{R}^5$. Here S $\cap$ T consists of the vectors that lie in both subspaces. Check that $x + y$ and cx are in S $\cap$ T if x and y are in both spaces.

16. Suppose P is a plane through $(0, 0, 0)$ and L is a line through $(0, 0, 0)$. The smallest vector space containing both P and L is either ____________ or ____________.

17. (a) Show that the set of invertible matrices in M is not a subspace.

(b) Show that the set of singular matrices in M is not a subspace.

18. True or false (check addition in each case by example):

(a) The symmetric matrices in M (with $A^{\mathrm{T}} = A$) form a subspace.

(b) The skew-symmetric matrices in M (with $A^{\mathrm{T}} = -A$) form a subspace.

(c) The unsymmetric matrices in M (with $A^{\mathrm{T}} \ne A$) form a subspace.

Questions 19–27 are about column spaces $C(A)$ and the equation $Ax = b$.

19. Describe the column spaces (lines or planes) of these particular matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

20. For which right sides (find a condition on b_1, b_2, b_3) are these systems solvable?

(a) $\begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 4 \\ 2 & 9 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

21. **Adding row 1 of A to row 2 produces B. Adding column 1 to column 2 produces C. A combination of the columns of (B or C?) is also a combination of the columns of A. Which two matrices have the same column space?**

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

22. **For which vectors (b_1, b_2, b_3) do these systems have a solution?**

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \textbf{and} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

23. **(Recommended) If we add an extra column b to a matrix A, then the column space gets larger unless ______________________. Give an example where the column space gets larger and an example where it doesn't. Why is $Ax = b$ solvable exactly when the column space doesn't get larger—it is the same for A and $[A\ b]$?**

24. **The columns of AB are combinations of the columns of A. This means: *The column space of AB is contained in* (possibly equal to) *the column space of A*. Give an example where the column spaces of A and AB are not equal.**

25. **Suppose $Ax = b$ and $Ay = b^*$ are both solvable. Then $Az = b + b^*$ is solvable. What is z? This translates into: If b and b^* are in the column space $C(A)$, then $b + b^*$ is in $C(A)$.**

26. **If A is any 5 by 5 invertible matrix, then its column space is ________________. Why?**

27. **True or false (with a counterexample if false):**
    - **(a)** **The vectors b that are not in the column space $C(A)$ form a subspace.**
    - **(b)** **If $C(A)$ contains only the zero vector, then A is the zero matrix.**
    - **(c)** **The column space of $2A$ equals the column space of A.**
    - **(d)** **The column space of $A - I$ equals the column space of A (test this).**

28. **Construct a 3 by 3 matrix whose column space contains $(1, 1, 0)$ and $(1, 0, 1)$ but not $(1, 1, 1)$. Construct a 3 by 3 matrix whose column space is only a line.**

29. **If the 9 by 12 system $Ax = b$ is solvable for every b, then $C(A) =$ ________________.**

Challenge Problems

30. **Suppose S and T are two subspaces of a vector space V.**
    - **(a)** **Definition: The sum $S+T$ contains all sums $s+t$ of a vector s in S and a vector t in T. Show that $S+T$ satisfies the requirements (addition and scalar multiplication) for a vector space.**
    - **(b)** **If S and T are lines in $\mathbb{R}^n$, what is the difference between $S+T$ and $S \cup T$? That union contains all vectors from S or T or both. Explain this statement: *The span of $S \cup T$ is $S+T$*. (Section 3.5 returns to this word "span".)**

31. **If S is the column space of A and T is $C(B)$, then $S+T$ is the column space of what matrix M? The columns of A and B and M are all in $\mathbb{R}^m$. (I don't think $A+B$ is always correct M.)**

32. **Show that the matrices A and $\begin{bmatrix} A & AB \end{bmatrix}$ (with extra columns) have the same column space. But find a square matrix with $C(A^2)$ smaller than $C(A)$.**
    Important point: An n by n matrix has $C(A) = \mathbb{R}^n$ exactly when A is an ____________ matrix.

3.2 The Nullspace of A : Solving $Ax = 0$ and $Rx = 0$

1. **Reduce A and B to their triangular echelon forms U. Which variables are free?**

   (a) $A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}$

   (b) $B = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 8 & 8 \end{bmatrix}$

2. **For the matrices in Problem 1, find a special solution for each free variable. (Set the free variable to 1. Set the other free variables to zero.)**

3. **By further row operations on each U in Problem 1, find the reduced echelon form R. True or false with a reason: The nullspace of R equals the nullspace of U.**

4. **For the same A and B, find the special solutions to $Ax = 0$ and $Bx = 0$. For an m by n matrix, the number of pivot variables plus the number of free variables is ____________. This is the COUNTING THEOREM: $r + (n - r) = n$.**

   (a) $A = \begin{bmatrix} -1 & 3 & 5 \\ -2 & 6 & 10 \end{bmatrix}$

   (b) $B = \begin{bmatrix} -1 & 3 & 5 \\ -2 & 6 & 7 \end{bmatrix}$

Questions 5-14 are about free variables and pivot variables.

5. **True or false (with reason if true or example to show it is false):**

   (a) **A square matrix has no free variables.**

   (b) **An invertible matrix has no free variables.**

   (c) **An m by n matrix has no more than n pivot variables.**

   (d) **An m by n matrix has no more than m pivot variables.**

6. **Put as many 1's as possible in a 4 by 7 echelon matrix U whose pivot columns are**

   (a) $2, 4, 5$

   (b) $1, 3, 6, 7$

   (c) 4 **and** 6.

7. **Put as many 1's as possible in a 4 by 8 *reduced* echelon matrix R so that the free columns are**

   (a) $2, 4, 5, 6$

   (b) $1, 3, 6, 7, 8$.

8. **Suppose column 4 of a 3 by 5 matrix is all zero. Then x_4 is certainly a ____________ variable. The special solution for this variable is the vector $x =$ ____________.**

9. **Suppose the first and last columns of a 3 by 5 matrix are the same (not zero). Then ____________ is a free variable. Find the special solution for this variable.**

10. **Suppose an m by n matrix has r pivots. The number of special solutions is ____________. The nullspace contains only $x = 0$ when $r =$ ____________. The column space is all of $\mathbb{R}^m$ when $r =$ ____________.**

11. **The nullspace of a 5 by 5 matrix contains only $x = 0$ when the matrix has ____________ pivots. The column space is $\mathbb{R}^5$ when there are ____________ pivots. Explain why.**

12. **The equation $x - 3y - z = 0$ determines a plane in $\mathbb{R}^3$. What is the matrix A in this equation? Which variables are free? The special solutions are ____________ and ____________.**

13. **(Recommended) The plane** $x - 3y - z = 12$ **is parallel to** $x - 3y - z = 0$. **One particular point on this plane is** $(12, 0, 0)$. **All points on the plane have the form**

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

14. **Suppose column 1 + column 3 + column 5 = 0 in a 4 by 4 matrix with four pivots. Which column has no pivot? What is the special solution? Describe** $N(A)$.

Questions 15–22 ask for matrices (if possible) with specific properties.

15. **Construct a matrix for which** $N(A) =$ **all combinations of** $(2, 2, 1, 0)$ **and** $(3, 1, 0, 1)$.
16. **Construct** A **so that** $N(A) =$ **all multiples of** $(4, 3, 2, 1)$. **Its rank is ____________.**
17. **Construct a matrix whose column space contains** $(1, 1, 5)$ **and** $(0, 3, 1)$ **and whose nullspace contains** $(1, 1, 2)$.
18. **Construct a matrix whose column space contains** $(1, 1, 0)$ **and** $(0, 1, 1)$ **and whose nullspace contains** $(1, 0, 1)$ **and** $(0, 0, 1)$.
19. **Construct a matrix whose column space contains** $(1, 1, 1)$ **and whose nullspace is the line of multiples of** $(1, 1, 1, 1)$.
20. **Construct a 2 by 2 matrix whose nullspace equals its column space. This is possible.**
21. **Why does no 3 by 3 matrix have a nullspace that equals its column space?**
22. **If** $AB = 0$ **then the column space of** B **is contained in the ____________ of** A. **Why?**
23. **The reduced form** R **of a 3 by 3 matrix with randomly chosen entries is almost sure to be ____________. What** R **is virtually certain if the random** A **is 4 by 3?**
24. **Show by example that these three statements are generally false:**

    (a) A **and** A^{T} **have the same nullspace.** **(b)** A **and** A^{T} **have the same free variables.**

    (c) If R **is the reduced form** $\mathrm{rref}(A)$ **then** R^{T} **is** $\mathrm{rref}(A^{\mathrm{T}})$.

25. **If** $N(A) =$ **all multiples of** $x = (2, 1, 0, 1)$, **what is** R **and what is its rank?**
26. **If the special solutions to** $Rx = 0$ **are in the columns of these nullspace matrices** N, **go backward to find the nonzero rows of the reduced matrices** R:

$$N = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad N = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad N = \begin{bmatrix} \ \end{bmatrix} \textbf{ (empty 3 by 1)}.$$

27. **(a) What are the five 2 by 2 reduced matrices** R **whose entries are all 0's and 1's?**

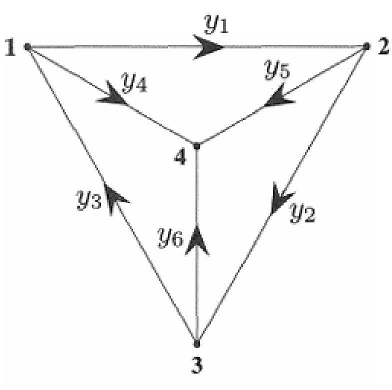
    (b) What are the eight 1 by 3 matrices containing only 0's and 1's? Are all eight of them reduced echelon matrices R?

28. **Explain why** A **and** $-A$ **always have the same reduced echelon form** R.
29. **If** A **is 4 by 4 and invertible, describe the nullspace of the 4 by 8 matrix** $B = [A \ A]$.
30. **How is the nullspace** $N(C)$ **related to the spaces** $N(A)$ **and** $N(B)$, **if** $C = \begin{bmatrix} A \\ B \end{bmatrix}$**?**
31. **Find the reduced row echelon forms** R **and the rank of these matrices:**

    (a) The 3 by 4 matrix with all entries equal to 4.

(b) The 3 by 4 matrix with $a_{ij} = i + j - 1$. **(c) The 3 by 4 matrix with $a_{ij} = (-1)^{i+j}$.**

32. Kirchhoff's Current Law $A^{\mathrm{T}}\mathbf{y} = \mathbf{0}$ says that *current in* = *current out* at every node. At node 1 this is $y_3 = y_1 + y_4$. Write the four equations for Kirchhoff's Law (arrows show the positive direction of each y). Reduce A^{T} to R and find three special solutions in the nullspace of A^{T} (4 by 6 matrix).



$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

33. Which of these rules gives a correct definition of the *rank* of A?

(a) The number of nonzero rows in R.

(b) The number of columns minus the total number of rows.

(c) The number of columns minus the number of free columns.

(d) The number of 1's in the matrix R.

34. Find the reduced R for each of these (block) matrices:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \qquad B = \begin{bmatrix} A & A \end{bmatrix}, \qquad C = \begin{bmatrix} A & A \\ A & 0 \end{bmatrix}.$$

35. Suppose all the pivot variables come *last* instead of first. Describe all four blocks in the reduced echelon form (the block B should be r by r):

$$R = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

What is the nullspace matrix N containing the special solutions?

36. (Silly problem) Describe all 2 by 3 matrices A_1 and A_2, with row echelon forms R_1 and R_2, such that $R_1 + R_2$ is the row echelon form of $A_1 + A_2$. Is it true that $R_1 = A_1$ and $R_2 = A_2$ in this case? Does $R_1 - R_2$ equal $\mathrm{rref}(A_1 - A_2)$?

37. If A has r pivot columns, how do you know that A^{T} has r pivot columns? Give a 3 by 3 example with different column numbers in `rref` for A and A^{T}.

38. What are the special solutions to $R\mathbf{x} = \mathbf{0}$ and $\mathbf{y}^{\mathrm{T}}R = \mathbf{0}$ for these R?

$$R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

39. Fill out these matrices so that they have rank 1:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & \_ & \_ \\ \_ & \_ & \_ \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 9 & \\ 2 & 6 & -3 \end{bmatrix}, \qquad M = \begin{bmatrix} a & b \\ c & \_ \end{bmatrix}.$$

40. If A is an m by n matrix with $r = 1$, its columns are multiples of one column and its rows are multiples of one row. The column space is a ____________ in $\mathbb{R}^m$. The nullspace is a ____________ in $\mathbb{R}^n$. The nullspace matrix N has shape ____________.

41. Choose vectors u and v so that $A = \mathbf{u}\mathbf{v}^{\mathrm{T}}$ = column times row:

$$A = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ 4 & 8 & 8 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} 2 & 2 & 6 & 4 \\ -1 & -1 & -3 & -2 \end{bmatrix}.$$

$A = \mathbf{u}\mathbf{v}^{\mathrm{T}}$ is the natural form for every matrix that has rank $r = 1$.

42. If A is a rank one matrix, the second row of R is ____________. Do an example.

Problems 43–45 are about r by r invertible matrices inside A.

43. If A has rank r, then it has an r by r submatrix S that is invertible. Remove $m - r$ rows and $n - r$ columns to find an invertible submatrix S inside A, B, and C. You could keep the pivot rows and pivot columns:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

44. Suppose P contains only the r pivot columns of an m by n matrix. Explain why this m by r submatrix P has rank r.

45. Transpose P in Problem 44. Find the r pivot columns of P^{T} (which is r by m). Transposing back, this produces an r by r invertible submatrix S inside P and A:

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 4 & 7 \end{bmatrix}$ produce P (3 by 2) and then the invertible S (2 by 2).

Problems 46–51 show that $\operatorname{rank}(AB)$ is not greater than $\operatorname{rank}(A)$ or $\operatorname{rank}(B)$.

46. Find the ranks of AB and AC (rank one matrix times rank one matrix):

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1.5 & 6 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & b \\ c & bc \end{bmatrix}.$$

47. The rank one matrix $\mathbf{u}\mathbf{v}^{\mathrm{T}}$ times the rank one matrix $\mathbf{w}\mathbf{z}^{\mathrm{T}}$ is u(____________)$^{\mathrm{T}}$ times the number ____________. This product $\mathbf{u}\mathbf{v}^{\mathrm{T}}\mathbf{w}\mathbf{z}^{\mathrm{T}}$ also has rank one unless ____________ $= 0$.

48. (a) Suppose column j of B is a combination of previous columns of B. Show that column j of AB is the same combination of previous columns of AB. Then AB cannot have new pivot columns, so $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$.

(b) Find A_1 and A_2 so that $\operatorname{rank}(A_1B) = 1$ and $\operatorname{rank}(A_2B) = 0$ for $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

49. Problem 48 proved that $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$. Then the same reasoning gives $\operatorname{rank}(B^{\mathrm{T}}A^{\mathrm{T}}) \leq \operatorname{rank}(A^{\mathrm{T}})$. How do you deduce that $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$?

50. (Important) Suppose A and B are n by n matrices, and $AB = I$. Prove from $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$ that the rank of A is n. So A is invertible and B must be its two-sided inverse (Section 2.5). Therefore $BA = I$ (which is not so obvious!).

51. If A is 2 by 3 and B is 3 by 2 and $AB = I$, show from its rank that $BA \neq I$. Give an example of A and B with $AB = I$. For $m < n$, a right inverse is not a left inverse.

52. Suppose A and B have the same reduced echelon form R.

(a) Show that A and B have the same nullspace and the same row space.

(b) We know $E_1A = R$ and $E_2B = R$. So A equals an ____________ matrix times B.

53. Express A and then B as the sum of two rank one matrices:

$$\text{rank} = 2, \qquad A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 8 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}.$$

54. Answer the same questions as in Worked Example 3.2 C for

$$A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 1 & c & 2 & 2 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 1-c & 2 \\ 0 & 2-c \end{bmatrix}.$$

55. What is the nullspace matrix N (containing the special solutions) for A, B, C?
Block matrices:

$$A = \begin{bmatrix} I & I \end{bmatrix}, \qquad B = \begin{bmatrix} I & I \\ 0 & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} I & I \\ I & I \end{bmatrix}.$$

56. Neat fact Every m by n matrix of rank r reduces to (m by r times) (r by n) by:

$$\boxed{A = (\textbf{pivot columns of } A)\ (\textbf{first } r \textbf{ rows of } R) = (\textbf{COL})(\textbf{ROW}).}$$

Write the 3 by 4 matrix A of all ones as the product of the 3 by 1 matrix from the pivot columns and the 1 by 4 matrix from R.

Challenge Problems

57. Suppose A is an m by n matrix of rank r. Its reduced echelon form is R. Describe exactly the matrix Z (its shape and all its entries) that comes from transposing the reduced row echelon form of R^{T}:

$$R = \text{ref}(A) \quad \textbf{and} \quad Z = \left(\text{ref}(A^{\mathrm{T}})\right)^{\mathrm{T}}.$$

58. (Recommended) Suppose R is m by n of rank r, with pivot columns first:

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}.$$

- **(a) What are the shapes of those four blocks?**
- **(b) Find a right-inverse B with $RB = I$ if $r = m$. The zero blocks are gone.**
- **(c) Find a left-inverse C with $CR = I$ if $r = n$. The F and 0 column is gone.**
- **(d) What is the reduced row echelon form of R^{T} (with shapes)?**
- **(e) What is the reduced row echelon form of $R^{\mathrm{T}}R$ (with shapes)?**

59. I think that the reduced echelon form of $R^{\mathrm{T}}R$ is always R (except for extra zero rows). Can you do an example when R is 2 by 3? Later we show that $A^{\mathrm{T}}A$ always has the same nullspace as A (a valuable fact).

60. Suppose you allow elementary column operations on A as well as elementary row operations (which get you to R). What is the "row-and-column reduced form" for an m by n matrix of rank r?

3.3 The Complete Solution to $Ax = b$

1. (Recommended) Execute the six steps of Worked Example 3.3 A to describe the column space and nullspace of A and the complete solution to $A\mathrm{x} = \mathrm{b}$:

$$A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}.$$

2. **Carry out the same six steps for this matrix A with rank one. You will find *two* conditions on b_1, b_2, b_3 for $A\mathrm{x} = \mathrm{b}$ to be solvable. Together these two conditions put b into the ____________ space (two planes give a line):**

$$A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \\ 4 & 2 & 6 \end{bmatrix}, \qquad \mathrm{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \\ 20 \end{bmatrix}.$$

Questions 3–15 are about the solution of $A\mathrm{x} = \mathrm{b}$. Follow the steps in the text to x_p and x_n. Start from the augmented matrix with last column b.

3. **Write the complete solution as x_p plus any multiple of s in the nullspace:**

$$\begin{aligned} x + 3y + 3z &= 1 \\ 2x + 6y + 9z &= 5 \\ -x - 3y + 3z &= 5. \end{aligned}$$

4. **Find the complete solution (also called the *general solution*) to**

$$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \\ 1 \end{bmatrix}.$$

5. **Under what condition on b_1, b_2, b_3 is this system solvable? Include b as a fourth column in elimination. Find all solutions when that condition holds:**

$$\begin{aligned} x + 2y - 2z &= b_1 \\ 2x + 5y - 4z &= b_2 \\ 4x + 9y - 8z &= b_3. \end{aligned}$$

6. **What conditions on b_1, b_2, b_3, b_4 make each system solvable? Find x in that case:**

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 2 & 5 \\ 3 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 5 & 7 \\ 3 & 9 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}.$$

7. **Show by elimination that (b_1, b_2, b_3) is in the column space if $b_3 - 2b_2 + 4b_1 = 0$.**

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 8 & 2 \\ 2 & 4 & 0 \end{bmatrix}.$$

   What combination of the rows of A gives the zero row?

8. **Which vectors (b_1, b_2, b_3) are in the column space of A? Which combinations of the rows of A give zero?**

   (a) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 6 & 3 \\ 0 & 2 & 5 \end{bmatrix}$ **(b)** $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}$

9. **(a)** **The Worked Example 3.3 A reached $[U \mid \mathrm{c}]$ from $[A \mid \mathrm{b}]$. Put the multipliers into L and verify that LU equals A and $L\mathrm{c}$ equals b.**

   (b) **Combine the pivot columns of A with the numbers -9 and 3 in the particular solution x_p. What is that linear combination and why?**

10. **Construct a 2 by 3 system $A\mathrm{x} = \mathrm{b}$ with particular solution $\mathrm{x}_p = (2, 4, 0)$ and homogeneous solution $\mathrm{x}_n =$ any multiple of $(1, 1, 1)$.**

11. **Why can't a 1 by 3 system have $\mathrm{x}_p = (2,4,0)$ and $\mathrm{x}_n =$ any multiple of $(1,1,1)$?**

12. **(a) If $A\mathrm{x} = \mathrm{b}$ has two solutions x_1 and x_2, find two solutions to $A\mathrm{x} = 0$.**

    (b) Then find another solution to $A\mathrm{x} = 0$ and another solution to $A\mathrm{x} = \mathrm{b}$.

13. **Explain why these are all false:**

    (a) The complete solution is any linear combination of x_p and x_n.

    (b) A system $A\mathrm{x} = \mathrm{b}$ has at most one particular solution.

    (c) The solution x_p with all free variables zero is the shortest solution (minimum length $\|\mathrm{x}\|$). Find a 2 by 2 counterexample.

    (d) If A is invertible there is no solution x_n in the nullspace.

14. **Suppose column 5 of U has no pivot. Then x_5 is a __________ variable. The zero vector (is) (is not) the only solution to $A\mathrm{x} = 0$. If $A\mathrm{x} = \mathrm{b}$ has a solution, then it has __________ solutions.**

15. **Suppose row 3 of U has no pivot. Then that row is __________. The equation $U\mathrm{x} = \mathrm{c}$ is only solvable provided __________. The equation $A\mathrm{x} = \mathrm{b}$ (is) (is not) (might not be) solvable.**

Questions 16–20 are about matrices of "full rank" $r = m$ or $r = n$.

16. **The largest possible rank of a 3 by 5 matrix is __________. Then there is a pivot in every __________ of U and R. The solution to $A\mathrm{x} = \mathrm{b}$ (always exists) (is unique). The column space of A is __________. An example is $A =$ ______________.**

17. **The largest possible rank of a 6 by 4 matrix is __________. Then there is a pivot in every __________ of U and R. The solution to $A\mathrm{x} = \mathrm{b}$ (always exists) (is unique). The nullspace of A is __________. An example is $A =$ ______________.**

18. **Find by elimination the rank of A and also the rank of A^{T}:**

$$A = \begin{bmatrix} 1 & 4 & 0 \\ 2 & 11 & 5 \\ -1 & 2 & 10 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & q \end{bmatrix} \quad \textbf{(rank depends on } q\textbf{)}.$$

19. **Find the rank of A and also of $A^{\mathrm{T}}A$ and also of AA^{T}:**

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 0 & 1 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

20. **Reduce A to its echelon form U. Then find a triangular L so that $A = LU$.**

$$A = \begin{bmatrix} 3 & 4 & 1 & 0 \\ 6 & 5 & 2 & 1 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 3 \\ 0 & 6 & 5 & 4 \end{bmatrix}.$$

21. **Find the complete solution in the form $\mathrm{x}_p + \mathrm{x}_n$ to these full rank systems:**

    (a) $x + y + z = 4$

    (b) $$\begin{aligned} x + y + z &= 4 \\ x - y + z &= 4 \end{aligned}$$

22. **If $A\mathrm{x} = \mathrm{b}$ has infinitely many solutions, why is it impossible for $A\mathrm{x} = B$ (new right side) to have only one solution? Could $A\mathrm{x} = B$ have no solution?**

23. **Choose the number q so that (if possible) the ranks are (a) 1, (b) 2, (c) 3:**

$$A = \begin{bmatrix} 6 & 4 & 2 \\ -3 & -2 & -1 \\ 9 & 6 & q \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 3 & 1 & 3 \\ q & 2 & q \end{bmatrix}.$$

24. Give examples of matrices A for which the number of solutions to $A\mathrm{x} = \mathrm{b}$ is

(a) 0 or 1, depending on b

(b) ∞, regardless of b

(c) 0 or ∞, depending on b

(d) 1, regardless of b

25. Write down all known relations between r and m and n if $A\mathrm{x} = \mathrm{b}$ has

(a) no solution for some b

(b) infinitely many solutions for every b

(c) exactly one solution for some b, no solution for other b

(d) exactly one solution for every b

Questions 26–33 are about Gauss-Jordan elimination (upwards as well as downwards) and the reduced echelon matrix R.

26. Continue elimination from U to R. Divide rows by pivots so the new pivots are all 1. Then produce zeros above those pivots to reach R:

$$U = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 5 \end{bmatrix}.$$

27. If A is a triangular matrix, when is $R = \mathrm{ref}(A)$ equal to I?

28. Apply Gauss-Jordan elimination to $U\mathrm{x} = \mathbf{0}$ and $U\mathrm{x} = \mathrm{c}$. Reach $R\mathrm{x} = \mathbf{0}$ and $R\mathrm{x} = \mathrm{d}$:

$$[U \mid \mathbf{0}] = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad [U \mid \mathbf{c}] = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solve $R\mathrm{x} = \mathbf{0}$ to find x_n (its free variable is $x_2 = 1$). Solve $R\mathrm{x} = \mathrm{d}$ to find x_p (its free variable is $x_2 = 0$).

29. Apply Gauss-Jordan elimination to reduce to $R\mathrm{x} = \mathbf{0}$ and $R\mathrm{x} = \mathrm{d}$:

$$[U \mid \mathbf{0}] = \begin{bmatrix} 3 & 0 & 6 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad [U \mid \mathbf{c}] = \begin{bmatrix} 3 & 0 & 6 & 9 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solve $U\mathrm{x} = \mathbf{0}$ or $R\mathrm{x} = \mathbf{0}$ to find x_n (free variable $= 1$). What are the solutions to $R\mathrm{x} = \mathrm{d}$?

30. Reduce to $U\mathrm{x} = \mathrm{c}$ (Gauss elimination) and then $R\mathrm{x} = \mathrm{d}$ (Gauss-Jordan):

$$A\mathrm{x} = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix} = \mathrm{b}.$$

Find a particular solution x_p and all homogeneous solutions x_n.

31. Find matrices A and B with the given property or explain why you can't:

(a) The only solution of $A\mathrm{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is $\mathrm{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

(b) The only solution of $B\mathrm{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is $\mathrm{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

32. Find the LU factorization of A and the complete solution to $A\mathrm{x} = \mathrm{b}$:

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 5 \end{bmatrix} \quad \textbf{and} \quad \mathrm{b} = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 5 \end{bmatrix}, \quad \textbf{and then} \quad \mathrm{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

33. The complete solution to $A\mathrm{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is $\mathrm{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Find A.

Challenge Problems

34. (Recommended!) Suppose you know that the 3 by 4 matrix A has the vector $\mathrm{s} = (2, 3, 1, 0)$ as the only special solution to $A\mathrm{x} = 0$.

(a) What is the *rank* of A and the complete solution to $A\mathrm{x} = 0$?

(b) What is the exact row reduced echelon form R of A?

(c) How do you know that $A\mathrm{x} = \mathrm{b}$ can be solved for all b?

35. Suppose K is the 9 by 9 second difference matrix (2's on the diagonal, -1's on the diagonal above and also below). Solve the equation $K\mathrm{x} = \mathrm{b} = (10, \ldots, 10)$. If you graph $x_1, \ldots, x_9$ above the points $1, \ldots, 9$ on the x axis, I think the nine points fall on a parabola.

36. Suppose $A\mathrm{x} = \mathrm{b}$ and $C\mathrm{x} = \mathrm{b}$ have the same (complete) solutions for every b. Is it true that A equals C?

37. Describe the column space of a reduced row echelon matrix R.

3.4 Independence, Basis and Dimension

Questions 1–10 are about linear independence and linear dependence.

1. Show that $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ are independent but $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4$ are dependent:

$$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \boldsymbol{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \boldsymbol{v}_4 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

Solve $c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + c_3\boldsymbol{v}_3 + c_4\boldsymbol{v}_4 = \mathbf{0}$ or $A\boldsymbol{x} = \mathbf{0}$. The $\boldsymbol{v}$'s go in the columns of A.

2. (Recommended) Find the largest possible number of independent vectors among

$$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} \quad \boldsymbol{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad \boldsymbol{v}_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \quad \boldsymbol{v}_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \quad \boldsymbol{v}_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

3. Prove that if $a = 0$ or $d = 0$ or $f = 0$ (3 cases), the columns of U are dependent:

$$U = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}.$$

4. If a, d, f in Question 3 are all nonzero, show that the only solution to $U\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0}$. Then the upper triangular U has independent columns.

5. Decide the dependence or independence of

(a) the vectors $(1, 3, 2)$ and $(2, 1, 3)$ and $(3, 2, 1)$

(b) the vectors $(1, -3, 2)$ and $(2, 1, -3)$ and $(-3, 2, 1)$.

6. **Choose three independent columns of U. Then make two other choices. Do the same for A.**

$$U = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 6 & 7 & 0 \\ 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 6 & 7 & 0 \\ 0 & 0 & 0 & 9 \\ 4 & 6 & 8 & 2 \end{bmatrix}.$$

7. **If w_1, w_2, w_3 are independent vectors, show that the differences $v_1 = w_2 - w_3$ and $v_2 = w_1 - w_3$ and $v_3 = w_1 - w_2$ are *dependent*. Find a combination of the v's that gives zero. Which matrix A in $[\, v_1 \quad v_2 \quad v_3 \,] = [\, w_1 \quad w_2 \quad w_3 \,] A$ is singular?**

8. **If w_1, w_2, w_3 are independent vectors, show that the sums $v_1 = w_2 + w_3$ and $v_2 = w_1 + w_3$ and $v_3 = w_1 + w_2$ are *independent*. (Write $c_1 v_1 + c_2 v_2 + c_3 v_3 = \mathbf{0}$ in terms of the w's. Find and solve equations for the c's, to show they are zero.)**

9. **Suppose v_1, v_2, v_3, v_4 are vectors in $\mathbf{R}^3$.**
   - **(a) These four vectors are dependent because ____________.**
   - **(b) The two vectors v_1 and v_2 will be dependent if ____________.**
   - **(c) The vectors v_1 and $(0, 0, 0)$ are dependent because ____________.**

10. **Find two independent vectors on the plane $x + 2y - 3z - t = 0$ in $\mathbf{R}^4$. Then find three independent vectors. Why not four? This plane is the nullspace of what matrix?**

Questions 11–14 are about the space *spanned* by a set of vectors. Take all linear combinations of the vectors.

11. **Describe the subspace of $\mathbf{R}^3$ (is it a line or plane or $\mathbf{R}^3$?) spanned by**
    - **(a) the two vectors $(1, 1, -1)$ and $(-1, -1, 1)$**
    - **(b) the three vectors $(0, 1, 1)$ and $(1, 1, 0)$ and $(0, 0, 0)$**
    - **(c) all vectors in $\mathbf{R}^3$ with whole number components**
    - **(d) all vectors with positive components.**

12. **The vector b is in the subspace spanned by the columns of A when ____________ has a solution. The vector c is in the row space of A when ____________ has a solution.**
    ***True or false*: If the zero vector is in the row space, the rows are dependent.**

13. **Find the dimensions of these 4 spaces. Which two of the spaces are the same? (a) column space of A, (b) column space of U, (c) row space of A, (d) row space of U:**

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \quad \textbf{and} \quad U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

14. **$v + w$ and $v - w$ are combinations of v and w. Write v and w as combinations of $v + w$ and $v - w$. The two pairs of vectors ____________ the same space. When are they a basis for the same space?**

Questions 15–25 are about the requirements for a basis.

15. **If $v_1, \ldots, v_n$ are linearly independent, the space they span has dimension ____________. These vectors are a ____________ for that space. If the vectors are the columns of an m by n matrix, then m is ____________ than n. If $m = n$, that matrix is ____________.**

16. **Find a basis for each of these subspaces of $\mathbf{R}^4$:**
    - **(a) All vectors whose components are equal.**
    - **(b) All vectors whose components add to zero.**
    - **(c) All vectors that are perpendicular to $(1, 1, 0, 0)$ and $(1, 0, 1, 1)$.**
    - **(d) The column space and the nullspace of I (4 by 4).**

17. **Find three different bases for the column space of $U = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$. Then find two different bases for the row space of U.**

18. **Suppose $v_1, v_2, \ldots, v_6$ are six vectors in $\mathbf{R}^4$.**

    (a) **Those vectors (do)(do not)(might not) span $\mathbf{R}^4$.**

    (b) **Those vectors (are)(are not)(might be) linearly independent.**

    (c) **Any four of those vectors (are)(are not)(might be) a basis for $\mathbf{R}^4$.**

19. **The columns of A are n vectors from $\mathbf{R}^m$. If they are linearly independent, what is the rank of A? If they span $\mathbf{R}^m$, what is the rank? If they are a basis for $\mathbf{R}^m$, what then? *Looking ahead*: The rank r counts the number of ____________ columns.**

20. **Find a basis for the plane $x - 2y + 3z = 0$ in $\mathbf{R}^3$. Then find a basis for the intersection of that plane with the xy plane. Then find a basis for all vectors perpendicular to the plane.**

21. **Suppose the columns of a 5 by 5 matrix A are a basis for $\mathbf{R}^5$.**

    (a) **The equation $Ax = 0$ has only the solution $x = 0$ because ____________.**

    (b) **If b is in $\mathbf{R}^5$ then $Ax = b$ is solvable because the basis vectors ____________ $\mathbf{R}^5$.**

    Conclusion: A is invertible. Its rank is 5. Its rows are also a basis for $\mathbf{R}^5$.

22. **Suppose S is a 5-dimensional subspace of $\mathbf{R}^6$. True or false (example if false):**

    (a) **Every basis for S can be extended to a basis for $\mathbf{R}^6$ by adding one more vector.**

    (b) **Every basis for $\mathbf{R}^6$ can be reduced to a basis for S by removing one vector.**

23. **U comes from A by subtracting row 1 from row 3:**

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

    Find bases for the two column spaces. Find bases for the two row spaces. Find bases for the two nullspaces. Which spaces stay fixed in elimination?

24. **True or false (give a good reason):**

    (a) **If the columns of a matrix are dependent, so are the rows.**

    (b) **The column space of a 2 by 2 matrix is the same as its row space.**

    (c) **The column space of a 2 by 2 matrix has the same dimension as its row space.**

    (d) **The columns of a matrix are a basis for the column space.**

25. **For which numbers c and d do these matrices have rank 2?**

$$A = \begin{bmatrix} 1 & 2 & 5 & 0 & 5 \\ 0 & 0 & c & 2 & 2 \\ 0 & 0 & 0 & d & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} c & d \\ d & c \end{bmatrix}.$$

Questions 26–30 are about spaces where the "vectors" are matrices.

26. **Find a basis (and the dimension) for each of these subspaces of 3 by 3 matrices:**

    (a) **All diagonal matrices.**

    (b) **All symmetric matrices ($A^{\mathrm{T}} = A$).**

    (c) **All skew-symmetric matrices ($A^{\mathrm{T}} = -A$).**

27. **Construct six linearly independent 3 by 3 echelon matrices $U_1, \ldots, U_6$.**

28. **Find a basis for the space of all 2 by 3 matrices whose columns add to zero. Find a basis for the subspace whose rows also add to zero.**

29. What subspace of 3 by 3 matrices is spanned (take all combinations) by

(a) the invertible matrices?

(b) the rank one matrices?

(c) the identity matrix?

30. Find a basis for the space of 2 by 3 matrices whose nullspace contains $(2, 1, 1)$**.**

Questions 31–35 are about spaces where the "vectors" are functions.

31. (a) Find all functions that satisfy $\frac{dy}{dx} = 0$.

(b) Choose a particular function that satisfies $\frac{dy}{dx} = 3$.

(c) Find all functions that satisfy $\frac{dy}{dx} = 3$.

32. The cosine space $\mathbf{F}_3$ **contains all combinations** $y(x) = A\cos x + B\cos 2x + C\cos 3x$**. Find a basis for the subspace with** $y(0) = 0$.

33. Find a basis for the space of functions that satisfy

(a) $\frac{dy}{dx} - 2y = 0$

(b) $\frac{dy}{dx} - \frac{y}{x} = 0$.

34. Suppose $y_1(x), y_2(x), y_3(x)$ **are three different functions of** x**. The vector space they span could have dimension 1, 2, or 3. Give an example of** y_1, y_2, y_3 **to show each possibility.**

35. Find a basis for the space of polynomials $p(x)$ **of degree** ≤ 3**. Find a basis for the subspace with** $p(1) = 0$.

36. Find a basis for the space S of vectors (a, b, c, d) **with** $a + c + d = 0$ **and also for the space T with** $a + b = 0$ **and** $c = 2d$**. What is the dimension of the intersection S ∩ T?**

37. If $AS = SA$ **for the *shift matrix*** S**, show that** A **must have this special form:**

$$\textbf{If} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \textbf{ then } A = \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}.$$

"The subspace of matrices that commute with the shift S **has dimension ______."**

38. Which of the following are bases for $\mathbf{R}^3$**?**

(a) $(1, 2, 0)$ **and** $(0, 1, -1)$

(b) $(1, 1, -1), (2, 3, 4), (4, 1, -1), (0, 1, -1)$

(c) $(1, 2, 2), (-1, 2, 1), (0, 8, 0)$

(d) $(1, 2, 2), (-1, 2, 1), (0, 8, 6)$

39. Suppose A **is 5 by 4 with rank 4. Show that** $Ax = \boldsymbol{b}$ **has no solution when the 5 by 5 matrix** $[\,A \quad \boldsymbol{b}\,]$ **is invertible. Show that** $Ax = \boldsymbol{b}$ **is solvable when** $[\,A \quad \boldsymbol{b}\,]$ **is singular.**

40. (a) Find a basis for all solutions to $d^4y/dx^4 = y(x)$.

(b) Find a particular solution to $d^4y/dx^4 = y(x) + 1$**. Find the complete solution.**

Challenge Problems

41. Write the 3 by 3 identity matrix as a combination of the other five permutation matrices! Then show that those five matrices are linearly independent. (Assume a combination gives $c_1P_1 + \cdots + c_5P_5 =$ **zero matrix, and check entries to prove that** c_1 **to** c_5 **must all be zero.) The five permutations are a basis for the subspace of 3 by 3 matrices with row and column sums all equal.**

42. Choose $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$ **in** $\mathbf{R}^4$**. It has 24 rearrangements like** (x_2, x_1, x_3, x_4) **and** (x_4, x_3, x_1, x_2)**. Those 24 vectors, including** $\boldsymbol{x}$ **itself, span a subspace S. Find specific vectors** $\boldsymbol{x}$ **so that the dimension of S is: (a) zero, (b) one, (c) three, (d) four.**

43. Intersections and sums have $\dim(\mathbf{V}) + \dim(\mathbf{W}) = \dim(\mathbf{V} \cap \mathbf{W}) + \dim(\mathbf{V} + \mathbf{W})$. Start with a basis $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ for the intersection $\mathbf{V} \cap \mathbf{W}$. Extend with $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_s$ to a basis for V, and separately with $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_t$ to a basis for W. Prove that the $\boldsymbol{u}$'s, $\boldsymbol{v}$'s and $\boldsymbol{w}$'s together are *independent*. The dimensions have $(r+s)+(r+t) = (r)+(r+s+t)$ as desired.

44. Mike Artin suggested a neat higher-level proof of that dimension formula in Problem 43. From all inputs $\boldsymbol{v}$ in V and $\boldsymbol{w}$ in W, the "sum transformation" produces $\boldsymbol{v}+\boldsymbol{w}$. Those outputs fill the space $\mathbf{V}+\mathbf{W}$. The nullspace contains all pairs $\boldsymbol{v} = \boldsymbol{u}$, $\boldsymbol{w} = -\boldsymbol{u}$ for vectors $\boldsymbol{u}$ in $\mathbf{V} \cap \mathbf{W}$. (Then $\boldsymbol{v}+\boldsymbol{w} = \boldsymbol{u}-\boldsymbol{u} = \mathbf{0}$.) So $\dim(\mathbf{V}+\mathbf{W}) + \dim(\mathbf{V} \cap \mathbf{W})$ equals $\dim(\mathbf{V}) + \dim(\mathbf{W})$ (*input dimension from* V *and* W) by the Counting Theorem.

$$\textit{dimension of outputs} + \textit{dimension of nullspace} = \textit{dimension of inputs.}$$

Problem For an m by n matrix of rank r, what are those 3 dimensions? Outputs = column space. This question will be answered in Section 3.5, can you do it now?

45. Inside $\mathbf{R}^n$, suppose dimension (V) + dimension (W) ¿ n. Show that some nonzero vector is in both V and W.

46. Suppose A is 10 by 10 and $A^2 = 0$ (zero matrix). So A multiplies each column of A to give the zero vector. This means that the column space of A is contained in the ____________. If A has rank r, those subspaces have dimension $r \leq 10 - r$. So the rank is $r \leq 5$.

3.5 Dimensions of the Four Subspaces

1. (a) If a 7 by 9 matrix has rank 5, what are the dimensions of the four subspaces? What is the sum of all four dimensions?
   (b) If a 3 by 4 matrix has rank 3, what are its column space and left nullspace?

2. Find bases and dimensions for the four subspaces associated with A and B:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 8 \end{bmatrix}.$$

3. Find a basis for each of the four subspaces associated with A:

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

4. Construct a matrix with the required property or explain why this is impossible:
   (a) Column space contains $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, row space contains $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.
   (b) Column space has basis $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, nullspace has basis $\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$.
   (c) Dimension of nullspace = 1 + dimension of left nullspace.
   (d) Nullspace contains $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$, column space contains $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.
   (e) Row space = column space, nullspace $\neq$ left nullspace.

5. If V is the subspace spanned by $(1,1,1)$ and $(2,1,0)$, find a matrix A that has V as its row space. Find a matrix B that has V as its nullspace. Multiply AB.

6. Without using elimination, find dimensions and bases for the four subspaces for

$$A = \begin{bmatrix} 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}.$$

7. Suppose the 3 by 3 matrix A is invertible. Write down bases for the four subspaces for A, and also for the 3 by 6 matrix $B = [\,A \quad A\,]$. (The basis for **Z** is empty.)

8. What are the dimensions of the four subspaces for A, B, and C, if I is the 3 by 3 identity matrix and 0 is the 3 by 2 zero matrix?

$$A = [\,I \quad 0\,] \quad \text{and} \quad B = \begin{bmatrix} I & I \\ \mathrm{T} & \mathrm{T} \end{bmatrix} \quad \text{and} \quad C = [\,0\,].$$

9. Which subspaces are the same for these matrices of different sizes?

   (a) $[\,A\,]$ and $\begin{bmatrix} A \\ A \end{bmatrix}$ (b) $\begin{bmatrix} A \\ A \end{bmatrix}$ and $\begin{bmatrix} A & A \\ A & A \end{bmatrix}$.

   Prove that all three of those matrices have the *same* rank r.

10. If the entries of a 3 by 3 matrix are chosen randomly between 0 and 1, what are the most likely dimensions of the four subspaces? What if the random matrix is 3 by 5?

11. (Important) A is an m by n matrix of rank r. Suppose there are right sides $\boldsymbol{b}$ for which $Ax = \boldsymbol{b}$ has *no solution.*

    (a) What are all inequalities ($<$ or $\leq$) that must be true between m, n, and r?

    (b) How do you know that $A^{\mathrm{T}}\boldsymbol{y} = \boldsymbol{0}$ has solutions other than $\boldsymbol{y} = \boldsymbol{0}$?

12. Construct a matrix with $(1, 0, 1)$ and $(1, 2, 0)$ as a basis for its row space and its column space. Why can't this be a basis for the row space and nullspace?

13. True or false (with a reason or a counterexample):

    (a) If $m = n$ then the row space of A equals the column space.

    (b) The matrices A and $-A$ share the same four subspaces.

    (c) If A and B share the same four subspaces then A is a multiple of B.

14. Without computing A, find bases for its four fundamental subspaces:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 9 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

15. If you exchange the first two rows of A, which of the four subspaces stay the same? If $\boldsymbol{v} = (1, 2, 3, 4)$ is in the left nullspace of A, write down a vector in the left nullspace of the new matrix after the row exchange.

16. Explain why $\boldsymbol{v} = (1, 0, -1)$ *cannot* be a row of A and also in the nullspace.

17. Describe the four subspaces of $\mathbf{R}^3$ associated with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad I + A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

18. (Left nullspace) Add the extra column $\boldsymbol{b}$ and reduce A to echelon form:

$$[\,A \quad \boldsymbol{b}\,] = \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 4 & 5 & 6 & b_2 \\ 7 & 8 & 9 & b_3 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 0 & -3 & -6 & b_2 - 4b_1 \\ 0 & 0 & 0 & b_3 - 2b_2 + b_1 \end{bmatrix}.$$

    A combination of the rows of A has produced the zero row. What combination is it? (Look at $b_3 - 2b_2 + b_1$ on the right side.) Which vectors are in the nullspace of A^{T} and which vectors are in the nullspace of A?

19. Following the method of Problem 18, reduce A to echelon form and look at zero rows. The $\boldsymbol{b}$ column tells which combinations you have taken of the rows:

(a) $\begin{bmatrix} 1 & 2 & b_1 \\ 3 & 4 & b_2 \\ 4 & 6 & b_3 \end{bmatrix}$ **(b)** $\begin{bmatrix} 1 & 2 & b_1 \\ 2 & 3 & b_2 \\ 2 & 4 & b_3 \\ 2 & 5 & b_4 \end{bmatrix}$

From the b column after elimination, read off $m - r$ basis vectors in the left nullspace. Those y's are combinations of rows that give zero rows in the echelon form.

20. **(a) Check that the solutions to $Ax = 0$ are perpendicular to the rows of A:**

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} = ER.$$

(b) How many independent solutions to $A^{\mathrm{T}}\boldsymbol{y} = 0$? Why does $\boldsymbol{y}^{\mathrm{T}}$ = row 3 of E^{-1}?

21. **Suppose A is the sum of two matrices of rank one: $A = \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}} + \boldsymbol{w}\boldsymbol{z}^{\mathrm{T}}$.**

(a) Which vectors span the column space of A?

(b) Which vectors span the row space of A?

(c) The rank is less than 2 if _________ or if _________.

(d) Compute A and its rank if $\boldsymbol{u} = \boldsymbol{z} = (1, 0, 0)$ and $\boldsymbol{v} = \boldsymbol{w} = (0, 0, 1)$.

22. **Construct $A = \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}} + \boldsymbol{w}\boldsymbol{z}^{\mathrm{T}}$ whose column space has basis $(1, 2, 4), (2, 2, 1)$ and whose row space has basis $(1, 0), (1, 1)$. Write A as (3 by 2) times (2 by 2).**

23. **Without multiplying matrices, find bases for the row and column spaces of A:**

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 3 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}.$$

How do you know from these shapes that A cannot be invertible?

24. **(Important) $A^{\mathrm{T}}\boldsymbol{y} = \boldsymbol{d}$ is solvable when $\boldsymbol{d}$ is in which of the four subspaces? The solution $\boldsymbol{y}$ is unique when the _________ contains only the zero vector.**

25. **True or false (with a reason or a counterexample):**

(a) A and A^{T} have the same number of pivots.

(b) A and A^{T} have the same left nullspace.

(c) If the row space equals the column space then $A^{\mathrm{T}} = A$.

(d) If $A^{\mathrm{T}} = -A$ then the row space of A equals the column space.

26. **If a, b, c are given with $a \neq 0$, how would you choose d so that $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has rank 1? Find a basis for the row space and nullspace. Show they are perpendicular!**

27. **Find the ranks of the 8 by 8 checkerboard matrix B and the chess matrix C:**

$$B = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \quad \textbf{and} \quad C = \begin{bmatrix} r & n & b & q & k & b & n & r \\ p & p & p & p & p & p & p & p \\ & & & \textbf{four zero rows} & & & & \\ p & p & p & p & p & p & p & p \\ r & n & b & q & k & b & n & r \end{bmatrix}.$$

The numbers r, n, b, q, k, p are all different. Find bases for the row space and left nullspace of B and C. Challenge problem: Find a basis for the nullspace of C.

28. **Can tic-tac-toe be completed (5 ones and 4 zeros in A) so that rank (A) = 2 but neither side passed up a winning move?**

Challenge Problems

29. **If $A = uv^{\mathrm{T}}$ is a 2 by 2 matrix of rank 1, redraw Figure 3.5 to show clearly the Four Fundamental Subspaces. If B produces those same four subspaces, what is the exact relation of B to A?**

30. **M is the space of 3 by 3 matrices. Multiply every matrix X in M by**

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}. \quad \textbf{Notice: } A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

(a) **Which matrices X lead to $AX =$ zero matrix?**

(b) **Which matrices have the form AX for some matrix X?**

(a) finds the "nullspace" of that operation AX and (b) finds the "column space". What are the dimensions of those two subspaces of M? Why do the dimensions add to $(n - r) + r = 9$?

31. **Suppose the m by n matrices A and B have *the same four subspaces*. If they are both in row reduced echelon form, prove that F must equal G:**

$$A = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} I & G \\ 0 & 0 \end{bmatrix}.$$

4 Chapter 4: Orthogonality

4.1 Orthogonality of the Four Subspaces

Questions 1–12 grow out of Figures 4.2 and 4.3 with four subspaces.

1. **Construct any 2 by 3 matrix of rank one. Copy Figure 4.2 and put one vector in each subspace (and put two in the nullspace). Which vectors are orthogonal?**

2. **Redraw Figure 4.3 for a 3 by 2 matrix of rank $r = 2$. Which subspace is $\mathbf{Z}$ (zero vector only)? The nullspace part of any vector x in $\mathbf{R}^2$ is $x_n =$ ____________.**

3. **Construct a matrix with the required property or say why that is impossible:**

   (a) **Column space contains** $\begin{bmatrix}1\\2\\-3\end{bmatrix}$ **and** $\begin{bmatrix}2\\-3\\5\end{bmatrix}$**, nullspace contains** $\begin{bmatrix}1\\1\\1\end{bmatrix}$

   (b) **Row space contains** $\begin{bmatrix}1\\2\\-3\end{bmatrix}$ **and** $\begin{bmatrix}2\\-3\\5\end{bmatrix}$**, nullspace contains** $\begin{bmatrix}1\\1\\1\end{bmatrix}$

   (c) $Ax = \begin{bmatrix}1\\1\\1\end{bmatrix}$ **has a solution and** $A^{\mathrm{T}}\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$

   (d) **Every row is orthogonal to every column (A is not the zero matrix)**

   (e) **Columns add up to a column of zeros, rows add to a row of 1's.**

4. **If $AB = 0$ then the columns of B are in the ____________ of A. The rows of A are in the ____________ of B. With $AB = 0$, why can't A and B be 3 by 3 matrices of rank 2?**

5. (a) **If $Ax = \boldsymbol{b}$ has a solution and $A^{\mathrm{T}}\boldsymbol{y} = \mathbf{0}$, is $(\boldsymbol{y}^{\mathrm{T}}\boldsymbol{x} = 0)$ or $(\boldsymbol{y}^{\mathrm{T}}\boldsymbol{b} = 0)$?**

   (b) **If $A^{\mathrm{T}}\boldsymbol{y} = (1, 1, 1)$ has a solution and $Ax = \mathbf{0}$, then ____________.**

6. **This system of equations $Ax = \boldsymbol{b}$ has *no solution* (they lead to $0 = 1$):**

$$\begin{aligned} x + 2y + 2z &= 5 \\ 2x + 2y + 3z &= 5 \\ 3x + 4y + 5z &= 9 \end{aligned}$$

   Find numbers y_1, y_2, y_3 to multiply the equations so they add to $0 = 1$. You have found a vector $\boldsymbol{y}$ in which subspace? Its dot product $\boldsymbol{y}^{\mathrm{T}}\boldsymbol{b}$ is 1, so no solution x.

7. **Every system with no solution is like the one in Problem 6. There are numbers $y_1, \ldots, y_m$ that multiply the m equations so they add up to $0 = 1$. This is called Fredholm's Alternative:**

   Exactly one of these problems has a solution
   $$Ax = \boldsymbol{b} \quad \text{OR} \quad A^{\mathrm{T}}\boldsymbol{y} = \mathbf{0} \quad \text{with} \quad \boldsymbol{y}^{\mathrm{T}}\boldsymbol{b} = 1.$$

   If $\boldsymbol{b}$ is not in the column space of A, it is not orthogonal to the nullspace of A^{T}. Multiply the equations $x_1 - x_2 = 1$ and $x_2 - x_3 = 1$ and $x_1 - x_3 = 1$ by numbers y_1, y_2, y_3 chosen so that the equations add up to $0 = 1$.

8. **In Figure 4.3, how do we know that Ax_r is equal to Ax? How do we know that this vector is in the column space? If $A = \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$ and $x = \begin{bmatrix}1\\0\end{bmatrix}$ what is x_r?**

9. **If $A^{\mathrm{T}}Ax = \mathbf{0}$ then $Ax = \mathbf{0}$. Reason: Ax is in the nullspace of A^{T} and also in the _________ of A and those spaces are _________. *Conclusion: $A^{\mathrm{T}}A$ has the same nullspace as A. This key fact is repeated in the next section.***

10. **Suppose A is a symmetric matrix ($A^{\mathrm{T}} = A$).**

(a) **Why is its column space perpendicular to its nullspace?**

(b) **If $Ax = 0$ and $Az = 5z$, which subspaces contain these "eigenvectors" x and z? Symmetric matrices have perpendicular eigenvectors $x^{\mathrm{T}}z = 0$.**

11. **(Recommended) Draw Figure 4.2 to show each subspace correctly for**

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}.$$

12. **Find the pieces x_r and x_n and draw Figure 4.3 properly if**

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Questions 13–23 are about orthogonal subspaces.

13. **Put bases for the subspaces V and W into the columns of matrices V and W. Explain why the test for orthogonal subspaces can be written $V^{\mathrm{T}}W =$ zero matrix. This matches $v^{\mathrm{T}}w = 0$ for orthogonal vectors.**

14. **The floor V and the wall W are not orthogonal subspaces, because they share a nonzero vector (along the line where they meet). No planes V and W in $\mathbf{R}^3$ can be orthogonal! Find a vector in the column spaces of both matrices:**

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 5 & 4 \\ 6 & 3 \\ 5 & 1 \end{bmatrix}$$

This will be a vector Ax and also $B\widehat{x}$. Think 3 by 4 with the matrix $[\,A \quad B\,]$.

15. **Extend Problem 14 to a p-dimensional subspace V and a q-dimensional subspace W of $\mathbf{R}^n$. What inequality on $p+q$ guarantees that V intersects W in a nonzero vector? These subspaces cannot be orthogonal.**

16. **Prove that every y in $\mathbf{N}(A^{\mathrm{T}})$ is perpendicular to every Ax in the column space, using the matrix shorthand of equation (2). Start from $A^{\mathrm{T}}y = 0$.**

17. **If S is the subspace of $\mathbf{R}^3$ containing only the zero vector, what is $\mathbf{S}^\perp$? If S is spanned by $(1,1,1)$, what is $\mathbf{S}^\perp$? If S is spanned by $(1,1,1)$ and $(1,1,-1)$, what is a basis for $\mathbf{S}^\perp$?**

18. **Suppose S only contains two vectors $(1,5,1)$ and $(2,2,2)$ (not a subspace). Then $\mathbf{S}^\perp$ is the nullspace of the matrix $A =$ ____________. $\mathbf{S}^\perp$ is a subspace even if S is not.**

19. **Suppose L is a one-dimensional subspace (a line) in $\mathbf{R}^3$. Its orthogonal complement $\mathbf{L}^\perp$ is the ____________ perpendicular to L. Then $(\mathbf{L}^\perp)^\perp$ is a ____________ perpendicular to $\mathbf{L}^\perp$. In fact $(\mathbf{L}^\perp)^\perp$ is the same as ____________.**

20. **Suppose V is the whole space $\mathbf{R}^4$. Then $\mathbf{V}^\perp$ contains only the vector ____________. Then $(\mathbf{V}^\perp)^\perp$ is ____________. So $(\mathbf{V}^\perp)^\perp$ is the same as ____________.**

21. **Suppose S is spanned by the vectors $(1,2,2,3)$ and $(1,3,3,2)$. Find two vectors that span $\mathbf{S}^\perp$. This is the same as solving $Ax = 0$ for which A?**

22. **If P is the plane of vectors in $\mathbf{R}^4$ satisfying $x_1 + x_2 + x_3 + x_4 = 0$, write a basis for $\mathbf{P}^\perp$. Construct a matrix that has P as its nullspace.**

23. **If a subspace S is contained in a subspace V, prove that $\mathbf{S}^\perp$ contains $\mathbf{V}^\perp$.**

Questions 24–30 are about perpendicular columns and rows.

24. **Suppose an n by n matrix is invertible: $AA^{-1} = I$. Then the first column of A^{-1} is orthogonal to the space spanned by which rows of A?**

25. **Find $A^{\mathrm{T}}A$ if the columns of A are unit vectors, all mutually perpendicular.**

26. **Construct a 3 by 3 matrix A with no zero entries whose columns are mutually perpendicular. Compute $A^{\mathrm{T}}A$. Why is it a diagonal matrix?**

27. **The lines $3x + y = b_1$ and $6x + 2y = b_2$ are ____________. They are the same line if ____________. In that case (b_1, b_2) is perpendicular to the vector ____________. The nullspace of the matrix is the line $3x + y =$ ____________. One particular vector in that nullspace is ____________.**

28. **Why is each of these statements false?**

    (a) $(1, 1, 1)$ **is perpendicular to** $(1, 1, -2)$ **so the planes** $x + y + z = 0$ **and** $x + y - 2z = 0$ **are orthogonal subspaces.**

    (b) The subspace spanned by $(1, 1, 0, 0, 0)$ **and** $(0, 0, 0, 1, 1)$ **is the orthogonal complement of the subspace spanned by** $(1, -1, 0, 0, 0)$ **and** $(2, -2, 3, 4, -4)$.

    (c) Two subspaces that meet only in the zero vector are orthogonal.

29. **Find a matrix with $\boldsymbol{v} = (1, 2, 3)$ in the row space and column space. Find another matrix with $\boldsymbol{v}$ in the nullspace and column space. Which pairs of subspaces can $\boldsymbol{v}$ *not* be in?**

Challenge Problems

30. **Suppose A is 3 by 4 and B is 4 by 5 and $AB = 0$. So $\mathbf{N}(A)$ contains $\mathbf{C}(B)$. Prove from the dimensions of $\mathbf{N}(A)$ and $\mathbf{C}(B)$ that $\operatorname{rank}(A) + \operatorname{rank}(B) \leq 4$.**

31. **The command $N = \operatorname{null}(A)$ will produce a basis for the nullspace of A. Then the command $B = \operatorname{null}(N')$ will produce a basis for the ____________ of A.**

32. **Suppose I give you four nonzero vectors $\boldsymbol{r}, \boldsymbol{n}, \boldsymbol{c}, \boldsymbol{l}$ in $\mathbf{R}^2$.**

    (a) What are the conditions for those to be bases for the four fundamental subspaces $\mathbf{C}(A^{\mathrm{T}}), \mathbf{N}(A), \mathbf{C}(A), \mathbf{N}(A^{\mathrm{T}})$ of a 2 by 2 matrix?

    (b) What is one possible matrix A?

33. **Suppose I give you eight vectors $\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{n}_1, \boldsymbol{n}_2, \boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{l}_1, \boldsymbol{l}_2$ in $\mathbf{R}^4$.**

    (a) What are the conditions for those pairs to be bases for the four fundamental subspaces of a 4 by 4 matrix?

    (b) What is one possible matrix A?

4.2 Projections

Questions 1–9 ask for projections $\boldsymbol{p}$ onto lines. Also errors $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ and matrices P.

1. **Project the vector $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$. Check that $\boldsymbol{e}$ is perpendicular to $\boldsymbol{a}$:**

$$\textbf{(a)}\quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad \textbf{(b)}\quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{a} = \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}.$$

2. ***Draw* the projection of $\boldsymbol{b}$ onto $\boldsymbol{a}$ and also compute it from $\boldsymbol{p} = \widehat{x}\boldsymbol{a}$:**

$$\textbf{(a)}\quad \boldsymbol{b} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \textbf{(b)}\quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{a} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

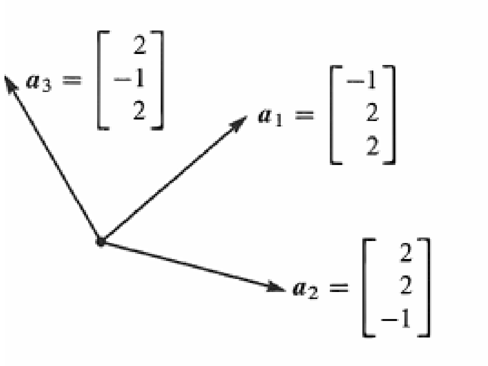
3. **In Problem 1, find the projection matrix $P = \boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ onto the line through each vector $\boldsymbol{a}$. Verify in both cases that $P^2 = P$. Multiply $P\boldsymbol{b}$ in each case to compute the projection $\boldsymbol{p}$.**

4. **Construct the projection matrices P_1 and P_2 onto the lines through the $\boldsymbol{a}$'s in Problem 2. Is it true that $(P_1 + P_2)^2 = P_1 + P_2$? This *would* be true if $P_1P_2 = 0$.**

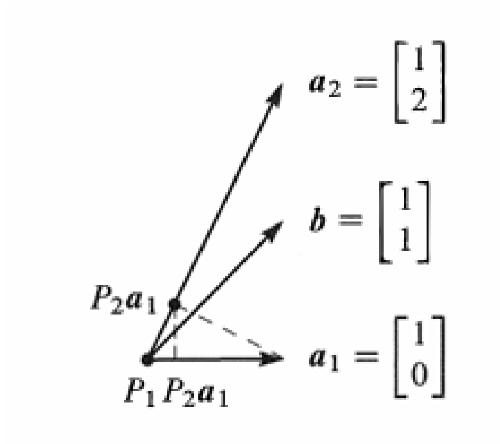
5. **Compute the projection matrices $\boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ onto the lines through $\boldsymbol{a}_1 = (-1, 2, 2)$ and $\boldsymbol{a}_2 = (2, 2, -1)$. Multiply those projection matrices and explain why their product P_1P_2 is what it is.**

6. **Project $b = (1,0,0)$ onto the lines through a_1 and a_2 in Problem 5 and also onto $a_3 = (2,-1,2)$. Add up the three projections $p_1 + p_2 + p_3$.**

7. **Continuing Problems 5–6, find the projection matrix P_3 onto $a_3 = (2,-1,2)$. Verify that $P_1 + P_2 + P_3 = I$. This is because the basis a_1, a_2, a_3 is orthogonal!**



Questions 5–6–7: orthogonal



Questions 8–9–10: not orthogonal

8. **Project the vector $b = (1,1)$ onto the lines through $a_1 = (1,0)$ and $a_2 = (1,2)$. Draw the projections p_1 and p_2 and add $p_1 + p_2$. The projections do not add to b because the a's are not orthogonal.**

9. **In Problem 8, the projection of b onto the *plane* of a_1 and a_2 will equal b. Find $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ for $A = [\, a_1 \quad a_2 \,] = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ = invertible matrix.**

10. **Project $a_1 = (1,0)$ onto $a_2 = (1,2)$. Then project the result back onto a_1. Draw these projections and multiply the projection matrices $P_1 P_2$: Is this a projection?**

Questions 11–20 ask for projections, and projection matrices, onto subspaces.

11. **Project b onto the column space of A by solving $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$ and $p = A\widehat{x}$:**

$$\textbf{(a)} \quad A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \textbf{and} \quad b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \qquad \textbf{(b)} \quad A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \textbf{and} \quad b = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}.$$

Find $e = b - p$. It should be perpendicular to the columns of A.

12. **Compute the projection matrices P_1 and P_2 onto the column spaces in Problem 11. Verify that $P_1 b$ gives the first projection p_1. Also verify $P_2^2 = P_2$.**

13. **(Quick and Recommended) Suppose A is the 4 by 4 identity matrix with its last column removed. A is 4 by 3. Project $b = (1,2,3,4)$ onto the column space of A. What shape is the projection matrix P and what is P?**

14. **Suppose b equals 2 times the first column of A. What is the projection of b onto the column space of A? Is $P = I$ for sure in this case? Compute p and P when $b = (0,2,4)$ and the columns of A are $(0,1,2)$ and $(1,2,0)$.**

15. **If A is doubled, then $P = 2A(4A^{\mathrm{T}}A)^{-1}2A^{\mathrm{T}}$. This is the same as $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$. The column space of $2A$ is the same as ____________. Is $\widehat{x}$ the same for A and $2A$?**

16. **What linear combination of $(1,2,-1)$ and $(1,0,1)$ is closest to $b = (2,1,1)$?**

17. ***(Important)*** **If $P^2 = P$ show that $(I - P)^2 = I - P$. When P projects onto the column space of A, $I - P$ projects onto the ______.**

18. **(a) If P is the 2 by 2 projection matrix onto the line through $(1, 1)$, then $I - P$ is the projection matrix onto ______.**

    (b) If P is the 3 by 3 projection matrix onto the line through $(1, 1, 1)$, then $I - P$ is the projection matrix onto ______.

19. **To find the projection matrix onto the plane $x - y - 2z = 0$, choose two vectors in that plane and make them the columns of A. The plane will be the column space of A! Then compute $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$.**

20. **To find the projection matrix P onto the same plane $x - y - 2z = 0$, write down a vector e that is perpendicular to that plane. Compute the projection $Q = ee^{\mathrm{T}}/e^{\mathrm{T}}e$ and then $P = I - Q$.**

Questions 21–26 show that projection matrices satisfy $\boldsymbol{P^2 = P}$ and $\boldsymbol{P^{\mathrm{T}} = P}$.

21. **Multiply the matrix $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ by itself. Cancel to prove that $P^2 = P$. Explain why $P(P\boldsymbol{b})$ always equals $P\boldsymbol{b}$: The vector $P\boldsymbol{b}$ is in the column space of A so its projection onto that column space is ______.**

22. **Prove that $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ is symmetric by computing P^{T}. Remember that the inverse of a symmetric matrix is symmetric.**

23. **If A is square and invertible, the warning against splitting $(A^{\mathrm{T}}A)^{-1}$ does not apply. It is true that $AA^{-1}(A^{\mathrm{T}})^{-1}A^{\mathrm{T}} = I$. When A is invertible, why is $P = I$? What is the error e?**

24. **The nullspace of A^{T} is ______ to the column space $\mathbf{C}(A)$. So if $A^{\mathrm{T}}\boldsymbol{b} = 0$, the projection of $\boldsymbol{b}$ onto $\mathbf{C}(A)$ should be $\boldsymbol{p} =$ ______. Check that $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ gives this answer.**

25. **The projection matrix P onto an n-dimensional subspace of $\mathbf{R}^m$ has rank $r = n$. Reason: The projections $P\boldsymbol{b}$ fill the subspace S. So S is the ______ of P.**

26. **If an m by m matrix has $A^2 = A$ and its rank is m, prove that $A = I$.**

27. **The important fact that ends the section is this: If $\boldsymbol{A^{\mathrm{T}}Ax = 0}$ then $\boldsymbol{Ax = 0}$.**
    New Proof: **The vector Ax is in the nullspace of ______. Ax is always in the column space of ______. To be in both of those perpendicular spaces, Ax must be zero.**

28. **Use $P^{\mathrm{T}} = P$ and $P^2 = P$ to prove that the length squared of column 2 always equals the diagonal entry P_{22}. This number is $\frac{2}{6} = \frac{4}{36} + \frac{4}{36} + \frac{4}{36}$ for**

$$P = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}.$$

29. **If B has rank m (full row rank, independent rows) show that BB^{T} is invertible.**

Challenge Problems

30. **(a) Find the projection matrix P_C onto the column space of A (after looking closely at the matrix!)**

$$A = \begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$$

    (b) Find the 3 by 3 projection matrix P_R onto the row space of A. Multiply $B = P_C A P_R$. Your answer B should be a little surprising—can you explain it?

31. **In $\mathbf{R}^m$, suppose I give you $\boldsymbol{b}$ and also a combination $\boldsymbol{p}$ of $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$. How would you test to see if $\boldsymbol{p}$ is the projection of $\boldsymbol{b}$ onto the subspace spanned by the $\boldsymbol{a}$'s?**

32. Suppose P_1 is the projection matrix onto the 1-dimensional subspace spanned by the first column of A. Suppose P_2 is the projection matrix onto the 2-dimensional column space of A. After thinking a little, compute the product P_2P_1.

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

33. Suppose you know the average $\widehat{x}_{\text{old}}$ of $b_1, b_2, \ldots, b_{999}$. When b_{1000} arrives, check that the new average is a combination of $\widehat{x}_{\text{old}}$ and the mismatch $b_{1000} - \widehat{x}_{\text{old}}$:

$$\widehat{\boldsymbol{x}}_{\text{new}} = \frac{b_1 + \cdots + b_{1000}}{1000} = \frac{b_1 + \cdots + b_{999}}{999} + \frac{1}{1000}\left(b_{1000} - \frac{b_1 + \cdots + b_{999}}{999}\right).$$

This is a "Kalman filter" $\widehat{x}_{\text{new}} = \widehat{x}_{\text{old}} + \frac{1}{1000}(b_{1000} - \widehat{x}_{\text{old}})$ with gain matrix $\frac{1}{1000}$.
The last page of the book extends the Kalman filter to matrix updates.

34. (2017) Suppose P_1 and P_2 are projection matrices ($P_i^2 = P_i = P_i^{\mathrm{T}}$). Prove this fact:

$$P_1P_2 \textbf{ is a projection matrix if and only if } P_1P_2 = P_2P_1.$$

4.3 Least Squares Approximations

Problems 1–11 use four data points $\boldsymbol{b} = (0, 8, 8, 20)$ to bring out the key ideas.

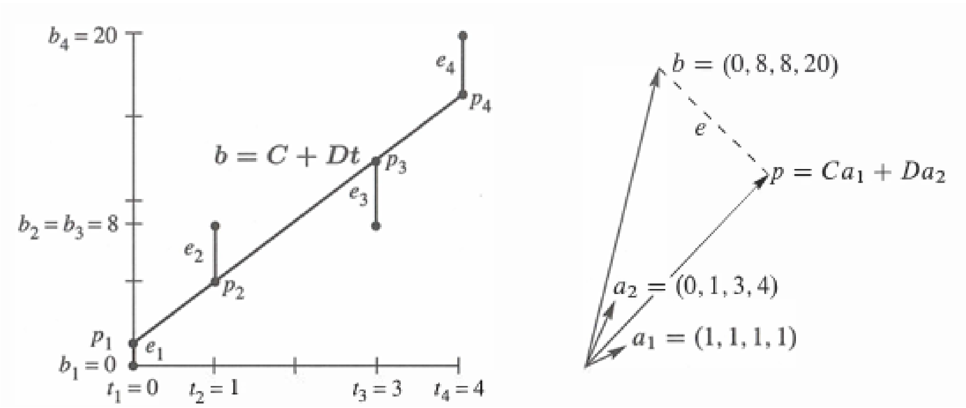


Figure 4.9: **Problems 1–11:** The closest line $C + Dt$ matches $C\boldsymbol{a}_1 + D\boldsymbol{a}_2$ in $\mathbf{R}^4$.

1. With $\boldsymbol{b} = (0, 8, 8, 20)$ at $t = 0, 1, 3, 4$, set up and solve the normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. For the best straight line in Figure 4.9a, find its four heights p_i and four errors e_i. What is the minimum value $E = e_1^2 + e_2^2 + e_3^2 + e_4^2$?

2. (Line $C + Dt$ does go through $\boldsymbol{p}$'s) With $\boldsymbol{b} = (0, 8, 8, 20)$ at times $t = 0, 1, 3, 4$, write down the four equations $A\boldsymbol{x} = \boldsymbol{b}$ (unsolvable). Change the measurements to $\boldsymbol{p} = (1, 5, 13, 17)$ and find an exact solution to $A\widehat{\boldsymbol{x}} = \boldsymbol{p}$.

3. Check that $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = (-1, 3, -5, 3)$ is perpendicular to both columns of the same matrix A. What is the shortest distance $\|e\|$ from $\boldsymbol{b}$ to the column space of A?

4. (By calculus) Write down $E = \|A\boldsymbol{x} - \boldsymbol{b}\|^2$ as a sum of four squares—the last one is $(C + 4D - 20)^2$. Find the derivative equations $\partial E/\partial C = 0$ and $\partial E/\partial D = 0$. Divide by 2 to obtain the normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

5. Find the height C of the best *horizontal line* to fit $\boldsymbol{b} = (0, 8, 8, 20)$. An exact fit would solve the unsolvable equations $C = 0, C = 8, C = 8, C = 20$. Find the 4 by 1 matrix A in these equations and solve $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. Draw the horizontal line at height $\widehat{x} = C$ and the four errors in $\boldsymbol{e}$.

6. Project $\boldsymbol{b} = (0, 8, 8, 20)$ onto the line through $\boldsymbol{a} = (1, 1, 1, 1)$. Find $\widehat{x} = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ and the projection $\boldsymbol{p} = \widehat{x}\boldsymbol{a}$. Check that $e = \boldsymbol{b} - \boldsymbol{p}$ is perpendicular to $\boldsymbol{a}$, and find the shortest distance $\|e\|$ from $\boldsymbol{b}$ to the line through $\boldsymbol{a}$.

7. Find the closest line $\boldsymbol{b} = Dt$, *through the origin*, to the same four points. An exact fit would solve $D \cdot 0 = 0, D \cdot 1 = 8, D \cdot 3 = 8, D \cdot 4 = 20$. Find the 4 by 1 matrix and solve $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. Redraw Figure 4.9a showing the best line $\boldsymbol{b} = Dt$ and the e's.

8. Project $\boldsymbol{b} = (0, 8, 8, 20)$ onto the line through $\boldsymbol{a} = (0, 1, 3, 4)$. Find $\widehat{x} = D$ and $\boldsymbol{p} = \widehat{x}\boldsymbol{a}$. The best C in Problems 5–6 and the best D in Problems 7–8 do *not* agree with the best (C, D) in Problems 1–4. That is because $(1, 1, 1, 1)$ and $(0, 1, 3, 4)$ are ____________ perpendicular.

9. For the closest parabola $b = C + Dt + Et^2$ to the same four points, write down the unsolvable equations $Ax = \boldsymbol{b}$ in three unknowns $\boldsymbol{x} = (C, D, E)$. Set up the three normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$ (solution not required). In Figure 4.9a you are now fitting a parabola to 4 points—what is happening in Figure 4.9b?

10. For the closest cubic $b = C + Dt + Et^2 + Ft^3$ to the same four points, write down the four equations $Ax = \boldsymbol{b}$. Solve them by elimination. In Figure 4.9a this cubic now goes exactly through the points. What are $\boldsymbol{p}$ and e?

11. The average of the four times is $\widehat{t} = \frac{1}{4}(0 + 1 + 3 + 4) = 2$. The average of the four b's is $\widehat{b} = \frac{1}{4}(0 + 8 + 8 + 20) = 9$.

(a) Verify that the best line goes through the center point $(\widehat{t}, \widehat{b}) = (2, 9)$.

(b) Explain why $C + D\widehat{t} = \widehat{b}$ comes from the first equation in $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

Questions 12–16 introduce basic ideas of statistics—the foundation for least squares.

12. (Recommended) This problem projects $\boldsymbol{b} = (b_1, \ldots, b_m)$ onto the line through $\boldsymbol{a} = (1, \ldots, 1)$. We solve m equations $\boldsymbol{a}x = \boldsymbol{b}$ in 1 unknown (by least squares).

(a) Solve $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}\widehat{x} = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}$ to show that $\widehat{x}$ is the *mean* (the average) of the b's.

(b) Find $e = \boldsymbol{b} - \boldsymbol{a}\widehat{x}$ and the *variance* $\|e\|^2$ and the *standard deviation* $\|e\|$.

(c) The horizontal line $\widehat{\boldsymbol{b}} = 3$ is closest to $\boldsymbol{b} = (1, 2, 6)$. Check that $\boldsymbol{p} = (3, 3, 3)$ is perpendicular to e and find the 3 by 3 projection matrix P.

13. First assumption behind least squares: $A\boldsymbol{x} = \boldsymbol{b} -$ (*noise e with mean zero*). Multiply the error vectors $e = \boldsymbol{b} - A\boldsymbol{x}$ by $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ to get $\widehat{\boldsymbol{x}} - \boldsymbol{x}$ on the right. The estimation errors $\widehat{\boldsymbol{x}} - \boldsymbol{x}$ also average to zero. The estimate $\widehat{\boldsymbol{x}}$ is *unbiased.*

14. Second assumption behind least squares: The m errors e_i are independent with variance σ^2, so the average of $(\boldsymbol{b} - A\boldsymbol{x})(\boldsymbol{b} - A\boldsymbol{x})^{\mathrm{T}}$ is $\sigma^2 I$. Multiply on the left by $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ and on the right by $A(A^{\mathrm{T}}A)^{-1}$ to show that the average matrix $(\widehat{\boldsymbol{x}} - \boldsymbol{x})(\widehat{\boldsymbol{x}} - \boldsymbol{x})^{\mathrm{T}}$ is $\sigma^2(A^{\mathrm{T}}A)^{-1}$. This is the covariance matrix W in Section 10.2.

15. A doctor takes 4 readings of your heart rate. The best solution to $x = b_1, \ldots, x = b_4$ is the average $\widehat{x}$ of $b_1, \ldots, b_4$. The matrix A is a column of 1's. Problem 14 gives the expected error $(\widehat{x} - x)^2$ as $\sigma^2(A^{\mathrm{T}}A)^{-1} =$ ____________. *By averaging, the variance drops from σ^2 to $\sigma^2/4$.*

16. If you know the average $\widehat{x}_9$ of 9 numbers $b_1, \ldots, b_9$, how can you quickly find the average $\widehat{x}_{10}$ with one more number b_{10}? The idea of *recursive* least squares is to avoid adding 10 numbers. What number multiplies $\widehat{x}_9$ in computing $\widehat{x}_{10}$?

$$\widehat{x}_{10} = \frac{1}{10}b_{10} + \underline{\qquad\qquad}\,\widehat{x}_9 = \frac{1}{10}(b_1 + \cdots + b_{10}) \quad \textbf{as in Worked Example 4.2 C.}$$

Questions 17–24 give more practice with $\widehat{\boldsymbol{x}}$ and $\boldsymbol{p}$ and e.

17. Write down three equations for the line $b = C + Dt$ to go through $b = 7$ at $t = -1$, $b = 7$ at $t = 1$, and $b = 21$ at $t = 2$. Find the least squares solution $\widehat{\boldsymbol{x}} = (C, D)$ and draw the closest line.

18. **Find the projection $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$ in Problem 17. This gives the three heights of the closest line. Show that the error vector is $e = (2, -6, 4)$. Why is $Pe = \mathbf{0}$?**

19. **Suppose the measurements at $t = -1, 1, 2$ are the errors $2, -6, 4$ in Problem 18. Compute $\widehat{\boldsymbol{x}}$ and the closest line to these new measurements. Explain the answer: $\boldsymbol{b} = (2, -6, 4)$ is perpendicular to ____________ so the projection is $\boldsymbol{p} = \mathbf{0}$.**

20. **Suppose the measurements at $t = -1, 1, 2$ are $\boldsymbol{b} = (5, 13, 17)$. Compute $\widehat{\boldsymbol{x}}$ and the closest line and e. The error is $e = \mathbf{0}$ because this $\boldsymbol{b}$ is ____________.**

21. **Which of the four subspaces contains the error vector e? Which contains $\boldsymbol{p}$? Which contains $\widehat{\boldsymbol{x}}$? What is the nullspace of A?**

22. **Find the best line $C + Dt$ to fit $b = 4, 2, -1, 0, 0$ at times $t = -2, -1, 0, 1, 2$.**

23. **Is the error vector e orthogonal to $\boldsymbol{b}$ or $\boldsymbol{p}$ or e or $\widehat{\boldsymbol{x}}$? Show that $\|e\|^2$ equals $e^{\mathrm{T}}\boldsymbol{b}$ which equals $\boldsymbol{b}^{\mathrm{T}}\boldsymbol{b} - \boldsymbol{p}^{\mathrm{T}}\boldsymbol{b}$. This is the smallest total error E.**

24. **The partial derivatives of $\|A\boldsymbol{x}\|^2$ with respect to $x_1, \ldots, x_n$ fill the vector $2A^{\mathrm{T}}A\boldsymbol{x}$. The derivatives of $2\boldsymbol{b}^{\mathrm{T}}A\boldsymbol{x}$ fill the vector $2A^{\mathrm{T}}\boldsymbol{b}$. So the derivatives of $\|A\boldsymbol{x} - \boldsymbol{b}\|^2$ are zero when ____________.**

Challenge Problems

25. ***What condition on* $(t_1, b_1), (t_2, b_2), (t_3, b_3)$ *puts those three points onto a straight line?* A column space answer is: (b_1, b_2, b_3) must be a combination of $(1, 1, 1)$ and (t_1, t_2, t_3). Try to reach a specific equation connecting the t's and b's. I should have thought of this question sooner!**

26. **Find the *plane* that gives the best fit to the 4 values $\boldsymbol{b} = (0, 1, 3, 4)$ at the corners $(1, 0)$ and $(0, 1)$ and $(-1, 0)$ and $(0, -1)$ of a square. The equations $C + Dx + Ey = b$ at those 4 points are $A\boldsymbol{x} = \boldsymbol{b}$ with 3 unknowns $\boldsymbol{x} = (C, D, E)$. What is A? At the center $(0, 0)$ of the square, show that $C + Dx + Ey =$ average of the b's.**

27. **(Distance between lines) The points $P = (x, x, x)$ and $Q = (y, 3y, -1)$ are on two lines in space that don't meet. Choose x and y to minimize the squared distance $\|P - Q\|^2$. The line connecting the closest P and Q is perpendicular to ____________.**

28. **Suppose the columns of A are not independent. How could you find a matrix B so that $P = B(B^{\mathrm{T}}B)^{-1}B^{\mathrm{T}}$ does give the projection onto the column space of A? (The usual formula will fail when $A^{\mathrm{T}}A$ is not invertible.)**

29. **Usually there will be exactly one hyperplane in $\mathbf{R}^n$ that contains the n given points $\boldsymbol{x} = \mathbf{0}, \boldsymbol{a}_1, \ldots, \boldsymbol{a}_{n-1}$. (Example for $n = 3$: There will be one plane containing $\mathbf{0}, \boldsymbol{a}_1, \boldsymbol{a}_2$ unless ____________.) What is the test to have exactly one plane in $\mathbf{R}^n$?**

30. **Example 2 shifted the times t_i to make them add to zero. We subtracted away the average time $\widehat{t} = (t_1 + \cdots + t_m)/m$ to get $T_i = t_i - \widehat{t}$. Those T_i add to zero.
With the columns $(1, \ldots, 1)$ and $(T_1, \ldots, T_m)$ now orthogonal, $A^{\mathrm{T}}A$ is diagonal. Its entries are m and $T_1^2 + \cdots + T_m^2$. Show that the best C and D have direct formulas:**

$$\boldsymbol{T} \text{ is } \boldsymbol{t} - \widehat{t} \qquad C = \frac{b_1 + \cdots + b_m}{m} \qquad \text{and} \qquad D = \frac{b_1T_1 + \cdots + b_mT_m}{T_1^2 + \cdots + T_m^2}.$$

The best line is* $C + D\boldsymbol{T}$ or $C + D(\boldsymbol{t} - \widehat{t})$. *The time shift that makes $A^{\mathrm{T}}A$ diagonal is an example of the Gram-Schmidt process: orthogonalize the columns of A in advance.

4.4 Orthonormal Bases and Gram-Schmidt

Problems 1–12 are about orthogonal vectors and orthogonal matrices.

1. **Are these pairs of vectors orthonormal or only orthogonal or only independent?**

$$\textbf{(a)}\ \begin{bmatrix}1\\0\end{bmatrix} \text{ and } \begin{bmatrix}-1\\1\end{bmatrix} \qquad \textbf{(b)}\ \begin{bmatrix}.6\\.8\end{bmatrix} \text{ and } \begin{bmatrix}.4\\-.3\end{bmatrix} \qquad \textbf{(c)}\ \begin{bmatrix}\cos\theta\\\sin\theta\end{bmatrix} \text{ and } \begin{bmatrix}-\sin\theta\\\cos\theta\end{bmatrix}.$$

Change the second vector when necessary to produce orthonormal vectors.

2. **The vectors $(2,2,-1)$ and $(-1,2,2)$ are orthogonal. Divide them by their lengths to find orthonormal vectors $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$. Put those into the columns of Q and multiply $Q^{\mathrm{T}}Q$ and QQ^{T}.**

3. **(a) If A has three orthogonal columns each of length 4, what is $A^{\mathrm{T}}A$?**
   (b) If A has three orthogonal columns of lengths 1, 2, 3, what is $A^{\mathrm{T}}A$?

4. **Give an example of each of the following:**
   (a) A matrix Q that has orthonormal columns but $QQ^{\mathrm{T}} \neq I$.
   (b) Two orthogonal vectors that are not linearly independent.
   (c) An orthonormal basis for $\mathbf{R}^3$, including the vector $\boldsymbol{q}_1 = (1,1,1)/\sqrt{3}$.

5. **Find two orthogonal vectors in the plane $x+y+2z=0$. Make them orthonormal.**

6. **If Q_1 and Q_2 are orthogonal matrices, show that their product Q_1Q_2 is also an orthogonal matrix. (Use $Q^{\mathrm{T}}Q = I$.)**

7. **If Q has orthonormal columns, what is the least squares solution $\widehat{\boldsymbol{x}}$ to $Q\boldsymbol{x} = \boldsymbol{b}$?**

8. **If $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$ are orthonormal vectors in $\mathbf{R}^5$, what combination ______ $\boldsymbol{q}_1+$ ______ $\boldsymbol{q}_2$ is closest to a given vector $\boldsymbol{b}$?**

9. **(a) Compute $P = QQ^{\mathrm{T}}$ when $\boldsymbol{q}_1 = (.8,.6,0)$ and $\boldsymbol{q}_2 = (-.6,.8,0)$. Verify that $P^2 = P$.**
   (b) Prove that always $(QQ^{\mathrm{T}})^2 = QQ^{\mathrm{T}}$ by using $Q^{\mathrm{T}}Q = I$. Then $P = QQ^{\mathrm{T}}$ is the projection matrix onto the column space of Q.

10. **Orthonormal vectors are automatically linearly independent.**
    (a) Vector proof: When $c_1\boldsymbol{q}_1 + c_2\boldsymbol{q}_2 + c_3\boldsymbol{q}_3 = \boldsymbol{0}$, what dot product leads to $c_1 = 0$? Similarly $c_2 = 0$ and $c_3 = 0$. Thus the $\boldsymbol{q}$'s are independent.
    (b) Matrix proof: Show that $Q\boldsymbol{x} = \boldsymbol{0}$ leads to $\boldsymbol{x} = \boldsymbol{0}$. Since Q may be rectangular, you can use Q^{T} but not Q^{-1}.

11. **(a) Gram-Schmidt: Find orthonormal vectors $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$ in the plane spanned by $\boldsymbol{a} = (1,3,4,5,7)$ and $\boldsymbol{b} = (-6,6,8,0,8)$.**
    (b) Which vector in this plane is closest to $(1,0,0,0,0)$?

12. **If $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$ is a basis for $\mathbf{R}^3$, any vector $\boldsymbol{b}$ can be written as**

$$\boldsymbol{b} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + x_3\boldsymbol{a}_3 \qquad \textbf{or} \qquad \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \boldsymbol{a}_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \boldsymbol{b}.$$

    (a) Suppose the $\boldsymbol{a}$'s are orthonormal. Show that $x_1 = \boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{b}$.
    (b) Suppose the $\boldsymbol{a}$'s are orthogonal. Show that $x_1 = \boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{b}/\boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{a}_1$.
    (c) If the $\boldsymbol{a}$'s are independent, x_1 is the first component of _________ times $\boldsymbol{b}$.

Problems 13–25 are about the Gram-Schmidt process and $A = QR$.

13. **What multiple of $\boldsymbol{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ should be subtracted from $\boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ to make the result $\boldsymbol{B}$ orthogonal to $\boldsymbol{a}$? Sketch a figure to show $\boldsymbol{a}, \boldsymbol{b}$, and $\boldsymbol{B}$.**

14. **Complete the Gram-Schmidt process in Problem 13 by computing $\boldsymbol{q}_1 = \boldsymbol{a}/\|\boldsymbol{a}\|$ and $\boldsymbol{q}_2 = \boldsymbol{B}/\|\boldsymbol{B}\|$ and factoring into QR:**

$$\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 \end{bmatrix} \begin{bmatrix} \|\boldsymbol{a}\| & ? \\ 0 & \|\boldsymbol{B}\| \end{bmatrix}.$$

15. **(a) Find orthonormal vectors $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ such that $\boldsymbol{q}_1, \boldsymbol{q}_2$ span the column space of**

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}.$$

(b) Which of the four fundamental subspaces contains q_3?

(c) Solve $Ax = (1, 2, 7)$ by least squares.

16. What multiple of $\boldsymbol{a} = (4, 5, 2, 2)$ is closest to $\boldsymbol{b} = (1, 2, 0, 0)$? Find orthonormal vectors $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$ in the plane of a and b.

17. Find the projection of b onto the line through a:

$$\boldsymbol{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{p} = ? \quad \textbf{and} \quad \boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = ?$$

Compute the orthonormal vectors $\boldsymbol{q}_1 = a/\|a\|$ and $\boldsymbol{q}_2 = e/\|e\|$.

18. (Recommended) Find orthogonal vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ by Gram-Schmidt from $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$:

$$\boldsymbol{a} = (1, -1, 0, 0) \qquad \boldsymbol{b} = (0, 1, -1, 0) \qquad \boldsymbol{c} = (0, 0, 1, -1).$$

$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ and a, b, c are bases for the vectors perpendicular to $\boldsymbol{d} = (1, 1, 1, 1)$.

19. If $A = QR$ then $A^{\mathrm{T}}A = R^{\mathrm{T}}R =$ ____________ triangular times ____________ triangular. *Gram-Schmidt on A corresponds to elimination on $A^{\mathrm{T}}A$.* The pivots for $A^{\mathrm{T}}A$ must be the squares of diagonal entries of R. Find Q and R by Gram-Schmidt for this A:

$$A = \begin{bmatrix} -1 & 1 \\ 2 & 1 \\ 2 & 4 \end{bmatrix} \qquad \textbf{and} \qquad A^{\mathrm{T}}A = \begin{bmatrix} 9 & 9 \\ 9 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 9 & 9 \\ 0 & 9 \end{bmatrix}.$$

20. True or false (give an example in either case):

(a) Q^{-1} is an orthogonal matrix when Q is an orthogonal matrix.

(b) If Q (3 by 2) has orthonormal columns then $\|Qx\|$ always equals $\|x\|$.

21. Find an orthonormal basis for the column space of A:

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{bmatrix} \qquad \textbf{and} \qquad \boldsymbol{b} = \begin{bmatrix} -4 \\ -3 \\ 3 \\ 0 \end{bmatrix}.$$

Then compute the projection of b onto that column space.

22. Find orthogonal vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ by Gram-Schmidt from

$$\boldsymbol{a} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{c} = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}.$$

23. Find $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ (orthonormal) as combinations of $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ (independent columns). Then write A as QR:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}.$$

24. (a) Find a basis for the subspace S in $\mathbf{R}^4$ spanned by all solutions of

$$x_1 + x_2 + x_3 - x_4 = 0.$$

(b) Find a basis for the orthogonal complement $\mathrm{S}^{\perp}$.

(c) Find b_1 in S and b_2 in $\mathrm{S}^{\perp}$ so that $b_1 + b_2 = \boldsymbol{b} = (1, 1, 1, 1)$.

25. If $ad - bc > 0$, the entries in $A = QR$ are

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{\sqrt{a^2+c^2}} \begin{bmatrix} a & -c \\ c & a \end{bmatrix} \frac{1}{\sqrt{a^2+c^2}} \begin{bmatrix} a^2+c^2 & ab+cd \\ 0 & ad-bc \end{bmatrix}.$$

Write $A = QR$ when $a, b, c, d = 2, 1, 1, 1$ and also $1, 1, 1, 1$. Which entry of R becomes zero when the columns are dependent and Gram-Schmidt breaks down?

Problems 26–29 use the QR code in equation (11). It executes Gram-Schmidt.

26. Show why C (found via C^* in the steps after (11)) is equal to C in equation (8).

27. Equation (8) subtracts from c its components along A and B. Why not subtract the components along a and along b?

28. Where are the mn^2 multiplications in equation (11)?

29. Apply the MATLAB qr code to $\boldsymbol{a} = (2, 2, -1)$, $\boldsymbol{b} = (0, -3, 3)$, $\boldsymbol{c} = (1, 0, 0)$. What are the $\boldsymbol{q}$'s?

Problems 30–35 involve orthogonal matrices that are special.

30. The first four *wavelets* are in the columns of this wavelet matrix W:

$$W = \frac{1}{2} \begin{bmatrix} 1 & 1 & \sqrt{2} & 0 \\ 1 & 1 & -\sqrt{2} & 0 \\ 1 & -1 & 0 & \sqrt{2} \\ 1 & -1 & 0 & -\sqrt{2} \end{bmatrix}.$$

What is special about the columns? Find the inverse wavelet transform W^{-1}.

31. (a) Choose c so that Q is an orthogonal matrix:

$$Q = c \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}.$$

(b) Project $\boldsymbol{b} = (1, 1, 1, 1)$ onto the first column. Then project $\boldsymbol{b}$ onto the plane of the first two columns.

32. If $\boldsymbol{u}$ is a unit vector, then $Q = I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$ is a reflection matrix (Example 3). Find Q_1 from $\boldsymbol{u} = (0, 1)$ and Q_2 from $\boldsymbol{u} = (0, \sqrt{2}/2, \sqrt{2}/2)$. Draw the reflections when Q_1 and Q_2 multiply the vectors $(1, 2)$ and $(1, 1, 1)$.

33. Find all matrices that are both orthogonal and lower triangular.

34. $Q = I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$ is a reflection matrix when $\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u} = 1$. Two reflections give $Q^2 = I$.

(a) Show that $Q\boldsymbol{u} = -\boldsymbol{u}$. The mirror is perpendicular to $\boldsymbol{u}$.

(b) Find $Q\boldsymbol{v}$ when $\boldsymbol{u}^{\mathrm{T}}\boldsymbol{v} = 0$. The mirror contains $\boldsymbol{v}$. It reflects to itself.

Challenge Problems

35. (MATLAB) Factor $[Q, R] = \mathbf{qr}(A)$ for $A = \mathbf{eye}(4) - \mathbf{diag}([1 \quad 1 \quad 1], -1)$. You are orthogonalizing the columns $(1, -1, 0, 0)$ and $(0, 1, -1, 0)$ and $(0, 0, 1, -1)$ and $(0, 0, 0, 1)$ of A. Can you scale the orthogonal columns of Q to get nice integer components?

36. If A is m by n with rank n, $\mathbf{qr}(A)$ produces a *square* Q and zeros below R:

$$\text{The factors from MATLAB are } (m \text{ by } m)(m \text{ by } n) \qquad A = [\, Q_1 \quad Q_2 \,] \begin{bmatrix} R \\ 0 \end{bmatrix}.$$

The n columns of Q_1 are an orthonormal basis for which fundamental subspace?
The $m - n$ columns of Q_2 are an orthonormal basis for which fundamental subspace?

37. We know that $P = QQ^{\mathrm{T}}$ is the projection onto the column space of Q (m by n). Now add another column $\boldsymbol{a}$ to produce $A = [\, Q \quad \boldsymbol{a} \,]$. Gram-Schmidt replaces $\boldsymbol{a}$ by what vector $\boldsymbol{q}$? Start with $\boldsymbol{a}$, subtract _________, divide by _________ to find $\boldsymbol{q}$.

5 Chapter 5: Determinants

5.1 The Properties of Determinants

Questions 1–12 are about the rules for determinants.

1. **If a 4 by 4 matrix has** $\det A = \frac{1}{2}$**, find** $\det(2A)$ **and** $\det(-A)$ **and** $\det(A^2)$ **and** $\det(A^{-1})$**.**

2. **If a 3 by 3 matrix has** $\det A = -1$**, find** $\det(\frac{1}{2}A)$ **and** $\det(-A)$ **and** $\det(A^2)$ **and** $\det(A^{-1})$**.**

3. **True or false, with a reason if true or a counterexample if false:**

   (a) **The determinant of** $I + A$ **is** $1 + \det A$**.**

   (b) **The determinant of** ABC **is** $|A||B||C|$**.**

   (c) **The determinant of** $4A$ **is** $4|A|$**.**

   (d) **The determinant of** $AB - BA$ **is zero. Try an example with** $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$**.**

4. **Which row exchanges show that these "reverse identity matrices"** J_3 **and** J_4 **have** $|J_3| = -1$ **but** $|J_4| = +1$**?**

$$\det \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = -1 \qquad \textbf{but} \qquad \det \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = +1.$$

5. **For** $n = 5, 6, 7$**, count the row exchanges to permute the reverse identity** J_n **to the identity matrix** I_n**. Propose a rule for every size** n **and predict whether** J_{101} **has determinant** $+1$ **or** -1**.**

6. **Show how Rule 6 (determinant** $= 0$ **if a row is all zero) comes from Rule 3.**

7. **Find the determinants of rotations and reflections:**

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad \textbf{and} \quad Q = \begin{bmatrix} 1 - 2\cos^2\theta & -2\cos\theta\sin\theta \\ -2\cos\theta\sin\theta & 1 - 2\sin^2\theta \end{bmatrix}.$$

8. **Prove that every orthogonal matrix (**$Q^{\mathrm{T}}Q = I$**) has determinant** 1 **or** -1**.**

   (a) **Use the product rule** $|AB| = |A||B|$ **and the transpose rule** $|Q| = |Q^{\mathrm{T}}|$**.**

   (b) **Use only the product rule. If** $|\det Q| > 1$ **then** $\det Q^n = (\det Q)^n$ **blows up. How do you know this can't happen to** Q^n**?**

9. **Do these matrices have determinant** $0, 1, 2,$ **or** 3**?**

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

10. **If the entries in every row of** A **add to zero, solve** $Ax = \mathbf{0}$ **to prove** $\det A = 0$**. If those entries add to one, show that** $\det(A - I) = 0$**. Does this mean** $\det A = 1$**?**

11. **Suppose that** $CD = -DC$ **and find the flaw in this reasoning: Taking determinants gives** $|C||D| = -|D||C|$**. Therefore** $|C| = 0$ **or** $|D| = 0$**. One or both of the matrices must be singular. (That is not true.)**

12. **The inverse of a** 2 **by** 2 **matrix seems to have determinant** $= 1$**:**

$$\det A^{-1} = \det \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{ad - bc}{ad - bc} = 1.$$

    What is wrong with this calculation? What is the correct $\det A^{-1}$**?**

Questions 13–27 use the rules to compute specific determinants.

13. Reduce A to U and find $\det A =$ product of the pivots:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \qquad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix}.$$

14. By applying row operations to produce an upper triangular U, compute

$$\det \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 6 & 6 & 1 \\ -1 & 0 & 0 & 3 \\ 0 & 2 & 0 & 7 \end{bmatrix} \quad \textbf{and} \quad \det \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}.$$

15. Use row operations to simplify and compute these determinants:

$$\det \begin{bmatrix} 101 & 201 & 301 \\ 102 & 202 & 302 \\ 103 & 203 & 303 \end{bmatrix} \quad \textbf{and} \quad \det \begin{bmatrix} 1 & t & t^2 \\ t & 1 & t \\ t^2 & t & 1 \end{bmatrix}.$$

16. Find the determinants of a rank one matrix and a skew-symmetric matrix :

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & -4 & 5 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} 0 & 1 & 3 \\ -1 & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix}.$$

17. A skew-symmetric matrix has $A^{\mathrm{T}} = -A$. Insert a, b, c for $1, 3, 4$ in Question 16 and show that $|A| = 0$. Write down a 4 by 4 example with $|A| = 1$.

18. Use row operations to show that the 3 by 3 "Vandermonde determinant" is

$$\det \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix} = (b-a)(c-a)(c-b).$$

19. Find the determinants of U and U^{-1} and U^2:

$$U = \begin{bmatrix} 1 & 4 & 6 \\ 0 & 2 & 5 \\ 0 & 0 & 3 \end{bmatrix} \quad \textbf{and} \quad U = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}.$$

20. Suppose you do two row operations at once, going from

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \textbf{to} \quad \begin{bmatrix} a - Lc & b - Ld \\ c - \ell a & d - \ell b \end{bmatrix}.$$

Find the second determinant. Does it equal $ad - bc$?

21. *Row exchange:* Add row 1 of A to row 2, then subtract row 2 from row 1. Then add row 1 to row 2 and multiply row 1 by -1 to reach B. Which rules show

$$\det B = \begin{vmatrix} c & d \\ a & b \end{vmatrix} \quad \textbf{equals} \quad -\det A = -\begin{vmatrix} a & b \\ c & d \end{vmatrix}?$$

Those rules could replace Rule 2 in the definition of the determinant.

22. From $ad - bc$, find the determinants of A and A^{-1} and $A - \lambda I$:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \textbf{and} \quad A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \quad \textbf{and} \quad A - \lambda I = \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}.$$

Which two numbers λ lead to $\det(A - \lambda I) = 0$? Write down the matrix $A - \lambda I$ for each of those numbers λ—it should not be invertible.

23. From $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ **find** A^2 **and** A^{-1} **and** $A - \lambda I$ **and their determinants. Which two numbers** λ **lead to** $\det(A - \lambda I) = 0$**?**

24. Elimination reduces A **to** U**. Then** $A = LU$**:**

$$A = \begin{bmatrix} 3 & 3 & 4 \\ 6 & 8 & 7 \\ -3 & 5 & -9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 4 & 1 \end{bmatrix} \begin{bmatrix} 3 & 3 & 4 \\ 0 & 2 & -1 \\ 0 & 0 & -1 \end{bmatrix} = LU.$$

Find the determinants of $L, U, A, U^{-1}L^{-1}$**, and** $U^{-1}L^{-1}A$**.**

25. If the i, j **entry of** A **is** i **times** j**, show that** $\det A = 0$**. (Exception when** $A = [1]$**.)**

26. If the i, j **entry of** A **is** $i + j$**, show that** $\det A = 0$**. (Exception when** $n = 1$ **or** 2**.)**

27. Compute the determinants of these matrices by row operations:

$$A = \begin{bmatrix} 0 & a & 0 \\ 0 & 0 & b \\ c & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 0 & a & 0 & 0 \\ 0 & 0 & b & 0 \\ 0 & 0 & 0 & c \\ d & 0 & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad C = \begin{bmatrix} a & a & a \\ a & b & b \\ a & b & c \end{bmatrix}.$$

28. True or false (give a reason if true or a 2 by 2 example if false):

(a) If A **is not invertible then** AB **is not invertible.**
(b) The determinant of A **is always the product of its pivots.**
(c) The determinant of $A - B$ **equals** $\det A - \det B$**.**
(d) AB **and** BA **have the same determinant.**

29. What is wrong with this proof that projection matrices have $\det P = 1$**?**

$$P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} \quad \textbf{so} \quad |P| = |A|\frac{1}{|A^{\mathrm{T}}||A|}|A^{\mathrm{T}}| = 1.$$

30. (Calculus question) Show that the partial derivatives of $\ln(\det A)$ **give** A^{-1}**!**

$$f(a, b, c, d) = \ln(ad - bc) \quad \textbf{leads to} \quad \begin{bmatrix} \partial f/\partial a & \partial f/\partial c \\ \partial f/\partial b & \partial f/\partial d \end{bmatrix} = A^{-1}.$$

31. (MATLAB) The Hilbert matrix hilb(n**) has** i, j **entry equal to** $1/(i + j - 1)$**. Print the determinants of hilb(1), hilb(2), ..., hilb(10). Hilbert matrices are hard to work with! What are the pivots of hilb (5)?**

32. (MATLAB) What is a typical determinant (experimentally) of rand(n**) and randn(**n**) for** $n = 50, 100, 200, 400$**? (And what does "Inf" mean in MATLAB?)**

33. (MATLAB) Find the largest determinant of a 6 by 6 matrix of 1's and -1**'s.**

34. If you know that $\det A = 6$**, what is the determinant of** B**?**

$$\textbf{From } \det A = \begin{vmatrix} \textbf{row } 1 \\ \textbf{row } 2 \\ \textbf{row } 3 \end{vmatrix} = 6 \textbf{ find } \det B = \begin{vmatrix} \textbf{row } 3 + \textbf{row } 2 + \textbf{row } 1 \\ \textbf{row } 2 + \textbf{row } 1 \\ \textbf{row } 1 \end{vmatrix}.$$

5.2 Permutations and Cofactors

Problems 1–10 use the big formula with $n!$ **terms:** $|A| = \sum \pm a_{1\alpha}a_{2\beta}\cdots a_{n\omega}$**. Every term uses each row and each column once.**

1. Compute the determinants of A, B, C **from six terms. Are their rows independent?**

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 4 & 4 \\ 5 & 6 & 7 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

2. **Compute the determinants of A, B, C, D. Are their columns independent?**

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \qquad C = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix} \qquad D = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}.$$

3. **Show that** $\det A = 0$**, regardless of the five nonzeros marked by** x**'s:**

$$A = \begin{bmatrix} x & x & x \\ 0 & 0 & x \\ 0 & 0 & x \end{bmatrix}.$$

What are the cofactors of row 1?
What is the rank of A?
What are the 6 terms in $\det A$?

4. **Find two ways to choose nonzeros from four different rows and columns:**

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{bmatrix} \quad (B \text{ has the same zeros as } A).$$

Is $\det A$ **equal to** $1+1$ **or** $1-1$ **or** $-1-1$**? What is** $\det B$?

5. **Place the smallest number of zeros in a 4 by 4 matrix that will guarantee** $\det A = 0$**. Place as many zeros as possible while still allowing** $\det A \neq 0$**.**

6. **(a) If** $a_{11} = a_{22} = a_{33} = 0$**, how many of the six terms in** $\det A$ **will be zero?**

   (b) If $a_{11} = a_{22} = a_{33} = a_{44} = 0$**, how many of the 24 products** $a_{1j}a_{2k}a_{3l}a_{4m}$ **are sure to be zero?**

7. **How many 5 by 5 permutation matrices have** $\det P = +1$**? Those are even permutations. Find one that needs four exchanges to reach the identity matrix.**

8. **If** $\det A$ **is not zero, at least one of the** $n!$ **terms in formula (8) is not zero. Deduce from the big formula that some ordering of the rows of** A **leaves no zeros on the diagonal. (Don't use** P **from elimination; that** PA **can have zeros on the diagonal.)**

9. **Show that 4 is the largest determinant for a 3 by 3 matrix of 1's and** -1**'s.**

10. **How many permutations of** $(1, 2, 3, 4)$ **are even and what are they? Extra credit: What are all the possible 4 by 4 determinants of** $I + P_{\text{even}}$?

Problems 11–22 use cofactors $C_{ij} = (-1)^{i+j} \det M_{ij}$**. Remove row** i **and column** j**.**

11. **Find all cofactors and put them into cofactor matrices** C, D**. Find** AC **and** $\det B$**.**

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 0 & 0 \end{bmatrix}.$$

12. **Find the cofactor matrix** C **and multiply** A **times** C^{T}**. Compare** AC^{T} **with** A^{-1}**:**

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \qquad A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

13. **The** n **by** n **determinant** C_n **has 1's above and below the main diagonal:**

$$C_1 = |0| \quad C_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \quad C_3 = \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} \quad C_4 = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix}.$$

   (a) What are these determinants C_1, C_2, C_3, C_4?

   (b) By cofactors find the relation between C_n **and** C_{n-1} **and** C_{n-2}**. Find** C_{10}**.**

14. The matrices in Problem 13 have 1's just above and below the main diagonal. Going down the matrix, which order of columns (if any) gives all 1's? Explain why that permutation is *even* for $n = 4, 8, 12, \dots$ and *odd* for $n = 2, 6, 10, \dots$ Then

$$C_n = 0 \text{ (odd } n) \qquad C_n = 1 \ (n = 4, 8, \cdots) \qquad C_n = -1 \ (n = 2, 6, \cdots).$$

15. The tridiagonal $1, 1, 1$ matrix of order n has determinant E_n:

$$E_1 = |1| \quad E_2 = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} \quad E_3 = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix} \quad E_4 = \begin{vmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{vmatrix}.$$

(a) By cofactors show that $E_n = E_{n-1} - E_{n-2}$.

(b) Starting from $E_1 = 1$ and $E_2 = 0$ find $E_3, E_4, \dots, E_8$.

(c) By noticing how these numbers eventually repeat, find E_{100}.

16. F_n is the determinant of the $1, 1, -1$ tridiagonal matrix of order n:

$$F_2 = \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 2 \quad F_3 = \begin{vmatrix} 1 & -1 & 0 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix} = 3 \quad F_4 = \begin{vmatrix} 1 & -1 & & \\ 1 & 1 & -1 & \\ & 1 & 1 & -1 \\ & & 1 & 1 \end{vmatrix} \neq 4.$$

Expand in cofactors to show that $F_n = F_{n-1} + F_{n-2}$. These determinants are *Fibonacci numbers* $1, 2, 3, 5, 8, 13, \dots$. The sequence usually starts $1, 1, 2, 3$ (with two 1's) so our F_n is the usual F_{n+1}.

17. The matrix B_n is the $-1, 2, -1$ matrix A_n except that $b_{11} = 1$ instead of $a_{11} = 2$. Using cofactors of the *last* row of B_4 show that $|B_4| = 2|B_3| - |B_2| = 1$.

$$B_4 = \begin{bmatrix} 1 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \qquad B_3 = \begin{bmatrix} 1 & -1 & \\ -1 & 2 & -1 \\ & -1 & 2 \end{bmatrix} \qquad B_2 = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

The recursion $|B_n| = 2|B_{n-1}| - |B_{n-2}|$ is satisfied when every $|B_n| = 1$. This recursion is the same as for the A's in Example 6. The difference is in the starting values $1, 1, 1$ for the determinants of sizes $n = 1, 2, 3$.

18. Go back to B_n in Problem 17. It is the same as A_n except for $b_{11} = 1$. So use linearity in the first row, where $\begin{bmatrix} 1 & -1 & 0 \end{bmatrix}$ equals $\begin{bmatrix} 2 & -1 & 0 \end{bmatrix}$ minus $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$:

$$|B_n| = \begin{vmatrix} 1 & -1 & 0 \\ -1 & & \\ & A_{n-1} & \\ 0 & & \end{vmatrix} = \begin{vmatrix} 2 & -1 & 0 \\ -1 & & \\ & A_{n-1} & \\ 0 & & \end{vmatrix} - \begin{vmatrix} 1 & 0 & 0 \\ -1 & & \\ & A_{n-1} & \\ 0 & & \end{vmatrix}.$$

Linearity gives $|B_n| = |A_n| - |A_{n-1}| =$ _____.

19. Explain why the 4 by 4 Vandermonde determinant contains x^3 but not x^4 or x^5:

$$V_4 = \det \begin{bmatrix} 1 & a & a^2 & a^3 \\ 1 & b & b^2 & b^3 \\ 1 & c & c^2 & c^3 \\ 1 & x & x^2 & x^3 \end{bmatrix}.$$

The determinant is zero at $x =$ _____, _____, and _____. The cofactor of x^3 is $V_3 = (b-a)(c-a)(c-b)$. Then $V_4 = (b-a)(c-a)(c-b)(x-a)(x-b)(x-c)$.

20. Find G_2 and G_3 and then by row operations G_4. Can you predict G_n?

$$G_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \qquad G_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} \qquad G_4 = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{vmatrix}.$$

21. **Compute S_1, S_2, S_3 for these $1, 3, 1$ matrices. By Fibonacci guess and check S_4.**

$$S_1 = |3| \qquad S_2 = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} \qquad S_3 = \begin{vmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{vmatrix}.$$

22. **Change 3 to 2 in the upper left corner of the matrices in Problem 21. Why does that subtract S_{n-1} from the determinant S_n? Show that the determinants of the new matrices become the Fibonacci numbers $2, 5, 13$ (always F_{2n+1}).**

Problems 23–26 are about block matrices and block determinants.

23. **With 2 by 2 blocks in 4 by 4 matrices, you cannot always use block determinants:**

$$\begin{vmatrix} A & B \\ 0 & D \end{vmatrix} = |A||D| \qquad \textbf{but} \qquad \begin{vmatrix} A & B \\ C & D \end{vmatrix} \neq |A||D| - |C||B|.$$

   (a) **Why is the first statement true? Somehow B doesn't enter.**

   (b) **Show by example that equality fails (as shown) when C enters.**

   (c) **Show by example that the answer $\det(AD - CB)$ is also wrong.**

24. **With block multiplication, $A = LU$ has $A_k = L_k U_k$ in the top left corner:**

$$A = \begin{bmatrix} A_k & * \\ * & * \end{bmatrix} = \begin{bmatrix} L_k & 0 \\ * & * \end{bmatrix} \begin{bmatrix} U_k & * \\ 0 & * \end{bmatrix}.$$

   (a) **Suppose the first three pivots of A are $2, 3, -1$. What are the determinants of L_1, L_2, L_3 (with diagonal 1's) and U_1, U_2, U_3 and A_1, A_2, A_3?**

   (b) **If A_1, A_2, A_3 have determinants $5, 6, 7$ find the three pivots from equation (3).**

25. **Block elimination subtracts CA^{-1} times the first row $\begin{bmatrix} A & B \end{bmatrix}$ from the second row $\begin{bmatrix} C & D \end{bmatrix}$. This leaves the *Schur complement* $D - CA^{-1}B$ in the corner:**

$$\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}.$$

   Take determinants of these block matrices to prove correct rules if A^{-1} exists:

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = |A|\,|D - CA^{-1}B| = |AD - CB| \quad \textbf{provided } AC = CA.$$

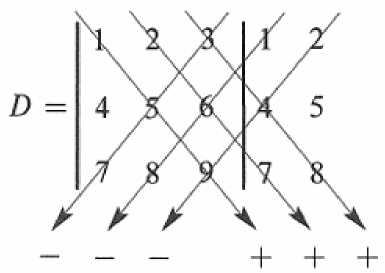
26. **If A is m by n and B is n by m, block multiplication gives $\det M = \det AB$:**

$$M = \begin{bmatrix} 0 & A \\ -B & I \end{bmatrix} = \begin{bmatrix} AB & A \\ 0 & I \end{bmatrix} \begin{bmatrix} I & 0 \\ -B & I \end{bmatrix}.$$

   If A is a single row and B is a single column what is $\det M$? If A is a column and B is a row what is $\det M$? Do a 3 by 3 example of each.

27. **(A calculus question) Show that the derivative of $\det A$ with respect to a_{11} is the cofactor C_{11}. The other entries are fixed—we are only changing a_{11}.**

28. **A 3 by 3 determinant has three products "down to the right" and three "down to the left" with minus signs. Compute the six terms like $(1)(5)(9) = 45$ to find D.**



Explain without determinants why this particular matrix is or is not invertible.

29. **For E_4 in Problem 15, five of the $4! = 24$ terms in the big formula (8) are nonzero. Find those five terms to show that $E_4 = -1$.**

30. **For the 4 by 4 tridiagonal second difference matrix (entries $-1, 2, -1$) find the five terms in the big formula that give** $\det A = 16 - 4 - 4 - 4 + 1$.

31. **Find the determinant of this cyclic P by cofactors of row 1 and then the "big formula". How many exchanges reorder $4, 1, 2, 3$ into $1, 2, 3, 4$? Is $|P^2| = 1$ or -1?**

$$P = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad P^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}.$$

Challenge Problems

32. **Cofactors of the $1, 3, 1$ matrices in Problem 21 give a recursion $S_n = 3S_{n-1} - S_{n-2}$. Amazingly that recursion produces every second Fibonacci number. Here is the challenge.**

Show that S_n is the Fibonacci number F_{2n+2} by proving $F_{2n+2} = 3F_{2n} - F_{2n-2}$. **Keep using Fibonacci's rule $F_k = F_{k-1} + F_{k-2}$ starting with $k = 2n + 2$.**

33. **The symmetric Pascal matrices have determinant 1. If I subtract 1 from the n, n entry, why does the determinant become zero? (Use rule 3 or cofactors.)**

$$\det \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix} = 1 \ (\textbf{known}) \qquad \det \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & \mathbf{19} \end{bmatrix} = \mathbf{0} \ (\textbf{to explain}).$$

34. **This problem shows in two ways that** $\det A = 0$ **(the x's are any numbers):**

$$A = \begin{bmatrix} x & x & x & x & x \\ x & x & x & x & x \\ 0 & 0 & 0 & x & x \\ 0 & 0 & 0 & x & x \\ 0 & 0 & 0 & x & x \end{bmatrix}.$$

(a) **How do you know that the rows are linearly dependent?**

(b) **Explain why all 120 terms are zero in the big formula for** $\det A$.

35. **If $|\det(A)| > 1$, prove that the powers A^n cannot stay bounded. But if $|\det(A)| \le 1$, show that some entries of A^n might still grow large. Eigenvalues will give the right test for stability, determinants tell us only one number.**

5.3 Cramer's Rule, Inverses, and Volumes

Problems 1–5 are about Cramer's Rule for $x = A^{-1}b$.

1. **Solve these linear equations by Cramer's Rule** $x_j = \det B_j / \det A$:

$$\textbf{(a)} \quad \begin{aligned} 2x_1 + 5x_2 &= 1 \\ x_1 + 4x_2 &= 2 \end{aligned} \qquad \qquad \textbf{(b)} \quad \begin{aligned} 2x_1 + x_2 \phantom{{}+x_3} &= 1 \\ x_1 + 2x_2 + x_3 &= 0 \\ x_2 + 2x_3 &= 0 \end{aligned}$$

2. **Use Cramer's Rule to solve for y (only). Call the 3 by 3 determinant D:**

$$\textbf{(a)} \quad \begin{aligned} ax + by &= 1 \\ cx + dy &= 0 \end{aligned} \qquad \qquad \textbf{(b)} \quad \begin{aligned} ax + by + cz &= 1 \\ dx + ey + fz &= 0 \\ gx + hy + iz &= 0 \end{aligned}$$

3. **Cramer's Rule breaks down when** $\det A = 0$. **Example (a) has no solution while (b) has infinitely many. What are the ratios** $x_j = \det B_j / \det A$ **in these two cases?**

$$\text{(a)} \quad \begin{aligned} 2x_1 + 3x_2 &= 1 \\ 4x_1 + 6x_2 &= 1 \end{aligned} \quad \text{(parallel lines)} \qquad \text{(b)} \quad \begin{aligned} 2x_1 + 3x_2 &= 1 \\ 4x_1 + 6x_2 &= 2 \end{aligned} \quad \text{(same line)}$$

4. **Quick proof of Cramer's rule. The determinant is a linear function of column 1. It is zero if two columns are equal. When** $b = Ax = x_1 a_1 + x_2 a_2 + x_3 a_3$ **goes into the first column of** A, **the determinant of this matrix** B_1 **is**

$$\begin{vmatrix} b & a_2 & a_3 \end{vmatrix} = \begin{vmatrix} x_1 a_1 + x_2 a_2 + x_3 a_3 & a_2 & a_3 \end{vmatrix} = x_1 \begin{vmatrix} a_1 & a_2 & a_3 \end{vmatrix} = x_1 \det A.$$

   (a) **What formula for** x_1 **comes from left side = right side?**

   (b) **What steps lead to the middle equation?**

5. **If the right side** b **is the first column of** A, **solve the 3 by 3 system** $Ax = b$. **How does each determinant in Cramer's Rule lead to this solution** x?

Problems 6–15 are about $A^{-1} = C^{\mathrm{T}} / \det A$. **Remember to transpose** C.

6. **Find** A^{-1} **from the cofactor formula** $C^{\mathrm{T}} / \det A$. **Use symmetry in part (b).**

$$\text{(a)} \quad A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 7 & 1 \end{bmatrix} \qquad \text{(b)} \quad A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

7. **If all the cofactors are zero, how do you know that** A **has no inverse? If none of the cofactors are zero, is** A **sure to be invertible?**

8. **Find the cofactors of** A **and multiply** AC^{T} **to find** $\det A$:

$$A = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 6 & -3 & 0 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \quad \text{and } AC^{\mathrm{T}} = \underline{\qquad}.$$

   If you change that 4 to 100, why is $\det A$ **unchanged?**

9. **Suppose** $\det A = 1$ **and you know all the cofactors in** C. **How can you find** A?

10. **From the formula** $AC^{\mathrm{T}} = (\det A)I$ **show that** $\det C = (\det A)^{n-1}$.

11. **If all entries of** A **are integers, and** $\det A = 1$ **or** -1, **prove that all entries of** A^{-1} **are integers. Give a 2 by 2 example with no zero entries.**

12. **If all entries of** A **and** A^{-1} **are integers, prove that** $\det A = 1$ **or** -1. **Hint: What is** $\det A$ **times** $\det A^{-1}$?

13. **Complete the calculation of** A^{-1} **by cofactors that was started in Example 5.**

14. L **is lower triangular and** S **is symmetric. Assume they are invertible:**

$$\begin{array}{l} \textbf{To invert} \\ \textbf{triangular } L \\ \textbf{symmetric } S \end{array} \quad L = \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix} \qquad S = \begin{bmatrix} a & b & d \\ b & c & e \\ d & e & f \end{bmatrix}.$$

   (a) **Which three cofactors of** L **are zero? Then** L^{-1} **is also lower triangular.**

   (b) **Which three pairs of cofactors of** S **are equal? Then** S^{-1} **is also symmetric.**

   (c) **The cofactor matrix** C **of an orthogonal** Q **will be ______.** ***Why?***

15. **For** $n = 5$ **the matrix** C **contains ______ cofactors. Each 4 by 4 cofactor contains ______ terms and each term needs ______ multiplications. Compare with** $5^3 = 125$ **for the Gauss-Jordan computation of** A^{-1} **in Section 2.4.**

Problems 16–26 are about area and volume by determinants.

16. (a) **Find the area of the parallelogram with edges $\boldsymbol{v} = (3, 2)$ and $\boldsymbol{w} = (1, 4)$.**

    (b) **Find the area of the triangle with sides $\boldsymbol{v}$, $\boldsymbol{w}$, and $\boldsymbol{v} + \boldsymbol{w}$. Draw it.**

    (c) **Find the area of the triangle with sides $\boldsymbol{v}$, $\boldsymbol{w}$, and $\boldsymbol{w} - \boldsymbol{v}$. Draw it.**

17. **A box has edges from $(0, 0, 0)$ to $(3, 1, 1)$ and $(1, 3, 1)$ and $(1, 1, 3)$. Find its volume. Also find the area of each parallelogram face using $\|\boldsymbol{u} \times \boldsymbol{v}\|$.**

18. (a) **The corners of a triangle are $(2, 1)$ and $(3, 4)$ and $(0, 5)$. What is the area?**

    (b) **Add a corner at $(-1, 0)$ to make a lopsided region (four sides). Find the area.**

19. **The parallelogram with sides $(2, 1)$ and $(2, 3)$ has the same area as the parallelogram with sides $(2, 2)$ and $(1, 3)$. Find those areas from 2 by 2 determinants and say why they must be equal. (I can't see why from a picture. Please write to me if you do.)**

20. **The Hadamard matrix H has orthogonal rows. The box is a hypercube!**

$$\textbf{What is} \quad |H| = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{vmatrix} = \textbf{volume of a hypercube in } \mathbf{R}^4?$$

21. **If the columns of a 4 by 4 matrix have lengths L_1, L_2, L_3, L_4, what is the largest possible value for the determinant (based on volume)? If all entries of the matrix are 1 or -1, what are those lengths and the maximum determinant?**

22. **Show by a picture how a rectangle with area $x_1 y_2$ minus a rectangle with area $x_2 y_1$ produces the same area as our parallelogram.**

23. **When the edge vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ are perpendicular, the volume of the box is $\|\boldsymbol{a}\|$ times $\|\boldsymbol{b}\|$ times $\|\boldsymbol{c}\|$. The matrix $A^{\mathrm{T}}A$ is _____. Find $\det A^{\mathrm{T}}A$ and $\det A$.**

24. **The box with edges $\boldsymbol{i}$ and $\boldsymbol{j}$ and $\boldsymbol{w} = 2\boldsymbol{i} + 3\boldsymbol{j} + 4\boldsymbol{k}$ has height _____. What is the volume? What is the matrix with this determinant? What is $\boldsymbol{i} \times \boldsymbol{j}$ and what is its dot product with $\boldsymbol{w}$?**

25. **An n-dimensional cube has how many corners? How many edges? How many $(n-1)$-dimensional faces? The cube in $\mathbf{R}^n$ whose edges are the rows of $2I$ has volume _____. A hypercube computer has parallel processors at the corners with connections along the edges.**

26. **The triangle with corners $(0, 0), (1, 0), (0, 1)$ has area $\frac{1}{2}$. The pyramid in $\mathbf{R}^3$ with four corners $(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)$ has volume _____. What is the volume of a pyramid in $\mathbf{R}^4$ with five corners at $(0, 0, 0, 0)$ and the rows of I?**

Problems 27–30 are about areas dA and volumes dV in calculus.

27. **Polar coordinates satisfy $x = r\cos\theta$ and $y = r\sin\theta$. Polar area is $J\,dr\,d\theta$:**

$$J = \begin{vmatrix} \partial x/\partial r & \partial x/\partial\theta \\ \partial y/\partial r & \partial y/\partial\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}.$$

    The two columns are orthogonal. Their lengths are _____. Thus $J =$ _____.

28. **Spherical coordinates ρ, ϕ, θ satisfy $x = \rho\sin\phi\cos\theta$ and $y = \rho\sin\phi\sin\theta$ and $z = \rho\cos\phi$. Find the 3 by 3 matrix of partial derivatives: $\partial x/\partial\rho, \partial x/\partial\phi, \partial x/\partial\theta$ in row 1. Simplify its determinant to $J = \rho^2\sin\phi$. Then dV in spherical coordinates is $\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$, the volume of an infinitesimal "coordinate box".**

29. **The matrix that connects r, θ to x, y is in Problem 27. Invert that 2 by 2 matrix:**

$$J^{-1} = \begin{vmatrix} \partial r/\partial x & \partial r/\partial y \\ \partial\theta/\partial x & \partial\theta/\partial y \end{vmatrix} = \begin{vmatrix} \cos\theta & ? \\ ? & ? \end{vmatrix} = ?$$

    It is surprising that $\partial r/\partial x = \partial x/\partial r$ (*Calculus*, Gilbert Strang, p. 501). Multiplying the matrices J and J^{-1} gives the chain rule $\frac{\partial x}{\partial x} = \frac{\partial x}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial x}{\partial\theta}\frac{\partial\theta}{\partial x} = 1$.

30. **The triangle with corners $(0,0),(6,0)$, and $(1,4)$ has area _____. When you rotate it by $\theta = 60°$ the area is _____. The determinant of the rotation matrix is**

$$J = \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & ? \\ ? & ? \end{vmatrix} = ?$$

Problems 31–38 are about the triple product $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}$ in three dimensions.

31. **A box has base area $\|\boldsymbol{u} \times \boldsymbol{v}\|$. Its perpendicular height is $\|\boldsymbol{w}\| \cos\theta$. Base area times height = volume = $\|\boldsymbol{u} \times \boldsymbol{v}\| \|\boldsymbol{w}\| \cos\theta$ which is $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}$. Compute base area, height, and volume for $\boldsymbol{u} = (2,4,0), \boldsymbol{v} = (-1,3,0), \boldsymbol{w} = (1,2,2)$.**

32. **The volume of the same box is given more directly by a 3 by 3 determinant. Evaluate that determinant.**

33. **Expand the 3 by 3 determinant in equation (13) in cofactors of its row u_1, u_2, u_3. This expansion is the dot product of u with the vector _____.**

34. **Which of the triple products $(\boldsymbol{u} \times \boldsymbol{w}) \cdot \boldsymbol{v}$ and $(\boldsymbol{w} \times \boldsymbol{u}) \cdot \boldsymbol{v}$ and $(\boldsymbol{v} \times \boldsymbol{w}) \cdot \boldsymbol{u}$ are the same as $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}$? Which orders of the rows $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ give the correct determinant?**

35. **Let $P = (1,0,-1)$ and $Q = (1,1,1)$ and $R = (2,2,1)$. Choose S so that $PQRS$ is a parallelogram and compute its area. Choose T, U, V so that $OPQRSTUV$ is a tilted box and compute its volume.**

36. **Suppose (x,y,z) and $(1,1,0)$ and $(1,2,1)$ lie on a plane through the origin. What determinant is zero? What equation does this give for the plane?**

37. **Suppose (x,y,z) is a linear combination of $(2,3,1)$ and $(1,2,3)$. What determinant is zero? What equation does this give for the plane of all combinations?**

38. **(a) Explain from volumes why $\det 2A = 2^n \det A$ for n by n matrices.**

    (b) For what size matrix is the false statement $\det A + \det A = \det(A + A)$ true?

Challenge Problems

39. **If you know all 16 cofactors of a 4 by 4 invertible matrix A, how would you find A?**

40. **Suppose A is a 5 by 5 matrix. Its entries in row 1 multiply determinants (cofactors) in rows 2–5 to give the determinant. Can you guess a "Jacobi formula" for $\det A$ using 2 by 2 determinants from rows 1–2 *times* 3 by 3 determinants from rows 3–5? Test your formula on the $-1, 2, -1$ tridiagonal matrix that has determinant $= 6$.**

41. **The 2 by 2 matrix $AB =$ (2 by 3)(3 by 2) has a "Cauchy-Binet formula" for $\det AB$:**

$$\det AB = \textbf{sum of } (\textbf{2 by 2 determinants in } A)(\textbf{2 by 2 determinants in } B)$$

    (a) Guess which 2 by 2 determinants to use from A and B.

    (b) Test your formula when the rows of A are $1,2,3$ and $1,4,7$ with $B = A^{\mathrm{T}}$.

42. **The big formula has $n!$ terms. But if an entry of A is zero, $(n-1)!$ terms disappear. If A has only *three diagonals*, how many terms are left?**

    For $n = 1,2,3,4$ the tridiagonal determinant has $1,2,3,5$ terms. Those are Fibonacci numbers in Section 6.2! Show why a tridiagonal 5 by 5 determinant has $5 + 3 = 8$ nonzero terms (Fibonacci again). Use the cofactors of a_{11} and a_{12}.

6 Chapter 6: Eigenvalues and Eigenvectors

6.1 Introduction to Eigenvalues

1. **The example at the start of the chapter has powers of this matrix A:**

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \quad \textbf{and} \quad A^2 = \begin{bmatrix} .70 & .45 \\ .30 & .55 \end{bmatrix} \quad \textbf{and} \quad A^\infty = \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}.$$

   Find the eigenvalues of these matrices. All powers have the same eigenvectors.

   (a) **Show from A how a row exchange can produce different eigenvalues.**

   (b) **Why is a zero eigenvalue *not* changed by the steps of elimination?**

2. **Find the eigenvalues and the eigenvectors of these two matrices:**

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \quad \textbf{and} \quad A + I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}.$$

   $A + I$ has the ______ eigenvectors as A. Its eigenvalues are ______ by 1.

3. **Compute the eigenvalues and eigenvectors of A and A^{-1}. Check the trace !**

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \quad \textbf{and} \quad A^{-1} = \begin{bmatrix} -1/2 & 1 \\ 1/2 & 0 \end{bmatrix}.$$

   A^{-1} has the ______ eigenvectors as A. When A has eigenvalues λ_1 and λ_2, its inverse has eigenvalues _________ .

Challenge Problems

6.2 Diagonalizing a Matrix

Challenge Problems

6.3 Systems of Differential Equations

Challenge Problems

6.4 Symmetric Matrices

Challenge Problems

6.5 Positive Definite Matrices

Challenge Problems

7 Chapter 7: The Singular Value Decomposition (SVD)

7.1 Image Processing by Linear Algebra

Challenge Problems

7.2 Bases and Matrices in the SVD

Challenge Problems

7.3 Principal Component Analysis (PCA by the SVD)

Challenge Problems

7.4 The Geometry of the SVD

Challenge Problems